# Measuring Firm Capabilities from Job Postings: An Application to Climate Industrial Policy

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**Abstract**

Firms differ in the activities they are prepared to undertake, yet conventional data record expenditures, assets, and realized outcomes rather than the work firms are hiring to do. I measure that work from the responsibilities that job postings describe. A language model labels a targeted sample of postings for each kind of work, a classifier trained on their text embeddings scores 91.8 million openings on the career sites of 7,128 listed firms over 2014–2023, and calibration turns the scores into the expected share of each firm's hiring devoted to environmental, social, governance, and innovation work. Where an expense line exists, the innovation share recovers R&D spending; where none does, each share predicts records of its own kind of work better than those of the others. On days of news about US climate industrial policy, firm value moves with the environmental share of hiring in the direction of the news, by 0.6 percentage points over three days per standard deviation relative to industry peers, with no pre-trend or reversal. The repricing follows clean-technology rather than environmental management work, is absent from the social and governance shares, survives ratings, emissions, climate talk, and green patents, and is largest among small and asset-light firms, where financial statements record least of what they build.

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*Keywords:* firm capabilities, job postings, text embeddings, machine learning, measurement, climate industrial policy.

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# 1 Introduction

Technological, regulatory, and market changes repeatedly alter the returns to specific activities, from the diffusion of new technologies to the climate policies that reward clean activity, and how a firm fares depends in part on the activities it was positioned to undertake before the change occurred.[1] As artificial intelligence advanced, firms investing in AI-skilled workers experienced greater growth in sales, employment, and market value (Babina et al., 2024), and when the COVID-19 pandemic disrupted economic activity, firms whose pre-pandemic hiring was more oriented toward work that could be performed remotely exhibited better operating performance than their industry peers (Bai et al., 2023). In both cases, the differences that mattered were visible in the composition of firms' workforces or hiring before they appeared in outcomes. Yet such differences remain difficult to observe systematically. Conventional firm-level data provide extensive information about firms' expenditures, assets, and realized outcomes, but much less about the particular expertise they are seeking to acquire and the activities they are preparing to undertake,[2] so that differences in how firms are positioning themselves may go unnoticed until those activities generate measurable outcomes. This paper asks whether the activities toward which firms direct their hiring can be measured systematically, and whether doing so reveals differences in firms' exposure to changes in the returns to those activities that conventional measures miss.

To answer these questions, I develop a flexible approach that measures the share of a firm's new job openings whose responsibilities involve a given kind of work. Job postings suit this purpose. A posting states the responsibilities of a position that the firm has decided to fill, at a cost that it bears once the position is filled, and it is observed before the work produces patents, products, or revenue (Hershbein and Kahn, 2018; Deming and Kahn, 2018; Acemoglu et al., 2022). It describes the work itself rather than the category of the worker, so that two firms hiring in the same occupation can be seen to assign it different work. Using 91.8 million openings that 7,128 listed firms posted on their own career sites over 2014–2023, collected by LinkUp, I apply the approach to the environmental, social, and governance work that makes up corporate sustainability, and to innovation work as a benchmark.

Sustainability is a domain in which such a measure may be particularly informative.

[1] Because the capabilities that underlie such activities are developed over time and give a firm the opportunity to expand into an activity when its returns rise, they can be viewed as real options (Kogut and Kulatilaka, 2001).

[2] Financial statements classify costs by their nature and function rather than by the activity that they serve. Only a few broad activities have expense lines of their own, such as research and development and advertising, and even these record how much a firm spends on an activity rather than which capabilities it develops, so that two firms with similar research and development spending may be developing very different technologies.

Investors increasingly assess firms along environmental, social, and governance dimensions (Hartzmark and Sussman, 2019; Gillan, Koch and Starks, 2021), yet these dimensions have no common definition, and the ratings that measure them disagree markedly across providers (Chatterji et al., 2016; Berg, Kölbel and Rigobon, 2022) and rest largely on what firms disclose (Christensen, Serafeim and Sikochi, 2022). Disclosure itself can be selective, particularly where firms face little scrutiny (Lyon and Maxwell, 2011; Marquis, Toffel and Zhou, 2016), and climate disclosures can amount to cheap talk (Bingler et al., 2022). Hiring differs from these sources in two respects. Because a posting commits the firm to a cost, it reveals where the firm is prepared to invest rather than what it chooses to say, and because it describes work that the firm is preparing to undertake, it looks forward, whereas ratings, emissions, and reported revenue record what the firm has disclosed, emitted, or sold. Earlier evidence suggests that this information is not redundant.[3]

I define each kind of work by the responsibilities it entails. Environmental work manages, measures, reduces, or reports a firm's environmental impacts, or researches, builds, installs, or operates clean technology;[4] social work protects the safety, welfare, and rights of the people whom the firm's operations and products affect;[5] and governance work establishes, operates, or assures the firm's controls and its compliance with the law.[6] To check that the procedure measures what it is meant to measure, I also apply it to an activity whose spending the accounts record. Research and development (R&D) is such an activity, and labor accounts for more than two-thirds of business R&D spending in the United States (Moris and Shackelford, 2023), so that a measure built from hiring should track it. I therefore measure innovation work, which creates new or substantially improved products, processes, or methods,[7] and which, defined as broadly as R&D itself, serves as a positive control before the procedure is

[3]The green skills that firms demand in their postings predict their future profitability and green patents, which environmental ratings do not reflect (Darendeli, Law and Shen, 2022), and the traditional energy firms that produce much of the green innovation in the United States receive low ESG scores (Cohen, Gurun and Nguyen, 2026).

[4]Its boundaries follow the EU taxonomy of sustainable activities (European Parliament and Council of the European Union, 2020), the European sustainability reporting standards on the environment (European Commission, 2023), and the green-jobs definition of the US Bureau of Labor Statistics (2010).

[5]Its scope spans the four social standards of the European sustainability reporting standards (European Commission, 2023), which concern the firm's own workforce, the workers in its value chain, the communities that it affects, and its consumers and end-users, and includes occupational safety, labor rights, supplier audits, community engagement, product safety, and the protection of personal data.

[6]Its scope spans the business conduct that the European standards cover, such as the prevention of corruption, the protection of whistleblowers, and political engagement (European Commission, 2023), and the internal control over operations, reporting, and compliance that the COSO framework defines (Committee of Sponsoring Organizations of the Treadway Commission, 2013).

[7]This is the broad definition of innovation of the Oslo Manual (OECD and Eurostat, 2018), which counts research and development, as the Frascati Manual defines it (OECD, 2015), among the activities that lead to innovation.

applied to activities that no account records.

Reading the responsibilities that postings describe consistently across millions of postings, for capabilities that no fixed vocabulary or occupational classification identifies, requires both judgment and scale, which I combine in a teacher–student design known as distillation (Hinton, Vinyals and Dean, 2015). Each capability is first given a written definition of the work it comprises, applied to the responsibilities that a position entails rather than to its title or to the way the firm presents itself. A sample of postings is labelled by a large language model according to this definition, and a regularized logistic regression on each posting's text embedding (Chen et al., 2024), trained on these labels and calibrated on a random sample, extends the definition to the whole corpus, so that the measure for each firm and year is the expected share of new openings that involve the capability. Because the labels are fixed once produced and the embedding model's weights are public, the whole corpus can be scored again without querying a language model, many of which are proprietary and constantly evolving (Ludwig, Mullainathan and Rambachan, 2026; Törnberg, 2025), and a new capability requires only a written definition and a small labelled sample.

I validate the measure only against data that the language model did not produce (Ludwig, Mullainathan and Rambachan, 2026; Egami et al., 2023), and I begin with innovation, where the answer can be checked. The classification agrees closely with human coding, with a Cohen's $\kappa$ of 0.79,[8] and it ranks postings better than occupation codes or keyword dictionaries do. The share is also highest where innovation is expected. A fifth to a quarter of openings involve innovation work in computer and electronic products, software, and information services, against less than 2 percent in hospitals, restaurants, and general merchandise stores, and among large recruiters the share peaks at semiconductor makers and the developers of chip-design software, such as Advanced Micro Devices, Nvidia, and Cadence, at 60 to 75 percent. More importantly, the share tracks the records of innovation. A firm whose share is one standard deviation higher than those of its industry peers in the same year spends 3.96 percentage points more of its assets on R&D, against an average of 4.6 percent, and files 46 percent more patent applications over the next two years, and both relations hold within firms over time. Markets value the share accordingly. The same difference is associated with a Tobin's $q$ higher by 0.31, about a seventh of the average, and R&D spending accounts for all of this association. Markets thus value the innovation orientation of hiring as they value R&D, as expected if the share records the labor that R&D pays for.

The three kinds of work that no account records pass the same tests. The classification agrees with human coding with a $\kappa$ of 0.84 for environmental work, 0.74 for social work,

[8] I coded a validation sample of 700 postings blind to the model's labels, drawn so as to over-represent postings likely to involve innovation work.

and 0.72 for governance work,[9] and for each it ranks postings better than occupation codes or keyword dictionaries do. Each share is also highest where its work is expected, the environmental share in waste management and utilities, the social share in mining and oil and gas, where work is most hazardous, and the governance share in chemical and pharmaceutical manufacturing and in finance, where quality assurance and regulatory compliance are central. Each share then predicts the records of its own work. Firms with a higher environmental share apply for 23 percent more climate-change patents over the next two years but for no more patents in other classes, so that the share predicts the direction of their invention rather than its amount, a relation that remains, at a little over half its size, within firms. The social and governance shares are higher where their work is most needed. Within a firm and year, the social share of a state's openings is higher where the firm's establishments record more workplace injuries, and firms with a higher governance share face 21 percent more securities suits over the next two years. These relations show where the work is placed rather than what it achieves. A rise in a state's social hiring is not followed by fewer injuries, and the governance share as a whole does not predict material weaknesses in internal control. Across the four kinds of work, each share predicts the records of its own work more strongly than those of the others, the pattern of convergent and discriminant validity (Campbell and Fiske, 1959). Each share is also persistent within firms, with a year-on-year correlation between 0.82 and 0.89.[10]

To address the second question, I turn to climate industrial policy, a setting in which news changes the returns to one specific activity, which I expect the environmental share of hiring to capture. The period that my data cover includes six days of major news about US climate policy, the days that earlier studies of such news examine, three favorable, the Paris Agreement, the call of the 2020 presidential election for Joe Biden, and the agreement on the Inflation Reduction Act (IRA) of 2022,[11] and three unfavorable, the elections of Donald Trump in 2016 and 2024 and the announcement of the withdrawal from the Paris Agreement.

[9] For each of the three, I coded a validation sample of 800 postings in the same way.

[10] Because the shares change slowly, comparisons of a firm with itself over time rest on little variation and are a demanding test. Some relations nonetheless hold within firms. The innovation share tracks R&D and patenting, the environmental share green patenting at a little over half the size of the relation across firms, the social share the injuries of a firm's establishments across the states in which it operates, and the governance share derivative and fraud suits.

[11] The agreement on the IRA is the cleanest of the six days, because it concerned a single policy and came as a surprise. On 14 July 2022, press reports that Senator Manchin, the pivotal vote in the Senate, would not support new climate spending made the legislation appear all but abandoned, and on 27 July he and the Democratic leadership announced their agreement on it, so that within two weeks the probability of its passage fell close to zero and then rose close to one (Bauer, Offner and Rudebusch, 2026). The IRA rewards firms for clean activity, largely through uncapped tax credits per unit of clean output or investment (Bistline, Mehrotra and Wolfram, 2023), until Congress curtailed several of these credits in July 2025 (U.S. Congress, 2025).

If hiring reveals the activities that firms are positioning themselves to undertake, firms whose hiring is oriented toward environmental work should gain in value relative to their peers when the prospects of climate policy improve and lose when they worsen. These studies reach conflicting conclusions depending on how they measure exposure. Those that rely on environmental ratings find that highly rated firms gained on days that moved climate policy in opposite directions, on the 2016 election (Ramelli et al., 2021) as on the agreement on the IRA (Bauer, Offner and Rudebusch, 2026), and energy firms with high ratings gained after the 2024 election (Martins et al., 2025). Those that rely on the clean products that firms sell find responses in the direction of the news (Kruse, Mohnen and Sato, 2024; Lu et al., 2025*a*), but such measures record activity that already generates revenue rather than the capability that a firm is developing, and green patents and reported clean segments, which record the output of clean work, exist for only 18 and 2 percent of the firm-years in my sample. Hiring, by contrast, records the capability that a firm is building ex ante and covers any firm that posts through its own career site; I therefore study how firms' stock returns over the three days around each event relate to the environmental orientation of their hiring in the year before, comparing each firm with the other firms of its industry.

The first result is that firm value moves with the prospects of the policy, in both directions. Across these six days, a firm whose environmental share is one standard deviation higher than those of its industry peers earns 0.59 percentage points more in the direction of the news over the three days around each event ($t = 6.6$), gaining relative to its peers when the prospects of the policy improve and losing when they worsen. Analysts also revise these firms' long-term growth forecasts in the direction of the news, but not their earnings forecasts for the next three years, consistent with a change in the cash flows expected from activity that lies years ahead, as the policy's payments do, rather than in current profits.[12] In regression-implied terms, these differences amount to a relative revaluation of $48 to $56 billion on the agreement on the IRA, depending on whether firms are weighted by market value or equally, mostly among the tenth of firms furthest above their industry's mean.[13]

A natural concern is that the environmental share proxies for other characteristics that the same news reprices, and several tests suggest that it does not. The share enters with the sign of the news on five of the six days, the exception being the 2020 election, which coincided with news about COVID-19 vaccines. I also run the pooled regression on 20,000 sets of placebo days, each drawing one day from the two years on either side of each event,

[12]Long-term growth forecasts exist for about 800 firms per day, and the revisions concentrate on the agreement on the IRA and the 2024 election, so these estimates are less precise than the price responses and cannot exclude that part of the repricing reflects the rates at which investors discount distant cash flows.

[13]These amounts are differences in value relative to industry peers, not subsidies or aggregate gains.

and only one of them[14] yields an estimate as large. Controls for environmental ratings, emissions, climate talk, accounting intangibles, green patents, clean revenue, and exposure to taxes and trade each lower the estimate by at most 30 percent, and more than two fifths of it remains when all of them are included within six-digit industries. The timing of the response also separates the policy news from other news released around the same days. On the agreement on the IRA, the cleanest of the days, the response peaks in the first session after the announcement rather than on the preceding day, when the Federal Reserve raised interest rates. On the four days whose news arrived after the market closed, most of the response appears in the return from that close to the next morning's open, so that news released during the following session cannot account for it.[15]

The second result is that the repricing is specific to the work that the policy pays for. I group the fifteen classes of environmental function, from the closed list that bounds the labelling of environmental work, into two parts, clean technology, whose work develops, builds, and operates clean-energy, storage, and clean-mobility assets, and environmental management, whose work manages and reports environmental impacts and provides environmental services, which allows me to test this specificity directly.[16] The clean-technology part of the share carries the repricing, whereas environmental management is repriced about a sixth as much. Holding the orientation of hiring fixed, neither emissions nor the environmental score of LSEG moves systematically in that direction. Clean-technology hiring also anticipates the activity that the policy rewards, since US firms whose clean-technology share was higher in 2019–2021 were more likely to mention the policy's tax credits in later annual reports and to announce clean manufacturing plants, and these firms invested more in the year after the agreement on the IRA, although their investment shows no reliable response to the news of other years. Together, these results suggest that hiring identifies the firms positioned to undertake the activity that the policy rewards, and that their value responds to changes in the prospects of that specific activity.

The third result is that the repricing is largest where the capability is a large part of the

[14] This corresponds to a placebo $p$-value of 0.0001.

[15] Further tests address product-market rivals, exposures to interest rates, oil, and the market, political alignment, the selection of firms into the data, firm-specific news, and two further days of policy news. Elections nevertheless carry news about many policies besides climate policy, which limits what any single day can show.

[16] Clean technology groups six classes (renewable energy and energy efficiency; electrification and storage; clean mobility; hydrogen, carbon capture, and clean fuels; nuclear power; and green buildings), and environmental management the other nine, from permitting, waste, water, pollution, and air emissions to greenhouse-gas accounting, sustainability disclosure, climate risk, and biodiversity. A separate regression scores each part, and each posting's calibrated probability is credited to the part whose regression predicts it more strongly, so that the two parts sum to the environmental share. Of the 5,748 positive postings in the labelled samples, 57 percent fall in the clean-technology classes.

firm, in asset-light and small firms, whose balance sheets record least of the capabilities that they develop. It is also more than twice as large among firms that analysts follow, even when size and tangible intensity are held fixed, consistent with analysts producing information about a capability that financial statements do not record. Nor is the repricing confined to the sectors in which clean activity is most visible. It holds within manufacturing and across the sectors outside manufacturing, utilities, and energy, which together account for nine in ten of the firm-events, so that the orientation of hiring distinguishes firms within the broad range of industries in which clean activity is a choice rather than the business itself.

**Related Literature.** The paper contributes to three strands of literature. First, it relates to measures of specific capabilities and exposures. Recent work measures specific capabilities from workers' skills or job postings, such as digital capital, artificial intelligence, and firms' stocks of data (Tambe et al., 2020; Babina et al., 2024; Abis and Veldkamp, 2024), and firm-level exposures from the content of earnings calls (Hassan et al., 2019; Sautner et al., 2023*a*). A related literature measures green jobs and skills from occupations and job postings (Vona et al., 2018; Popp et al., 2021; Saussay, Sato and Vona, 2026; Curtis and Marinescu, 2023; Darendeli, Law and Shen, 2022; Hagendorff, Nguyen and Sila, 2025; Ben Dor et al., 2022; Chen, 2025; Liao et al., 2026), and Vona et al. (2018) distinguish the engineering skills of green technologies from the managerial skills of environmental practices at the level of occupations. I draw the same distinction within occupations, from the responsibilities that each posting describes, and show that it matters for valuation.

Second, the paper relates to studies of how climate-policy news and political events affect firm value (Meng, 2017; Wagner, Zeckhauser and Ziegler, 2018; Ramelli et al., 2021; Kruse, Mohnen and Sato, 2024; Bauer, Offner and Rudebusch, 2026; Martins et al., 2025; Lu et al., 2025*a*) and to measures of firms' climate exposure, from emissions (Bolton and Kacperczyk, 2021, 2023; Hsu, Li and Tsou, 2023) and ratings (Pástor, Stambaugh and Taylor, 2022; Engle et al., 2020) to earnings-call talk (Sautner et al., 2023*a*,*b*), green patents (Hege, Zhang and Pouget, 2025; Cohen, Gurun and Nguyen, 2026), and the climate-solution products that firms describe in their 10-K filings (Lu et al., 2025*b*). I view the hiring measure as a complement to these measures rather than a substitute. Unlike talk, it records commitments to hire. Unlike patents and reported segments, it is observed before clean work produces output and more uniformly across firms. Holding it fixed, the LSEG environmental score does not move in the direction of the news, which may help reconcile the evidence that highly rated firms gained on days that moved the prospects of climate policy in opposite directions. The evidence also separates the channel through which this news reaches firm value from the two that this literature emphasizes, investors' tastes for green assets (Pástor, Stambaugh and Taylor, 2021, 2022) and the risk of carbon pricing (Bolton and Kacperczyk, 2021, 2023), since neither

ratings, nor the overweights of environmentally minded funds, nor emissions account for the repricing. The repricing follows instead the growth opportunities in the activity that the policy pays for, and it depends on their direction rather than, as with the shocks to the return on new investment that Kogan and Papanikolaou (2014) study, on how large a part of firm value growth opportunities represent.

Finally, the paper relates to the measurement of intangible investment, much of which consists of the labor of employees who develop new products, processes, and organizational capabilities. For the intangible assets that firms produce for their own use, ranging from the software counted in the national accounts to the organizational capital that research adds to them, the investment is valued from the compensation of the occupations involved and the share of their working time assumed to be devoted to the asset (Corrado, Hulten and Sichel, 2005; Corrado et al., 2022). Recent work at the US Bureau of Economic Analysis extends this approach to data assets and estimates part of this share from online job advertisements, as the fraction of jobs in each occupation whose required skills involve data-related activities (Santiago Calderón and Rassier, 2025). The share of a firm's new openings whose responsibilities involve a given activity provides this fraction firm by firm and activity by activity, rather than occupation by occupation.

Overall, the evidence suggests that hiring demand can be informative about firms' productive orientation, and the measure may be useful to several audiences. For researchers, it offers a reproducible way to measure activities that no account records, which extends to any activity that can be defined by the work it requires. For investors and analysts, it complements ratings as a measure of exposure to the reversal of a policy, and it is available before patents or reported segments appear, since the estimate remains among firms without a green patent and among firms without a reported clean segment. For policymakers, it shows which firms an industrial policy reaches at a stage when conventional measures still show little difference between them.

The rest of the paper proceeds as follows. Section 2 describes the data, Section 3 the measure, and Section 4 validates it across four kinds of work. Section 5 introduces the application to climate industrial policy and the earlier evidence on it. Section 6 relates the environmental share to firm value on policy days, Section 7 examines what explains the relation, and Section 8 where it is largest. Section 9 concludes.

# 2 Data

The analysis combines job postings with financial statements, market prices, patents, public records of firms' environmental, workplace, and legal activity, and existing measures of

climate exposure. Because these sources identify firms in different ways, I link each of them to issuers in Compustat through dated links, so that each record is assigned to the issuer that consolidated the relevant entity at the time. Internet Appendix IA.III documents each source and each step of the construction, and Appendix Table A1 defines every variable.

## 2.1 Job Postings

The postings come from LinkUp, which indexes job postings exclusively from employers' own career websites, never from job boards, aggregators, or the listings that staffing agencies publish for other employers (LinkUp, 2020). LinkUp checks each career site at least every 48 hours and identifies each posting by its web address, so that a job advertised once is recorded once, with the date on which it first appeared and the date on which it was removed (LinkUp, 2020, 2026). This design suits a measure that reads the work a firm assigns, for three reasons that LinkUp's documentation describes (LinkUp, 2020). The data contain no duplicate listings, whereas data aggregated from job boards see the same advertisement syndicated many times and must remove duplicates statistically (Campello, Gao and Xu, 2024). A posting is dated by its removal from the employer's own site rather than by an expiry date fixed in advance by the fee paid to a board (Campello, Kankanhalli and Muthukrishnan, 2024). And every posting is written by the employer itself, in its own terms, free of the scams and lead-generation listings that pollute job boards.

Table 1 traces the postings from LinkUp's files to the estimation sample. The raw data files hold 257.9 million postings created between 2007 and 30 September 2023 on 61,519 career sites, 200.4 million of which carry a description. I keep the 126.3 million postings with a description on the 16,482 career sites that LinkUp links, at some point, to the stock ticker of a listed company, and I clean their text of markup, encoding errors, web and e-mail addresses, and telephone numbers, without dropping any posting for its length, language, or location. Two corrections make each opening correspond to a single job that the employer itself advertised. Some career sites give a job a new web address at every visit, so that one vacancy reappears as a string of new postings, and I merge each string back into one opening. The correction removes 5.0 million postings but is concentrated on a few sites and leaves the counts of the others almost unchanged. A few sites were also crawled for a time through job boards that carry other employers' jobs, and I drop those postings. Finally, the data begin in 2014. No posting in the data is recorded as removed before that year, so the 0.25 million postings created earlier appear only because they were still open in 2014, without the many jobs advertised alongside them that were filled sooner. I drop them, so that every opening is both advertised and removed within the period that the data cover.

I date each opening by the year in which it is removed from the career site. Removal is the observed event closest to the filling of the position, whereas creation records when the firm decided to recruit, and dating by removal also assigns each opening to the issuer that owns the site when the position is filled, the firm that makes the hire. A removed posting may have been withdrawn rather than filled, which the data cannot tell apart, but the choice of date rarely changes the year, since the median posting stays open for three to four weeks and about one in seven is removed in a later calendar year than it was created. Dating by removal requires two restrictions, and I apply both conservatively. I drop the 2.8 million postings still open at the end of the data, whose removal is not observed, and the 7.6 million open for more than 180 days, which LinkUp advises treating as evergreen requisitions, kept open on purpose, rather than as vacancies (LinkUp, 2020). None of these choices drives the measure. Computed with creation dates, without the 180-day limit, with a 90-day limit, with the postings still open at the end of the data, or with a job advertised for several locations counted once, the four shares correlate with their baseline values at 0.99 across firm-years, and the main estimates of Sections 4 and 6 change little when postings of any duration, and those still open at the end of the data, are kept (Appendix Table E9).

These steps leave 110.6 million postings. The crosswalk described in Section 2.2 assigns 100.4 million of them to Compustat issuers, and 94.5 million to issuers that consolidate the employer, in 45,751 firm-years of 9,161 issuers.

In the aggregate, the postings follow the US labor market. The US postings of the 1,488 sample firms that post in the United States in every year from 2015, counted at the end of each month, correlate with the private-sector job openings of the Job Openings and Labor Turnover Survey at 0.87 in levels and at 0.85 in twelve-month changes. Relative to US employment, they over-represent computer, sales, engineering, and management occupations and under-represent education, construction, and production occupations. Part of this gap comes from employers that rarely advertise on their own sites, such as public agencies and small contractors, and matters little here, because the paper compares listed firms within industries. Part may come from jobs that the firms themselves never post on their sites, such as those filled through staffing agencies, whose listings LinkUp excludes. The share therefore describes the jobs that a firm advertises, and Section 4 tests whether it nonetheless tracks the activity that independent records report. The measure uses all of a firm's postings, wherever the job is located, because the financial statements with which it is compared consolidate the firm's operations worldwide.

**Table 1.** From LinkUp's Records to the Estimation Sample

| Step | Postings | Career sites | Firm-years | Issuers |
|---|---|---|---|---|
| LinkUp records, created 2007 to 30 September 2023 | 257,908,254 | 61,519 | | |
| with a description | 200,410,559 | 50,291 | | |
| on career sites linked to a listed company's ticker | 126,305,916 | 16,482 | | |
| created from 2014 | 126,059,949 | 16,381 | | |
| with a measure score | 126,056,845 | | | |
| less: copies of jobs on the site and job-board postings | −5,034,452 | | | |
| removed by 30 September 2023 | 118,234,531 | | | |
| open at most 180 days | 110,588,197 | 15,795 | | |
| assigned to an issuer at the removal date | 100,437,722 | | | |
| less: issuers that do not consolidate the employer | −5,910,241 | | | |
| openings in consolidated firm-years | 94,527,481 | | 45,751 | 9,161 |
| Estimation sample: ≥20 openings and Compustat fundamentals | 91,774,781 | | 35,772 | 7,128 |
| of which with a CRSP security (market sample) | 74,222,591 | | 24,610 | 4,497 |

*Notes.* The first two rows count all records in LinkUp's files. The third keeps the postings with a description on the career sites that LinkUp's company reference links, at some point, to the primary ticker of a listed company, with their text cleaned; the crosswalk of Section 2.2 links nearly all of these sites to an issuer. Copies of a job already on the site are postings that reappear with the same title and location within three days of an earlier posting's removal and are themselves removed within three days; each is merged into the posting it continues. Job-board postings are those that five career sites drew for a time from boards listing other employers' jobs. Openings are dated by the calendar year of their removal from the career site. Postings not assigned to an issuer belong to private employers, to unlisted subsidiaries without a consolidating parent, or to career sites that the crosswalk could not link. Issuers that do not consolidate the employer are holders of minority stakes and joint ventures, landlords of operating companies, and managers of investment funds; the same line removes franchisees' openings on franchisors' career sites. Career sites: sites with at least one posting in the row, where counted. Consolidated firm-years: all issuer-years with at least one opening, before the requirements of the estimation sample, which covers 2014–2023. Market sample: firms of the estimation sample in Compustat North America with a CRSP security.

## 2.2 Linking Career Sites to Issuers

LinkUp identifies each career site by its own company identifier, but the employer behind a site can change, and a listed firm with several brands or subsidiaries can have several sites. I therefore link each site to Compustat issuers through dated intervals, each a period during which the site is linked to one issuer. For each site, LinkUp records the listed company with which it is associated, through the dated history of its primary ticker, legal entity identifiers, and Refinitiv PermIDs. I match the identifiers to Compustat through the security identifiers that Refinitiv and the Global Legal Entity Identifier Foundation associate with them, through S&P Capital IQ, and through the CRSP–Compustat Merged link, and I match the tickers to the dated ticker histories of Capital IQ, Compustat, CRSP, Datastream, and Worldscope. Each candidate is graded by the similarity of the site's name and web domain to those of the issuer, and a single issuer is retained for each period over which LinkUp associates the site

with the same listed company. The resulting crosswalk comprises 18,412 intervals that link 15,585 career sites to 10,514 issuers. Links that rest on an exact match of identifiers carry 58 percent of the postings assigned to issuers, and links that rest on a dated ticker confirmed by names or web domains a further 27 percent, while I ruled by hand on the pairs that rest on a ticker alone and reviewed the links most exposed to error against corporate events and each site's own postings.

Each posting is assigned to the issuer whose interval contains its removal date, so that a site's openings follow its owner through splits, spin-offs, acquisitions, and sales. I exclude postings when the issuer does not consolidate the employer, as for minority stakes, joint ventures, and the portfolio companies of investment funds, and I exclude the positions of independent franchisees that franchisors list on their own sites. The openings of a firm-year are therefore those of the entities whose costs its financial statements consolidate.

## 2.3 Financial Statements and the Estimation Sample

The unit of observation is the issuer as Compustat defines it. Financial statements come from Compustat North America and, for issuers listed elsewhere, from Compustat Global. I match the fiscal year with the most months in calendar year $t$ to year $t$, convert values reported in other currencies to US dollars at the exchange rate of the month in which the fiscal year ends, and classify each issuer by its historical NAICS code or, where that is missing, by its current one. When an issuer appears in both databases in the same year, its North American record is used. Intangible capital is the knowledge and organization capital that Peters and Taylor (2017) accumulate from past R&D and SG&A spending, from the data that WRDS distributes.

The estimation sample contains the firm-years with at least 20 openings, below which a share is too noisy to be informative, and with Compustat fundamentals. It comprises 35,772 firm-years of 7,128 firms over 2014–2023, with 91.8 million openings (Table 2). The threshold removes 13.8 percent of issuer-years but only 0.05 percent of openings. Most of these openings, 79.7 million, come from the 4,986 firms in Compustat North America, of which 3,987 are headquartered in the United States. The other 2,172 firms are listed elsewhere, most often in the United Kingdom, Australia, Japan, France, and China. Coverage rises over the period. The sample includes 17 percent of the firms in Compustat North America in 2014 and 42 percent in 2023, but these firms account for 61 percent of the employees and 65 percent of the market value of all firms in Compustat North America in 2014, and for 90 and 95 percent in 2023. The firms that the sample misses are therefore mostly small. The median firm in the North American sample has 4,100 employees and $3.1 billion in assets, against 600 employees

and $0.5 billion for the median firm in Compustat North America (Table 3). Manufacturing accounts for 38 percent of the firm-years, and retail trade, which hires many workers per firm, for 22 percent of the openings. A firm's openings also scale with its size and move with its growth, with an elasticity of 0.67 to employment across firms and of 0.77 from one year to the next within firms, as they should if they record the firm's own hiring.

**Table 2.** The Sample by Year

| | Sample firms | | | Openings | Compustat | Share in the sample (%) | | |
|---|---|---|---|---|---|---|---|---|
| Year | All | NA | Global | (million) | NA firms | firms | employees | mkt. value |
| 2014 | 1,632 | 1,490 | 142 | 1.3 | 8,976 | 16.6 | 60.7 | 65.2 |
| 2015 | 2,300 | 2,000 | 300 | 2.7 | 8,644 | 23.1 | 69.0 | 74.6 |
| 2016 | 2,561 | 2,142 | 419 | 6.0 | 8,371 | 25.6 | 73.5 | 80.7 |
| 2017 | 3,105 | 2,544 | 561 | 7.7 | 8,188 | 31.1 | 78.6 | 84.5 |
| 2018 | 3,831 | 2,959 | 872 | 9.4 | 8,028 | 36.9 | 82.6 | 88.9 |
| 2019 | 3,963 | 3,004 | 959 | 10.8 | 8,025 | 37.4 | 83.5 | 90.8 |
| 2020 | 4,052 | 2,973 | 1,079 | 9.2 | 8,119 | 36.6 | 86.1 | 92.3 |
| 2021 | 4,620 | 3,314 | 1,306 | 16.0 | 8,167 | 40.6 | 88.1 | 93.7 |
| 2022 | 4,858 | 3,504 | 1,354 | 17.9 | 8,130 | 43.1 | 89.7 | 94.5 |
| 2023 | 4,850 | 3,307 | 1,543 | 10.6 | 7,947 | 41.6 | 89.6 | 94.6 |
| *2014–2023* | 7,128 | 4,986 | 2,172 | 91.8 | 12,947 | 33.0 | 80.3 | 87.7 |

*Notes.* Estimation sample: firm-years 2014–2023 with at least 20 new openings and Compustat fundamentals. NA: Compustat North America; Global: Compustat Global, for firms listed elsewhere, mainly in the United Kingdom (313 firms), Australia (228), Japan (195), France (178), and China (172). Openings are dated by the year of removal from the career site, and the data end on 30 September 2023. Compustat NA firms have positive total assets. The last three columns give the share of these firms, of their employees, and of their market value that the sample includes; the 2014–2023 row counts firms once and pools firm-years.

**Table 3.** Firm Characteristics

| | Sample, NA | | Sample, Global | | Compustat NA | |
|---|---|---|---|---|---|---|
| | mean | median | mean | median | mean | median |
| Assets ($ billion) | 35.7 | 3.1 | 12.0 | 3.7 | 15.1 | 0.5 |
| Sales ($ billion) | 9.1 | 1.6 | 8.4 | 2.5 | 3.7 | 0.1 |
| Employees (thousands) | 22.1 | 4.1 | 27.7 | 9.9 | 10.8 | 0.6 |
| Market value ($ billion) | 16.2 | 2.8 | | | 6.7 | 0.3 |
| Age (years in Compustat) | 24.7 | 21.0 | 21.7 | 22.0 | 16.9 | 11.0 |
| Tobin's $q$ | 2.09 | 1.46 | | | 6.80 | 1.41 |
| Leverage | 0.29 | 0.26 | 0.25 | 0.23 | 0.45 | 0.20 |
| Return on assets | 0.07 | 0.09 | 0.12 | 0.11 | -0.44 | 0.03 |
| R&D / assets | 0.05 | 0.00 | 0.02 | 0.00 | 0.09 | 0.00 |
| SG&A / assets | 0.21 | 0.12 | 0.19 | 0.13 | 0.56 | 0.11 |
| Capex / assets | 0.04 | 0.02 | 0.04 | 0.03 | 0.05 | 0.02 |
| Openings per firm-year | 2,924 | 368 | 1,420 | 235 | | |
| Openings per 100 employees | 20.7 | 13.0 | 9.5 | 3.8 | | |
| Firm-years | 27,237 | | 8,535 | | 82,595 | |

*Notes.* Estimation sample, NA, and Global as in Table 2. Firm-years 2014–2023, in US dollars. Tobin's $q$ is assets plus market value minus book equity and deferred taxes, over assets (NA only); leverage is debt over assets; return on assets is operating income before depreciation, and R&D, SG&A, and capital expenditure are expenditures, over lagged assets. Age counts the years since the firm's first fiscal year in Compustat. Ratios are winsorized at 1 and 99 percent within year.

## 2.4 Patents

Patent applications are those published by the US Patent and Trademark Office, from Google Patents, with their assignees and technology classes. I attribute them to issuers through three existing patent–firm links, those of Ma (2026), Dyèvre and Seager (2023), and WRDS, each dated against the crosswalk between career sites and issuers (Section 2.2), so that patents and postings are attributed to issuers through the same dated links. Applications that these links miss are matched through the dated names of issuers and their subsidiaries . I count applications by year of filing in two classes of green technology. The first is the Y02 class of the Cooperative Patent Classification, which tags technologies for the mitigation of and adaptation to climate change. The second covers the clean-energy and electric-vehicle technologies that the IRA subsidizes, which I call IRA-class applications.[17] I code firm-years without an attributed application as zero. The economic value of granted patents is the stock market reaction to their grant, from the extended data of Kogan et al. (2017).

[17]The IRA classes are Y02E, Y02T, Y02C, H01M, H02S, F03D, C25B, and B60L of the Cooperative Patent Classification. The 2023 cohort covers applications filed through September 2023, the end of the posting data.

## 2.5 Records of Environmental, Workplace, and Legal Activity

The validation relies on records that are independent of the job postings. Where a record carries an identifier that a data vendor or regulator assigns, I link it to issuers through that identifier. Otherwise, I link it through a panel of dated names of issuers and their subsidiaries, built from Compustat, CRSP, the career sites in LinkUp, and the subsidiaries that issuers list in their Exhibit 21 filings.

For environmental work, the Toxics Release Inventory of the Environmental Protection Agency records the toxic chemicals that industrial facilities release each year, and I match facilities to issuers through the parent company that each facility reports. Formal enforcement cases at these facilities come from the agency's Enforcement and Compliance History Online, and environmental incidents reported in the news come from RepRisk. Two further sources, which Section 7.2 uses, concern the activity that the clean-energy tax credits reward. Mentions of these credits in 10-K filings come from the full-text search of SEC EDGAR, and clean manufacturing, energy, and industrial facilities come from the announcements in the Clean Investment Monitor, matched by the names of their developers. For social work, the Injury Tracking Application of the Occupational Safety and Health Administration records the injuries and hours worked that establishments report each year from 2016, which I link through employer identification numbers and names. For governance work, securities suits are the federal cases against SEC registrants that Audit Analytics records. In each source, I code firm-years of covered firms without an event as zero.

## 2.6 Existing Measures of Climate Exposure

I compare the measure with the main existing measures of climate exposure. Greenhouse gas emissions come from Trucost, matched through the Compustat identifier that S&P Global records for each company, and cover 87 percent of the firm-years of the estimation sample; Trucost estimates emissions where firms do not report them, and reported values are available mostly from 2020 onward. Following Ramelli et al. (2021), I measure carbon intensity as the logarithm of Scope 1 and 2 emissions over market equity. Environmental ratings are the environmental pillar scores of LSEG, formerly Refinitiv, whose history LSEG recomputes under its current methodology (Berg, Fabisik and Sautner, 2021), matched through ISIN or CUSIP and covering 82 percent of the firm-years. Climate talk is the share of a firm's earnings calls devoted to climate change, overall and by topic, measured by Sautner et al. (2023*a*), whose data carry the Compustat identifier and cover 83 percent of the firm-years. Measures of green work built from occupations use the green task statements and green occupations of O*NET, which I carry to the occupation codes of the postings through O*NET's crosswalk

between its taxonomies.

I also measure the clean businesses that firms report. From the business segments in Compustat, I classify a segment as clean technology if its name refers to renewable power, solar, wind, batteries or energy storage, electric vehicles or charging, hydrogen or fuel cells, biofuels, renewable natural gas, carbon capture, or clean or low-carbon energy, or if its industry code is that of renewable power generation, storage batteries, or ethanol. The clean revenue share is the revenue of these segments over the revenue of all segments.

### 2.7 Market Data

The application uses market data for the issuers of Compustat North America with a CRSP security, which I link to Compustat through the CRSP–Compustat Merged link, using the primary link valid at the end of each fiscal year for annual variables and on the day of each event for event returns. This market sample comprises 24,610 firm-years of 4,497 issuers. Abnormal returns are measured against the five factors of Fama and French (2015) and momentum (Carhart, 1997). Analysts' forecasts come from IBES, option prices from the volatility surfaces of OptionMetrics, the holdings of mutual funds and institutions from the CRSP Mutual Fund database and from the 13F filings that Thomson Reuters compiles, product-market rivals from the text-based network industries of Hoberg and Phillips (2016), and interest rates and oil prices from FRED. Because policy news can also reprice firms through their exposure to taxes and trade, as Wagner, Zeckhauser and Ziegler (2018) show for the 2016 election, I measure this exposure, following them and Ramelli et al. (2021), by the firm's cash effective tax rate, the foreign share of its pretax income, and the foreign share of its sales, from Compustat and its geographic segments.

## 3 Measuring Capabilities from Job Postings

This section describes how the measure is built. The procedure applies to any capability that can be defined by the work it requires, and I apply it to four: environmental, social, governance, and innovation work.[18] I describe each step for environmental work, the capability of the application in Section 5. I first define it and its two parts, then describe how postings are classified, how accurately, and how the classifications add up to a firm-year share. Finally, I describe how the environmental share varies across firms, industries, and time. Section 4 validates the measure.

[18]The labelling instructions used for each of the four capabilities are reproduced in Internet Appendix IA.I.

## 3.1 What Counts as Environmental Work

Existing measures of specific capabilities draw on the workforce, classifying workers or postings by their occupations, skills, or keywords (Tambe et al., 2020; Babina et al., 2024; Saussay, Sato and Vona, 2026; Darendeli, Law and Shen, 2022; Chen, 2025). The procedure of this paper departs from them in three ways. First, it infers the capability from the responsibilities that each position involves rather than from its occupation, skills, or keywords, which matters because a capability is often developed by occupations that also perform other work; the engineers who design a battery pack and those who manage a plant's environmental permits, for instance, can hold the same title. Second, it applies a written definition of the capability rather than a list of words, so that the same procedure extends to other capabilities. Third, it assigns every posting a calibrated probability, so that the firm-year measure is an expected share rather than a count of postings that pass a threshold (Section 3.4).

The procedure starts from a written definition of the capability. For environmental work, it asks a single question of each posting: does the position assign the hire work that manages, measures, reduces, or reports environmental impacts, or that researches, builds, installs, or operates clean technology? The answer comes from the duties that the posting assigns, following the task-based view of jobs, in which a job is a bundle of tasks that differs across firms even under the same title (Autor, Levy and Murnane, 2003). Three shortcuts are ruled out. The employer does not count, since an accountant at an electric-vehicle maker does no environmental work. The title does not count, since a "project engineer" can do almost anything. And what the firm says about itself does not count, since mission statements and values describe the firm rather than the job. A duty qualifies only if it names both an action and its object, as in "conduct spill inspections," so that generic instructions such as "ensure compliance" are not enough.

Reporting and regulatory frameworks fix where environmental work begins and ends: the EU taxonomy of sustainable activities (European Parliament and Council of the European Union, 2020), the European sustainability reporting standards on the environment (European Commission, 2023), and the green-jobs definition of the Bureau of Labor Statistics (U.S. Bureau of Labor Statistics, 2010). The frameworks also settle the boundary cases. Collecting or treating waste is environmental work when it is the job's own service. Selling green products is not, unless the seller designs or sizes the system. Postings with the same title show how the rule separates them (Table 4). A process engineer at Siemens Gamesa who develops the "manufacturing processes required for Wind Turbines" does clean-technology work, whereas a process engineer at Sealed Air, whose posting describes how the firm helps its customers "achieve their sustainability goals" but assigns extrusion and printing processes, does no environmental work. Likewise, a chemist at American Water who tests "various water

matrices" against the standards of the Environmental Protection Agency does environmental management work, whereas a chemist who analyzes finished products at a beverage maker does not. The table shows similar pairs for the other three capabilities.

**Table 4.** Similar Titles, Different Work

| | Title | Employer, year | Score | What decides it |
|---|---|---|---|---|
| E | Process Engineer | Siemens Gamesa Renewable Energy, 2021 | 2 | "developing and improving new products and manufacturing processes required for Wind Turbines" (class: energy) |
| E | Process Engineer | Sealed Air, 2017 | 0 | company description of "helping our customers achieve their sustainability goals"; duties in "extrusion, printing, and converting processes" |
| E | Chemist II | American Water, 2020 | 2 | "testing for various water matrices" so that data "meets EPA and accrediting authority standards" (class: water) |
| E | Chemist II | Cott, 2017 | 0 | routine analysis of finished products and concentrates; no environmental duty |
| S | Site Supervisor | Lehigh Hanson, 2019 | 1 | "Lead the safety culture daily at the plant and lead safety meetings" (class: occupational health and safety) |
| S | Maintenance Supervisor | Durham School Services, 2017 | 0 | employer "committed to exceptional safety"; no assigned safety duty |
| G | Senior Accountant | Iron Bow Technologies, 2018 | 1 | "Assist in the installation, modification and documentation of accounting processes, systems and internal control procedures" (class: financial risk controls) |
| G | Senior Accountant | Invesco, 2015 | 0 | reconciliations and support to the controller; no control duty |
| I | Mechanical Engineer | L3 Technologies (Unmanned Systems), 2017 | 2 | "Design mechanical components and assemblies to meet program requirements" (class: physical product) |
| I | Mechanical Engineer | Eastman Chemical (Flexsys), 2015 | 0 | "accountable for the mechanical maintenance and reliability engineering function"; troubleshooting and equipment repairs |

*Notes.* Postings from the labelled samples, with the score of the language model under the instructions in use (0: no qualifying duty; 1: a peripheral qualifying duty; 2: the capability is a main purpose of the position) and, for positives, the class of the work (Internet Appendix IA.I). Quotations are verbatim, including the employers' wording. E: environmental; S: social; G: governance; I: innovation.

Every posting receives a score of 0, 1, or 2. A score of 2 means that environmental work is the main purpose of the position, 1 that the posting assigns at least one qualifying duty peripheral to its purpose, and 0 that it assigns none. The measure counts any qualifying work, because peripheral duties, such as those of a plant manager who runs environmental inspections, are a large part of how firms carry out such work. Each positive posting is also assigned one of fifteen classes that describe its main environmental function, from a closed list in the labeller's instruction (Internet Appendix Table IA.13). I group these classes into

the two parts of environmental work used throughout the paper, according to whether the work concerns the technologies that generate, store, or use low-carbon energy, or the firm's environmental impacts and the environmental services it provides. Clean technology is the union of six classes, renewable energy and energy efficiency, electrification and storage, clean mobility, hydrogen, carbon capture, and clean fuels, nuclear power, and green buildings, whose work researches, builds, installs, or operates these technologies. Environmental management comprises the other nine, environmental management systems, permitting, and compliance; waste and circularity; water; pollution and chemicals; air emissions; greenhouse-gas accounting and reduction; corporate sustainability and environmental disclosure; climate risk; and biodiversity and land, whose work measures, manages, reduces, or reports a firm's environmental impacts or provides environmental services, together with the four positives whose function fits no listed class. Of the 5,748 labelled positives, 3,275, or 57 percent, fall in the clean-technology classes. The other three capabilities are defined in the same way, each by its own written definition: social work, which protects the safety, welfare, and rights of the people that the firm's operations and products affect; governance work, which establishes, operates, or assures the firm's controls and compliance with the law; and innovation work, which creates new or substantially improved products, processes, or methods (Section 4.4).

## 3.2 From Text to Probabilities

Applying such a definition requires reading what a posting asks the hire to do. Large language models now annotate text at or above the accuracy of crowd workers and trained coders (Gilardi, Alizadeh and Kubli, 2023; Törnberg, 2025), but applying one to 126 million postings would be very costly, as I show below. I therefore follow a teacher–student design, known as distillation (Hinton, Vinyals and Dean, 2015). A language model applies the definition to samples of postings, and a far cheaper classifier, trained on these labels, carries the definition to the whole corpus.[19] Nothing in the design depends on the particular labeller. Human coders or another model could supply the labels. The language model labels purposive training samples, which over-represent postings likely to involve the work so that the classifier sees enough positives to learn from. For environmental work, the training sample holds 31,995 postings, 4,768 of them positive (Internet Appendix IA.II.1 describes the training samples of all four kinds of work). Labels generated by a model require validation data that the model did not produce (Ludwig, Mullainathan and Rambachan, 2026; Egami et al., 2023).

[19]The labeller is Claude Sonnet 5.5 (`claude-sonnet-5-5`), run in batches of postings. A second labeller, GPT-5.4 mini (`gpt-5.4-mini-2026-03-17`), labels the validation sample, also run in batches. The labeller's instruction states the definitions, the scoring rule, the closed list of classes, and twenty to thirty worked examples. The instruction is reproduced in the Internet Appendix.

Two such checks are built into the design. A simple random sample of 4,000 postings, which is not used for training, serves to test the classifier on the population that it scores, and, because environmental work is rare, 2,000 postings drawn at random within a band of the classifier's score, weighted by their probability of selection, complement it in calibration. And I coded a validation sample of 800 postings blind to the model's labels, which tests the labels themselves (Section 3.3).

The classifier reads each posting through a public multilingual text-embedding model, BGE-M3, which maps up to 8,192 tokens into a vector of 1,024 numbers (Chen et al., 2024). Two logistic regressions on this vector predict the presence of each part of environmental work, the first trained to recognize postings labelled with one of the six clean-technology classes and the second postings labelled with any other class, and their predictions are combined into a single score $r_j$, the probability that a posting involves either part. Their regularization is chosen by cross-validation in which all postings of a firm fall into the same fold, so that the classifier is always evaluated on firms it has not seen, and the same cross-validation, with the area under the precision–recall curve as its criterion, chooses between a single regression and one for each part. Because the training samples are enriched in positives, the score ranks postings but overstates how often environmental work occurs. Calibration on the random sample converts it into a probability,

$$\pi_j = \frac{1}{1 + \exp\Big[-a \operatorname{logit}(r_j) - b\Big]}, \tag{1}$$

a transformation that preserves the ranking of postings (Platt, 1999). Internet Appendix IA.II reports the selection of each classifier, the training data it needs, and its calibration.

There are two alternative approaches to obtaining such measures from postings. The first is to label every posting directly with a generative large language model. With postings averaging about 500 words, labelling the roughly 100 million postings of the corpus once, for a single capability, would cost about $176,000 to $195,000 using the same labelling model as in this paper. The second approach is to fine-tune a language model for each capability, as Hansen et al. (2026) do to measure remote work in nearly 600 million postings from 30,000 human classifications. Each capability, and each revision of its definition, then requires a labelled sample of similar size, a separately trained model, and a new pass over the whole corpus, whereas in my design the corpus is embedded once,[20] and the logistic regressions of

[20]Embedding the corpus is the most demanding step of the design in computation, and it is done only once. The same embeddings serve every capability, so that measuring a new capability, or revising the definition of an existing one, requires only a new labelled sample and logistic regressions trained on it. Fine-tuning a language model, or adapting it to the domain of job postings, must instead be repeated for each capability and each revision, and the adapted model must then read the whole corpus again.

any capability score its 126 million postings locally on any standard computer.

### 3.3 How Well the Classifier Reads Postings

The classifier performs well on postings it was not trained on. Against my blind codes of 800 postings, the language model's labels agree with a Cohen's $\kappa$ of 0.91, and the classifier's with a $\kappa$ of 0.84 (95 percent confidence interval 0.77 to 0.90), and the classifier's score ranks these postings with an area under the ROC curve of 0.966 (Appendix Table B1). Against the language model's labels on the random samples, in which environmental work is rare, the classifier's F1 score is 0.59. On held-out halves of the random sample, the mean probability equals the labelled rate, 2.0 percent of postings, and the expected calibration error is 0.43 percentage points. A randomly chosen positive outranks a randomly chosen negative 95.5 percent of the time, and the area under the precision–recall curve is 37 times the base rate (Table 5, Panel A).

The same procedure performs similarly for the other three capabilities, each measured from its own written definition. On held-out postings, the mean probability equals the labelled rate for each of them, and the classifier ranks postings far better than chance (Table 5, Panel A).

The classifier also improves on simpler measures for all four capabilities, scored on the same random sample, which was not used to build any of them (Appendix Table B2). A hand-curated dictionary of ESG terms, and for innovation a pattern of research and development terms, flags 9 to 30 percent of postings, of which only 9 to 23 percent carry the capability. A dictionary of the 100 terms most indicative of positives in the training labels ranks postings with an area under the precision–recall curve of 0.22 to 0.63, against 0.43 to 0.77 for the classifier, and the average probability of a posting's occupation, the best that occupation codes alone can do, also ranks postings worse for every capability. Nor do the shares hinge on the particular postings from which the classifier learned. Refitting it on bootstrap draws of its training firms yields environmental shares that correlate with those in use at 0.97 to 0.99, and shares of the other capabilities that correlate at 0.95 or more (Table 5, Panel A).

**Table 5.** Accuracy and Calibration: Four Capabilities

| | Environmental | Social | Governance | Innovation |
|---|---|---|---|---|
| *A. Postings* | | | | |
| Labelled rate / mean probability (%) | 2.0 / 2.0 | 4.4 / 4.4 | 10.5 / 10.5 | 8.5 / 8.6 |
| Expected calibration error (points) | 0.43 | 0.79 | 0.96 | 0.71 |
| ROC-AUC | 0.955 | 0.889 | 0.896 | 0.969 |
| PR-AUC relative to the base rate | 36.5 | 9.6 | 5.9 | 8.8 |
| $\kappa$, classifier and blind human codes | 0.84 | 0.74 | 0.72 | 0.79 |
| Correlation with bootstrap refits | 0.97–0.99 | 0.95–0.96 | 0.98 | 0.99 |
| *B. Firm-year shares* | | | | |
| Mean share (% of openings) | 3.03 | 5.89 | 15.70 | 11.82 |
| Calibration slope, firm-year share | 1.16 (0.14) | 0.98 (0.19) | 1.03 (0.10) | 1.08 (0.06) |
| with industry-by-year effects | 1.11 (0.24) | 0.54 (0.21) | 1.03 (0.15) | 0.88 (0.11) |

*Notes.* Each capability is measured by the procedure of this section from its own written definition (Internet Appendix IA.I). Panel A: the labelled rate and the mean calibrated probability, the expected calibration error, and the ranking statistics refer to the random sample of 4,000 postings and, for calibration, its score-band supplements, with the calibration fitted on one half and evaluated on the other; ROC-AUC is the probability that a random positive outranks a random negative, and PR-AUC is the area under the precision–recall curve, whose chance level is the base rate. Cohen's $\kappa$ compares the classifier, at the threshold that maximizes agreement with the labeller, with blind human codes of a validation sample of 800 postings (700 for innovation). The last row of the panel gives the range of the correlations between the shares in use and those of classifiers refitted on five bootstrap draws of the training firms (Internet Appendix Table IA.20). Panel B: mean over the 35,772 firm-years of the estimation sample; the labels of the 2,949 postings of the random sample in firm-years of the estimation sample regressed on the expected share of their firm-year computed without the posting itself, without and with NAICS-3-by-year effects (Section 3.4), with standard errors clustered by firm in parentheses. Details: Appendix Table B1.

## 3.4 The Firm-Year Share

Let $\mathcal{J}_{ft}$ denote the $N_{ft}$ openings that firm $f$ removes in year $t$, and let $y_j$ equal one if opening $j$ assigns environmental work under the definition of Section 3.1. The environmental share of the firm's hiring is

$$s_{ft} = \frac{1}{N_{ft}} \sum_{j \in \mathcal{J}_{ft}} y_j, \tag{2}$$

the share that labelling every one of its postings would give, which I call its true share. The labels are not observed for the corpus, but the text of every posting is, and the classifier turns it into a calibrated probability $\pi_j$. Calibrated means that the probabilities can be read as frequencies: among postings that receive a probability of 0.3, about three in ten involve environmental work, so that $\mathbb{E}[y_j \mid \pi_j] = \pi_j$. I replace each label by its probability and use

the expected share,[21]

$$S_{ft} = \frac{1}{N_{ft}} \sum_{j \in \mathcal{J}_{ft}} \pi_j, \qquad \pi_j = \Pr\Big(y_j = 1 \;\Big|\; \text{text}_j\Big). \tag{3}$$

The expected share differs from the true share, but in a way that does not bias regressions. Because each probability is the average label among postings like it, the true share of a firm-year averages to its expected share, whatever the value of that expected share. The difference between the two,

$$u_{ft} = s_{ft} - S_{ft}, \qquad \mathbb{E}\Big[u_{ft} \;\Big|\; S_{ft}\Big] = 0, \tag{4}$$

therefore averages zero at every value of the measure. The truth is scattered around the measure, as when every pupil of a school is assigned the school's average score, rather than the measure around the truth, as when a scale adds a random error to every weighing. Errors of the second, classical kind bias the coefficient of a regression on the measure toward zero. Errors of the first kind, known as Berkson errors, leave it unbiased and only widen its standard error (Carroll et al., 2006).[22] The argument requires that the probabilities be calibrated given the other variables of each regression as well. If the classifier overstated environmental work in one industry, for instance, its errors would move with the industry effects. I cannot verify this for every set of controls and fixed effects, but two tests check it where it matters: whether firm-year shares are calibrated, with and without industry-by-year effects, and whether the errors vary with the year, the sector, or the characteristics of the firm.

Calibration of single postings does not by itself ensure calibration of firm-year shares, because the errors of postings from the same employer need not be independent. A firm's boilerplate, for example, could mislead the classifier on all its postings alike. I therefore test calibration at the level of the firm-year, with the random sample of 4,000 postings, which was labelled but not used for training. A posting drawn at random from a firm-year involves environmental work with a probability equal to the firm-year's true share, the share of its postings that involve such work. If the expected share is calibrated, the label of such a

[21] A common alternative counts the share of postings whose score exceeds a threshold. Such a flagged share is zero in 47 percent of firm-years, because a firm with few environmental postings often has none flagged. Its level depends on the threshold, which is poorly determined, since agreement with the labeller barely changes over a wide range of thresholds, and its errors vary across firm-years. In the estimation sample, the flagged share rises by 1.32 for each unit of the expected share, so that it would scale the coefficients of a regression in which it is the dependent variable by about 1.3, and those of a regression in which it is an explanatory variable by about 0.73 across firms and 0.63 within firms. Internet Appendix Table IA.19 reports the threshold, set where the classifier agrees most with the labeller, and these comparisons for the four kinds of work.

[22] If an outcome satisfies $Y = \alpha + \beta s_{ft} + \varepsilon$, substituting $s_{ft} = S_{ft} + u_{ft}$ gives $Y = \alpha + \beta S_{ft} + (\beta u_{ft} + \varepsilon)$, and since $u_{ft}$ averages zero at every value of $S_{ft}$, a regression of $Y$ on $S_{ft}$ recovers $\beta$.

posting should therefore rise one for one with the expected share of the firm-year that posted it, computed without the posting itself. Regressed on these expected shares, the labels of the random sample follow a slope of 1.16 (standard error 0.14), and of 1.11 (0.24) with industry-by-year effects, so that neither differs significantly from one (Table 5, Panel B). For the other three capabilities, the slopes lie between 0.98 and 1.08, none significantly different from one. Within industry-years, the slope for social work falls to 0.54, so that comparisons of social shares within industries are attenuated by about half.

The errors of the probabilities are also unrelated to the firm that posted the job and to when it did so. On the random sample, the difference between a posting's calibrated probability and its label does not differ significantly across years for any of the four capabilities, nor is it related to the size, labor productivity, Tobin's $q$, R&D intensity, or patenting of the posting's firm, with joint $p$-values between 0.42 and 0.89. Across NAICS-2 sectors, the error of the environmental share does not differ significantly ($p = 0.13$), nor, at the 5 percent level, do those of the social and innovation shares. For governance work, the error differs across sectors ($p = 0.01$), a difference that the industry-by-year effects of every regression absorb (Internet Appendix Figure IA.4). The errors that remain are thus unlikely to masquerade as the associations with firm outcomes that the rest of the paper reports.

I split the share into its two parts by crediting each posting's calibrated probability to the part whose regression predicts it more strongly. The clean-technology share and the environmental management share therefore sum to the environmental share. To separate the work that firms assign from the occupations they hire, let $\mu_{o,ft}$ be the fraction of the firm's openings in occupation $o$, the six-digit occupation of the O*NET-SOC code that LinkUp assigns to each posting, $s_{o,ft}$ the share among them, and $\bar{s}_{ot}$ the share among all openings in occupation $o$ in year $t$. The share then decomposes as

$$s_{ft} = \underbrace{\sum_{o} \mu_{o,ft}\, \bar{s}_{ot}}_{\text{occupation mix}} + \underbrace{\sum_{o} \mu_{o,ft} \left( s_{o,ft} - \bar{s}_{ot} \right)}_{\text{work within occupations}}. \tag{5}$$

The first term is what an occupation-based measure can capture. The second captures how much more or less environmental work a firm assigns to the occupations it hires than other firms do.

The occupation mix alone captures less than half of the variation in the share. In a linear projection of the environmental share on the first term of equation (5), the occupation mix accounts for 43 percent of the variance (Table 8). The rest is variation that the mix of occupations does not capture, and that measures built from occupations therefore miss.

In panel regressions, the share enters standardized within year,

$$H_{ft} = \frac{S_{ft} - \bar{S}_t}{\mathrm{sd}_t(S)}, \tag{6}$$

so that a coefficient measures the association with a one-standard-deviation difference between firms in the same year. In the event studies, the share is standardized among the firms with returns around each event. Because the expected share varies less than the share of labels it predicts (equation (4)), a one-standard-deviation difference in the measure is a smaller difference in the true share, so that magnitudes stated per standard deviation are, if anything, conservative.

I interpret the share as the orientation of a firm's hiring, the direction in which the firm is adding work, which should be kept apart from two related objects. A capability is a firm's ability to carry out an activity, built over time from the work of its employees and from other inputs, and intangible investment is the spending that develops such capabilities, which financial statements record across several lines but in none by the activity it serves (Internet Appendix IA.IV). The share therefore indicates where a firm's investment in a capability is directed, not the stock of the capability, the amount spent on it, or the part of firm value that it accounts for. It also records hires that a firm intends to make rather than those it has made, and it exists only for firms that recruit through their own career sites. Within these limits, the orientation of hiring is informative precisely where financial statements are silent. It reflects a commitment whose cost the firm bears once a position is filled, and it distinguishes kinds of work within an occupation. Sections 4.3 and 6.6 examine whether it also anticipates the patents and the revenue that the work produces.

## 3.5 The Share across Firms, Industries, and Time

Environmental work is a small and right-skewed part of hiring (Table 6). In the average firm-year, 3.0 percent of new openings involve environmental work, 1.4 percent clean technology and 1.6 percent environmental management. The median share is 0.8 percent and the 90th percentile 7.8 percent. The share doubled over the decade, from 2.1 percent of a firm-year's openings in 2014 to 4.2 percent in 2023, and its composition changed with it (Figure 1). About 70 percent of the rise occurs within firms, and among the 1,122 firms present in both years the share rose from 2.3 to 3.6 percent; the rest reflects the firms that enter the data as its coverage grows. Environmental management grew steadily, whereas clean technology accelerated after 2019 and overtook it in 2022, so that clean technology rose from 30 percent of all environmental openings in 2014 to 51 percent in 2023.

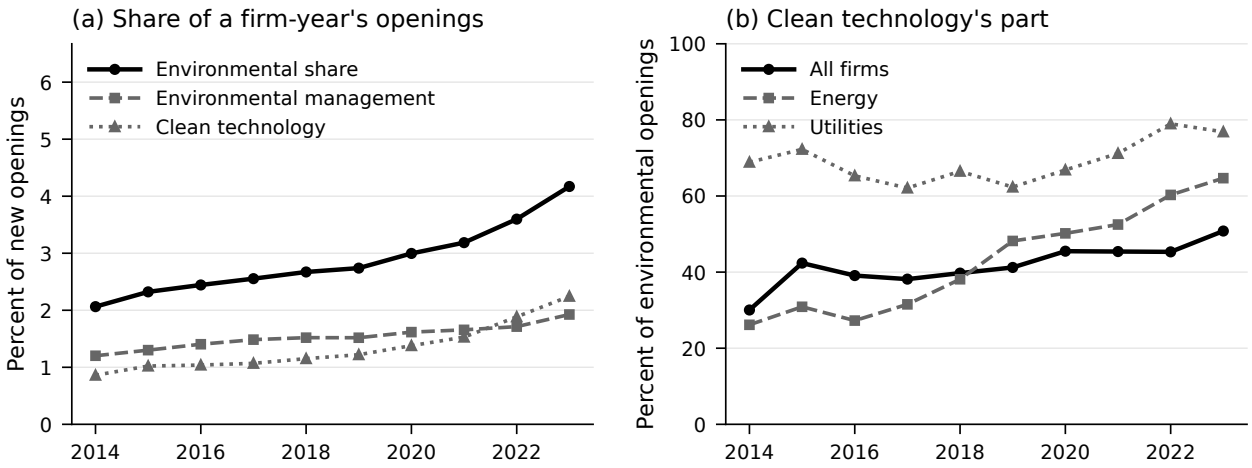



**Figure 1.** The Environmental Share over Time. Panel (a): the mean over firm-years of the environmental share and of its two parts, in percent of a firm-year's new openings. Panel (b): clean-technology openings as a percentage of all environmental openings, weighting firm-years by their openings, for all firms and for the firms of the GICS energy and utilities sectors. Estimation sample of Table 6. The four capability shares over time are shown in Appendix Figure B1.

Environmental hiring is highest in utilities, energy, and materials. It reaches 16.2 percent of openings in utilities, followed by energy (8.5 percent), materials (6.8 percent), and industrials (5.2 percent), and it is lowest in financials and communication services (0.6 percent). The composition differs sharply across industries. In waste management, which has the highest share of any industry (37.7 percent), almost all environmental work is environmental management. In electrical equipment, about 90 percent is clean technology. Clean technology also gained ground in the sectors most exposed to the transition, from 26 to 65 percent of environmental openings in energy between 2014 and 2023. The firms with the highest clean-technology shares are recognizable, such as FuelCell Energy, Vestas, Ameresco, Enphase, First Solar, EVgo, QuantumScape, and Sunnova.

Appendix B.3 describes which firms orient their hiring toward environmental work. Within industries, these firms talk more about climate change, have higher environmental ratings, and are younger and less profitable. Clean-technology hiring goes with higher capital expenditures and with young firms that pay no dividends, whereas environmental management hiring goes with tangible assets and higher ratings.

**Table 6.** The Environmental Share of Hiring

*A. Distribution across firm-years, percent of new openings*

| | Environmental | Clean technology | Environmental management |
|---|---|---|---|
| Mean | 3.03 | 1.44 | 1.59 |
| Median | 0.82 | 0.20 | 0.41 |
| Standard deviation | 6.36 | 4.60 | 4.10 |
| 90th percentile | 7.79 | 2.89 | 3.80 |

| | | | Mean share (% of openings) | | | SD | Clean tech in E (%) | |
|---|---|---|---|---|---|---|---|---|
| | Firm-years | Firms | Environ. | Clean technology | Environ. management | Environ. | 2014 | 2023 |
| *B. GICS sectors* | | | | | | | | |
| Utilities | 1,029 | 184 | 16.23 | 11.29 | 4.94 | 11.96 | 69 | 77 |
| Energy | 1,527 | 310 | 8.51 | 3.16 | 5.35 | 6.97 | 26 | 65 |
| Materials | 2,186 | 430 | 6.77 | 1.79 | 4.98 | 5.82 | 13 | 29 |
| Industrials | 6,382 | 1,212 | 5.18 | 2.46 | 2.72 | 9.55 | 25 | 46 |
| Consumer staples | 1,918 | 369 | 1.93 | 0.33 | 1.59 | 2.34 | 10 | 15 |
| Real estate | 867 | 153 | 1.90 | 0.93 | 0.97 | 2.26 | 43 | 60 |
| Information technology | 5,046 | 1,097 | 1.68 | 1.14 | 0.54 | 4.17 | 67 | 66 |
| Consumer discretionary | 5,381 | 1,043 | 1.53 | 0.95 | 0.58 | 3.86 | 34 | 63 |
| Health care | 4,740 | 1,117 | 1.07 | 0.46 | 0.61 | 1.27 | 29 | 32 |
| Communication services | 2,147 | 429 | 0.58 | 0.25 | 0.34 | 0.84 | 69 | 49 |
| Financials | 4,541 | 782 | 0.58 | 0.14 | 0.44 | 1.30 | 22 | 24 |
| All firms | 35,772 | 7,128 | 3.03 | 1.44 | 1.59 | 6.36 | 30 | 51 |
| *C. The ten NAICS-3 industries with the highest environmental share* | | | | | | | | |
| Waste management and remediation services (562) | 136 | 27 | 37.66 | 1.13 | 36.53 | 18.53 | 1 | 1 |
| Utilities (221) | 1,014 | 184 | 16.68 | 11.51 | 5.17 | 11.90 | 68 | 77 |
| Pipeline transportation (486) | 112 | 29 | 11.09 | 3.49 | 7.60 | 6.10 | 19 | 52 |
| Oil and gas extraction (211) | 420 | 86 | 9.80 | 3.22 | 6.58 | 6.44 | 22 | 49 |
| Electrical equipment, appliances and components (335) | 545 | 118 | 9.37 | 8.38 | 0.99 | 14.23 | 93 | 91 |
| Petroleum and coal products (324) | 269 | 50 | 9.22 | 4.59 | 4.64 | 6.21 | 36 | 72 |
| Mining (except oil and gas) (212) | 495 | 120 | 8.84 | 1.52 | 7.32 | 6.00 | 7 | 23 |
| Heavy and civil engineering construction (237) | 267 | 59 | 8.70 | 5.02 | 3.68 | 9.84 | 49 | 59 |
| Primary metals (331) | 316 | 66 | 6.11 | 2.32 | 3.79 | 7.28 | 19 | 29 |
| Transportation equipment (336) | 1,223 | 235 | 5.96 | 4.44 | 1.52 | 7.19 | 65 | 75 |

*Notes.* Estimation sample: 35,772 firm-years of 7,128 firms over 2014–2023 with at least 20 new openings and Compustat fundamentals, with 91.8 million openings. The environmental share is the expected share of a firm-year's new openings whose responsibilities involve environmental work, the mean of the calibrated probabilities of its postings, in percent. It is the sum of the clean-technology share (energy, clean mobility, electrification and storage, hydrogen and carbon capture, nuclear, green building) and the environmental management share (environmental management systems, waste, water, pollution, air emissions, greenhouse-gas accounting, sustainability reporting, climate risk, land). Panel B groups firms by GICS sector (the eight firm-years without a sector are left out but counted in the row for all firms), and Panel C lists the ten NAICS-3 industries with the highest mean environmental share among those with at least 100 firm-years. Shares are means over firm-years, and SD is the standard deviation of the environmental share across firm-years. The last two columns give the clean-technology share of all environmental openings of the group's firms in 2014 and in 2023, weighting firm-years by their openings.

# 4 Validating the Measure

This section asks whether the measure captures the work it is meant to capture. I first ask whether each share is highest where its kind of work is concentrated. I then relate each share to outcomes that are tied directly to the work it counts and measured from sources independent of the job postings. These outcomes are of two kinds. Some are what the work produces: innovation work produces the R&D that financial statements report and the patents that firms file, and environmental work produces green technologies. Others are needs to which the work responds, so that the share should be higher where the need is greater: environmental work where the firm's footprint is larger, safety work where its operations are more hazardous, and compliance and control work where its legal exposure is greater. For environmental work, I also compare the share with existing measures of environmental activity. I then ask whether each share predicts the outcomes of its own kind of work more strongly than the other shares do. Finally, I decompose the variation of the share and estimate how much of it is sampling noise.

## 4.1 Where Each Share Is Highest

A share that measures a kind of work should be highest in the industries, firms, occupations, and jobs where that work is concentrated. Section 3.5 showed this for environmental work across sectors, and Appendix Table C1 shows it for all four shares. The environmental share is highest in waste management and utilities, among waste collectors and the wind-turbine maker Vestas, and among wind turbine service technicians and solar installers. The social share is highest in mining and oil and gas extraction, among mining companies such as BHP and Freeport-McMoRan, and among health and safety engineers and specialists. The governance share is highest in chemical manufacturing, which includes pharmaceuticals, and in securities and banking, and among financial examiners, information security analysts, and compliance officers. The innovation share is highest in computer and electronic products, among chip designers such as Advanced Micro Devices and Nvidia, and among computer research scientists and hardware and software engineers. Among job titles posted by many firms, those whose postings most often involve each kind of work are environmental technicians, specialists, and engineers, along with the helpers and sorters of waste collection and recycling; safety managers, environmental, health, and safety staff, and human-resources managers; quality engineers, managers, and inspectors; and software and data engineers. These patterns cannot validate the shares by themselves, since industry and occupation codes reproduce some of them, but a share that failed them would not measure the work it is meant to measure.

## 4.2 The Benchmark: Innovation and R&D

Innovation serves as the benchmark because it is the one capability whose development financial statements record in a line of their own. The measurement of intangible investment treats R&D as investment in knowledge, in the national accounts and at the level of the firm (Corrado, Hulten and Sichel, 2009; Peters and Taylor, 2017), and most of this spending pays for the labor of the people who do the work, since labor accounts for more than two-thirds of business R&D spending in the United States (Moris and Shackelford, 2023). If the share of a firm's openings that involve innovation records where the firm directs its labor, it should therefore track the R&D that this labor generates, even though R&D does not distinguish the technologies being developed. No other capability allows such a test, because financial statements record the labor of environmental, social, and governance work across several lines and in none of its own (Internet Appendix IA.IV).

I regress R&D in year $t$, in percent of assets at the end of $t-1$, on the four standardized shares of year $t$, with NAICS-3-by-year effects and the logarithms of assets, employees, and openings, and then with firm effects, which ask whether a firm that raises its innovation share also raises its R&D. The innovation share passes this test. A firm whose innovation share is one standard deviation higher spends 3.96 percentage points more of its lagged assets on R&D ($t = 26.0$), against a mean of 4.6 percent, and the relation is close to linear across the distribution of the share (Appendix Figure C1). A firm whose share rises within its own history also raises its R&D (0.29, $t = 2.2$; Appendix Table C14). The procedure thus recovers the broad activity on which R&D reports the spending, although R&D does not reveal which technologies that spending develops (Appendix C.5). Markets also value the share much as they value R&D. A firm whose innovation share is one standard deviation higher has a Tobin's $q$ higher by 0.31, an association that R&D spending accounts for entirely and SG&A spending not at all (Appendix Table C12).

## 4.3 Environmental Work

Environmental work leaves two kinds of traces outside the postings. It develops technologies, which firms patent, and part of it manages the firm's impact on the environment, which is larger where the firm emits more. The first is an output of the work, which the share should predict, and the second a need to which the work responds, so that the share should be higher where the footprint is larger. The two parts of environmental work should also differ in the technologies they produce: clean technology the inventions that climate policies target, and environmental management the technologies of pollution abatement and monitoring.

I measure invention with the patent applications that a firm files in the two years after the

year of hiring: those in the Y02 scheme, which classifies technologies that mitigate climate change, those in the classes of the technologies that the clean-energy credits support (IRA classes), and the environmental technologies of the OECD's ENV-TECH list that Y02 does not cover (Section 2.4). Each regression relates the outcome to the four standardized shares of year $t$, with NAICS-3-by-year effects and the logarithms of assets, employees, and openings, by Poisson pseudo-maximum likelihood, so that coefficients approximate proportional differences. The environmental share predicts the technology that environmental work produces (Table 7). Firms whose share is one standard deviation higher file 23 percent more applications in the Y02 class over the following two years ($t = 4.2$) and 34 percent more in the IRA classes ($t = 4.8$). Green applications also make up 7.6 percentage points more of their patenting, against an average of 7.8 percent, while these firms do not file more applications in other classes. The share thus predicts the direction of a firm's invention rather than its amount. The clean-technology part carries the change in direction: it predicts a higher green share of applications (0.067, $t = 14.7$), whereas the environmental management part does so far less (0.015, $t = 2.1$), a difference with a $t$-statistic of 6.2. For counts of green applications, the clean-technology coefficient is the larger one, but the difference between the two parts is not significant. The environmental management part predicts its own technologies instead, namely those of pollution abatement, waste, water, and environmental monitoring that the OECD classifies outside Y02 ($t = 5.3$, against 0.4 for clean technology). The associations of the environmental share with green applications remain, at a little over half their size, when a firm is compared with itself over time (Appendix Table C2).

The share may also move before the technology does, although this evidence rests on few firms. Among patenting firms that have never filed a green application, a one-standard-deviation higher environmental share is associated with a probability of filing a first application in the IRA classes over the next two years that is higher by 1.8 percentage points ($t = 2.3$), against a base rate of 3.2 percent (Table 7, Panel B). For a first Y02 application, the difference is 1.6 points ($t = 1.8$), against 6.8 percent. For the clean-technology part, the differences are 5.2 points for a first IRA-class application and 3.2 points for a first Y02 application. Among all firms without a green application, however, most of which never patent, the share predicts only a first IRA-class application, and neither once utilities and the industries whose patenting was already most green are excluded (Appendix Table C3), and under alternative rules for keeping postings the coefficient for a first Y02 application falls just short of significance at the 10 percent level (Appendix Table E9). Hiring may thus precede a change in the direction of a firm's technology, but the evidence is suggestive, and it is an association, not a causal effect. Each part of the share is thus validated by an outcome of its own work, as the share is, clean technology by the green direction of a firm's patenting and environmental management by

**Table 7.** The Environmental Share and Green Technology

| | Environmental share (1) | Clean technology (2) | Environmental management (3) | Firm-years (4) |
|---|---|---|---|---|
| *A. Patent applications, $t+1$ to $t+2$* | | | | |
| Y02 applications | 0.205*** | 0.144*** | 0.044 | 20,349 |
| | (0.049) | (0.033) | (0.140) | |
| IRA-class applications | 0.296*** | 0.203*** | 0.089 | 18,461 |
| | (0.062) | (0.042) | (0.197) | |
| Environmental technologies outside Y02 | 0.214*** | 0.030 | 0.461*** | 16,997 |
| | (0.082) | (0.084) | (0.087) | |
| Green share of applications[a] | 0.076*** | 0.067*** | 0.015** | 11,266 |
| | (0.007) | (0.005) | (0.007) | |
| Other applications | −0.218* | −0.111 | −0.343** | 22,400 |
| | (0.127) | (0.095) | (0.148) | |
| *B. A first green application, patenting firms without one (percentage points)* | | | | |
| First Y02 application (base rate 6.8%) | 1.64* | 3.19** | 0.58 | 6,212 |
| | (0.90) | (1.44) | (0.61) | |
| First IRA-class application (base rate 3.2%) | 1.85** | 5.21*** | 0.46 | 7,423 |
| | (0.80) | (1.58) | (0.43) | |

*Notes.* Panel A: estimation sample, 2014–2023. Each row regresses the outcome on the shares of year $t$, entered together with the social, governance, and innovation shares, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects. Column (1) reports the coefficient of the standardized environmental share; columns (2) and (3) those of its two parts, each standardized and entered together in place of the share; column (4) gives the firm-years of column (1). Counts are estimated by Poisson pseudo-maximum likelihood, so that coefficients approximate proportional changes, and the outcome marked [a] by OLS. Environmental technologies outside Y02 are the classes of the OECD's ENV-TECH list that Y02 does not cover. Panel B: patenting firms that have filed no application of the class before year $t$; the outcome equals 100 if the firm files its first application of the class in $t+1$ to $t+2$, by OLS with the same controls (Appendix Table C3, which also reports all firms without a green application). Within-firm estimates are in Appendix Table C2. Standard errors clustered by firm in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

the technologies of pollution abatement, waste, water, and monitoring that Y02 leaves out.

Part of environmental work responds to the firm's impact on the environment: it monitors emissions and discharges, holds permits, and complies with environmental regulation. The share should therefore be higher where that impact is larger. I relate it to the Scope 1 emissions per dollar of revenue that the firm reports for the following year and to the environmental incidents in which the news involves the firm over the following two years, as RepRisk compiles them. Firms with a higher environmental share report greenhouse gas emissions about 15 percent higher per dollar of revenue, and they are involved in 15 percent more environmental incidents. Both parts share these associations across firms, but none remains when a firm is compared with itself over time (Appendix Table C2). The footprint is

a lasting attribute of the firm, whereas the share also follows the direction in which the firm is changing, as its association with the firm's own green patenting shows.

Finally, the share is related to existing measures of environmental activity but distinct from them (Table 8). Across firm-years, it correlates at 0.66 with the occupation mix of equation (5), at 0.36 with the green task intensity of the firm's occupations in O*NET, at 0.54 with the exposure to climate change that firms discuss on their earnings calls (Sautner et al., 2023*a*), and at 0.45 with the green share of the firm's patent applications. Within industry-years, these correlations fall to between 0.24 and 0.50, and those with emissions intensity, environmental incidents, and the LSEG environmental score lie between 0.09 and 0.13. The two parts differ as the rubric intends. Within industry-years, climate talk, especially about opportunities, and green patenting go with clean technology, at 0.37 to 0.41, and hardly with environmental management, at −0.05 to 0.07. The LSEG score is the rating of a single provider, and environmental ratings diverge widely across providers (Berg, Kölbel and Rigobon, 2022), so that its low correlation with the share describes this score rather than ratings in general. Appendix Table C5 enters these measures beside the share. The share keeps its associations with emissions and incidents beside each of them, and its association with green patents beside every measure except the occupation mix, which carries much of the information about green patenting across firms. Appendix Table C6 shows that clean-technology hiring and climate talk move together over time rather than one leading the other.

**Table 8.** The Environmental Share and Existing Measures

| | Across firm-years | | | Within industry-years | | | |
|---|---|---|---|---|---|---|---|
| | Environ-mental | Clean technology | Environ. management | Environ-mental | Clean technology | Environ. management | Firm-years |
| *Measures built from occupations* | | | | | | | |
| Occupation mix, equation (5) | 0.66 | 0.42 | 0.55 | 0.50 | 0.27 | 0.44 | 35,772 |
| O*NET green task intensity | 0.36 | 0.26 | 0.27 | 0.24 | 0.16 | 0.17 | 35,767 |
| *Ratings and disclosure* | | | | | | | |
| LSEG environmental pillar score | 0.21 | 0.14 | 0.17 | 0.11 | 0.06 | 0.10 | 29,462 |
| Climate-change exposure (calls) | 0.54 | 0.58 | 0.18 | 0.34 | 0.41 | 0.02 | 29,599 |
| Opportunity exposure | 0.46 | 0.56 | 0.09 | 0.28 | 0.38 | −0.05 | 29,599 |
| Regulatory exposure | 0.33 | 0.28 | 0.20 | 0.15 | 0.13 | 0.07 | 29,599 |
| *Footprint and technology* | | | | | | | |
| Log scope 1 emissions / revenue | 0.36 | 0.21 | 0.31 | 0.09 | 0.04 | 0.09 | 30,982 |
| Log (1 + environmental incidents) | 0.26 | 0.18 | 0.21 | 0.13 | 0.09 | 0.09 | 35,303 |
| Green (Y02) share of applications | 0.45 | 0.44 | 0.16 | 0.36 | 0.37 | 0.07 | 15,021 |

*Notes.* Pearson correlations of the environmental share and its two parts with each measure, over the firm-years of the estimation sample in which the measure is available (last column), across firm-years and after removing the means of NAICS-3-by-year cells from both variables. The occupation mix credits each opening with the mean environmental probability of its six-digit occupation in the year. O*NET green task intensity is the opening-weighted share of green task statements of the firm's occupations (Vona et al., 2018). The LSEG score is the environmental pillar score of a single provider. Climate-change exposure is the share of a firm's earnings calls devoted to climate change, overall and to its opportunities and regulation (Sautner et al., 2023*a*). Scope 1 emissions per dollar of revenue come from Trucost, including its estimates; environmental incidents from RepRisk; and the green share of applications is Y02 applications over all applications filed in the year, among firm-years with at least one application.

## 4.4 Social and Governance Work

The same procedure measures social and governance work, each from its own written definition. The social share counts work that protects or improves the rights, safety, welfare, or opportunities of the people whom the firm's operations and products affect, and its classifier has three heads, for worker safety, product and patient safety, and people and rights. The governance share counts work that establishes, operates, or assures the firm's accountability, oversight, regulatory compliance, and internal controls, and its classifier has four heads, for quality assurance, risk and financial controls, information security and privacy, and compliance and integrity. Governance thus refers here to the work that operates a firm's controls rather than to the structure of its board or the rights of its shareholders that governance indices score (Gompers, Ishii and Metrick, 2003). Neither kind of work produces an output that can be counted. What it achieves is an absence, fewer injuries or fewer failures of control, which is confounded with the need for the work, since firms hire more of it where hazards and exposure are greater. The clearest outcomes are therefore the needs to which the work responds: the hazards of a firm's operations for safety work, and its legal and regulatory exposure for compliance and control work. These outcomes show where the work is needed

rather than what it achieves.

Social work is the harder to validate, because the outcomes available for it, injuries and safety violations, are those that its work should prevent, and the share can relate to them in either direction. A firm hires more safety staff where its operations are more hazardous, and that work, where it succeeds, lowers injuries. Comparisons across firms mix the two, and they are weak for two further reasons. Worker safety, the part of social work that these outcomes concern, is among the most industry-bound parts that the paper measures, since industries and years explain 65 percent of the variance of the worker-safety share, against 56 to 63 percent for the four shares, and comparisons of social shares within industries are attenuated by about half (Section 3.4). Across firms, those whose social share is one standard deviation higher incur fewer serious violations in the inspections that the safety regulator schedules for high-hazard establishments, an imprecise estimate carried by worker safety, and their injury rates in the following year are lower by about 7 percent of the mean, although both associations vanish with firm effects. The sharper test uses the location that postings record. A firm with establishments in several states hires for work whose hazards differ from one establishment to another, so that comparing the same firm's hiring across the US states in which it operates, in the same year, holds fixed everything about the firm and asks only where it places its safety work. Within a firm and year, the social share of a state's openings is higher where the firm's establishments report more injuries, by 0.09 cases per 100 workers per standard deviation of the share ($t = 5.0$) and by 0.13 for worker safety, and the pattern is the same for the injuries of the two preceding years, so that it reflects a stable placement of safety work in the firm's more hazardous operations (Appendix Tables C8 and C9).

Governance work, as measured here, operates the firm's controls and its compliance with the law, and much of it responds to legal exposure. Federal litigation against the firm measures that exposure, so that the governance share should be higher where litigation is more frequent, especially its compliance and integrity part. Firms whose governance share is one standard deviation higher face 21 percent more securities suits over the next two years ($t = 3.5$), 26 percent more class actions, and 36 percent more derivative and fraud suits, and the compliance and integrity part carries most of these associations. Material weaknesses in internal control, which the risk and financial controls part should anticipate, are related to that part only (Appendix Table C10).

## 4.5 Does Each Share Pick Out Its Own Kind of Work?

A share could pass the tests above and still measure something common to several kinds of work, such as how technical or professional a firm's hiring is. Validity therefore also requires

that each share predict the outcomes of its own kind of work more strongly than the other shares do, the pattern of convergent and discriminant validity (Campbell and Fiske, 1959). Because the four shares enter every regression together, each coefficient measures what a share adds beyond the other three. Table 9 takes, for each kind of work, the outcome most directly tied to it: the green share of patent applications, the injury rate across a firm's US states, securities suits, and R&D. For environmental work, the green share measures the direction of a firm's invention rather than its amount, which the innovation share also predicts, since firms that innovate broadly file more applications of every kind, green or not (Appendix Table C18).

In every row, the coefficient of the share whose work the outcome concerns is positive, significant at the 1 percent level, and the largest. Green applications make up 7.6 percentage points more of the patenting of firms whose environmental share is one standard deviation higher, against about one point or less, in either direction, for the other shares. Within a firm and year, the social share is higher in the states whose establishments report more injuries, whereas the governance and innovation shares are lower there. The governance share predicts 21 percent more securities suits, against 14 percent for the innovation share, whereas the environmental and social shares do not predict them, and the innovation share predicts R&D higher by 3.96 percentage points of assets, against $-0.69$ to 0.65 for the other shares. The pattern holds across a wider set of outcomes. When each share is related to three outcomes of each kind of work, 8 of the 12 own-capability coefficients are significant at the 5 percent level, against 10 of the 36 others. A bootstrap that resamples firms 2,000 times rejects that the own-capability coefficients are jointly zero ($p = 0.0005$), and in only one draw do the other coefficients load as strongly as the own ($p = 0.001$).[23]

[23]The twelve outcomes are green patent applications, environmental incidents, and the change in toxic releases; serious violations, social incidents, and the injury rate; securities suits, non-standard audit opinions, and governance incidents; and patent applications, the value per granted patent, and R&D (Appendix Tables C18 and C19). The first test is a Wald statistic on the twelve own-capability coefficients with the bootstrap covariance, compared with its distribution over the recentred draws. The second is the mean absolute ratio of coefficient to clustered standard error in the own-capability cells minus that in the other cells, 1.19 in the sample, with a 95 percent bootstrap interval of 0.56 to 1.78; its $p$-value is the share of draws in which it is not positive, with the correction $(1 + k)/(B + 1)$.

**Table 9.** Each Share and the Outcomes of Each Kind of Work

| | Environmental share (1) | Social share (2) | Governance share (3) | Innovation share (4) | Observations (5) |
|---|---|---|---|---|---|
| Environmental: green share of applications, $t+1$ to $t+2$ (pp)[a] | **7.63**$^{***}$ | $-1.04^{**}$ | 0.26 | $-0.13$ | 11,266 |
| | (0.73) | (0.49) | (0.37) | (0.29) | |
| Social: injury rate in the firm's US states, $t+1$ to $t+2$[a,b] | 0.042 | **0.089**$^{***}$ | $-0.103^{***}$ | $-0.178^{***}$ | 68,098 |
| | (0.034) | (0.018) | (0.021) | (0.025) | |
| Governance: securities suits, $t+1$ to $t+2$ | 0.023 | $-0.081$ | **0.195**$^{***}$ | $0.127^{**}$ | 16,682 |
| | (0.055) | (0.076) | (0.055) | (0.050) | |
| Innovation: R&D over lagged assets, $t$ (pp)[a] | $-0.69^{***}$ | $0.36^{***}$ | $0.65^{***}$ | **3.96**$^{***}$ | 26,542 |
| | (0.13) | (0.12) | (0.15) | (0.15) | |

*Notes.* Each row regresses one outcome on the four standardized shares of year $t$, entered together, and the coefficients in bold relate each share to the outcome of its own kind of work. Rows 1, 3, and 4: firm-years of the estimation sample, with the logarithms of assets, employees, and openings and NAICS-3-by-year effects. The green share of applications is Y02 applications over all applications filed in $t+1$ to $t+2$, in percentage points, for firm-years with at least one application; securities suits are federal cases against the firm, estimated by Poisson pseudo-maximum likelihood; R&D is in percent of assets at the end of $t-1$. [a]OLS. [b]Firm, US state, and year cells with at least 10 openings, the shares of each cell's openings standardized across cells, with firm-by-year effects and the logarithm of the cell's openings; the injury rate counts the cases with days away, restricted, or transferred per 100 full-time workers at the firm's establishments in the state (Appendix Table C9). Standard errors clustered by firm in parentheses. $^{*}$, $^{**}$, and $^{***}$ denote significance at the 10, 5, and 1 percent levels.

## 4.6 Persistence, Variance, and Measurement Error

Industries and years explain most of the variation in the share, and what remains is mostly persistent within firms (Table 10). Six-digit industries and years explain 63 percent of the variance of the environmental share, 52 percent for clean technology and 68 percent for environmental management. The remaining firm-level variation, 37, 48, and 32 percent of the total, comes mostly from firm effects, which account for 83, 87, and 75 percent of it. Clean technology is thus the most firm-specific part of environmental work, and environmental management the most industry-driven. Sampling noise is small. A firm-year's share depends on which openings the firm happens to post, and comparing the shares of two random halves of each firm-year's openings shows how much. The two halves give a reliability of 0.98 across firm-years and 0.97 within six-digit industries and years, so that sampling noise accounts for 4.3 percent of the firm-level variance of the environmental share, 2.9 percent for clean technology, and 7.5 percent for environmental management. The comparison cannot detect

errors that the classifier makes alike on all of a firm's postings, since such errors appear in both halves. The comparisons with independent outcomes in Sections 4.2 to 4.5 address those.

**Table 10.** Variance Decomposition and Measurement Error

| | Environmental | Clean technology | Environmental management |
|---|---|---|---|
| Year | 0.9 | 0.9 | 0.2 |
| NAICS-6 × year | 62.7 | 51.7 | 67.6 |
| Firm level | 37.3 | 48.3 | 32.4 |
| of which persistent (firm effects) | 30.9 | 42.2 | 24.4 |
| of which transitory | 6.3 | 6.1 | 8.0 |
| Reliability: all / within NAICS-6 × year | 0.984 / 0.967 | 0.986 / 0.976 | 0.976 / 0.944 |
| Reliability within firm and NAICS-6 × year | 0.870 | 0.885 | 0.843 |
| Sampling noise, % of firm-level variance | 4.3 | 2.9 | 7.5 |

*Notes.* Estimation sample of Table 6. The first rows give the shares of variance, in percent of the total, explained by year, NAICS-6-by-year, and added firm effects; the firm level is what NAICS-6-by-year effects leave. Reliability correlates the shares of two random halves of each firm-year's openings (at least five in each) and applies the Spearman–Brown correction, averaged over two divisions; sampling noise is one minus the reliability across all firm-years, as a percentage of the firm-level variance.

The persistence of the share accords with what it measures. Capabilities are built by accumulating investment over time, and the advantage they confer rests on the difficulty of compressing that accumulation (Dierickx and Cool, 1989) and on the paths that a firm's earlier investments open for its later ones (Teece, Pisano and Shuen, 1997). A firm that builds an environmental capability should therefore direct its hiring toward environmental work year after year, and the share behaves in this way. Sampling noise, at 4 percent of the firm-level variance, does little to dampen this persistence. Persistence alone cannot validate the share, since any stable feature of a firm's postings would also persist, such as a company description repeated in every posting, which is why the definition disregards what firms say about themselves. Together with the outcomes that the share predicts across firms, however, it indicates that the share records a lasting orientation of hiring rather than year-to-year fluctuations in recruiting.

The same decomposition explains why the validation compares firms with one another. Within firms over time, the reliability of the share is lower, 0.87, and the variation that remains, 6 percent of the total for the environmental share, mixes changes in hiring with noise. Comparisons within firms nevertheless support each capability where the data allow them. The environmental share predicts green applications within firms at a little over half its association across firms, the social share is higher in the US states where the same firm's

establishments report more injuries, the governance share predicts derivative and fraud suits within firms, and the innovation share predicts R&D within firms. Associations with lasting attributes of the firm, such as its emissions and incidents, vanish within firms, as they should if those attributes change little while hiring moves.

Because a firm's postings can come from several career sites, a change in its share could also arise mechanically, when a site enters or leaves the firm's set after an acquisition or a change in the sites that LinkUp collects. I check this on the largest changes. A surge is a rise in a firm's share that is among the largest of the year, follows a year with little change, and is still at least half in place a year later.[24] Surges occur at 48 firms for environmental work, 108 for social work, 136 for governance work, and 88 for innovation. I decompose each surge into the change within the career sites that posted in both years and the contribution of the sites that entered or left. The change within continuing sites accounts for at least half of the rise in 94 to 98 percent of the surges. Most surges, however, occur at firms whose openings come from a single career site in both years, for which this holds by construction. Among the firms with several sites, the proportion is 81 to 89 percent, and the sites that are new in the surge year account on average for 9 to 16 percent of its openings (Appendix Table C20). Large changes in a firm's share thus mostly reflect changes in hiring at the same sites rather than changes in the sites observed.

# 5 Environmental Hiring and Exposure to Climate Industrial Policy

Having shown that the shares of hiring devoted to each of the four kinds of work track the activities that this work produces, I now use the measure to ask whether the orientation of hiring helps explain how firm value responds when outside shocks change the returns to the activities toward which that hiring is directed. I focus on the environmental share, because climate industrial policy provides a setting in which this question can be answered. News about elections and legislation changed, on identifiable days and in both directions, the payments that the United States offered for clean activity. I relate firms' returns on these days to the environmental share of their hiring in the year before, comparing each firm with the other firms of its industry. I conjecture that, if hiring reveals the activities that firms are positioning themselves to undertake, firms whose hiring is oriented toward environmental work should gain on news that raises the expected payments and lose on news that lowers

[24] A surge requires at least 50 openings in both years and a rise in the top decile of that year's increases, after a change in the previous year no larger than the year's median absolute change. Only a firm's first surge is counted.

them. I further conjecture that this response should come from the clean-technology work that the policy pays for rather than from environmental management. The remainder of this part describes the setting and the earlier evidence, states these conjectures more precisely, and tests them in turn.

## 5.1 The Setting

Climate policy offers a setting in which dated news changes the payoff of one specific capability. Over 2015–2024, the United States did not adopt a federal price on carbon. Its main climate policy, the Inflation Reduction Act (IRA) of August 2022 (U.S. Congress, 2022), instead pays firms for clean activity, mostly through uncapped tax credits per unit of clean output or investment, which account for $271 billion of the $392 billion that its climate provisions cost through 2031 (Bistline, Mehrotra and Wolfram, 2023, pp. 78–86). A firm benefits to the extent that it can carry out the supported activity, so the policy raises the payoff of a capability that has no expense line of its own.

The value of the credits also depends on whether the policy stays in force, and its direction changed with the presidency during the sample period (Kruse, Mohnen and Sato, 2024; Martins et al., 2025).[25] News about elections and legislation therefore changes the expected payoff of the capability in both directions.

I study the market's response on seven days of news about the policy (Appendix D). Three days brought news favorable to climate policy. On Saturday, 12 December 2015, the parties to the United Nations Framework Convention on Climate Change agreed to the Paris Agreement (Kruse, Mohnen and Sato, 2024). Over the weekend of 7 and 8 November 2020, the major networks called the presidential election for Joe Biden (Ramelli et al., 2021). On 27 July 2022, after the close of trading, Senator Manchin and the Democratic leadership announced their agreement on the IRA (Bauer, Offner and Rudebusch, 2026). Four days brought unfavorable news: the election of Donald Trump on 8 November 2016; the announcement of the withdrawal from the Paris Agreement on 1 June 2017; press reports on the evening of 14 July 2022 that Senator Manchin, the pivotal vote in the Senate, would not support new climate spending (Bauer, Offner and Rudebusch, 2026); and the re-election of Donald Trump on 5 November 2024 (Martins et al., 2025). Day 0 is the day on which the news broke, or the next trading day if it broke on a weekend: 14 December 2015, 9 November 2016 (the day after the vote), 1 June 2017, 9 November 2020, 14 and 27 July 2022, and 6 November 2024

[25] In July 2025, after the last of the days studied, a budget reconciliation law ended the credit for clean vehicles acquired after September 2025, restricted the clean hydrogen credit, and ended the clean electricity credits for wind and solar facilities placed in service after 2027, unless their construction began by 4 July 2026 (U.S. Congress, 2025, secs. 70502, 70511–70513).

(the day after the vote; Internet Appendix Table IA.38). I measure returns over trading days $-1$ to $+1$, so that each window includes the first full trading session after the news, including on the two days in July 2022 when the news broke after the close.

Three sets of days appear in the paper. The six days other than 14 July 2022 are the main sample, three favorable and three unfavorable, and I report every result with and without 14 July 2022, which I add following Bauer, Offner and Rudebusch (2026). Sections 6.6 and 7 add two days of news about the regulation of emissions and climate disclosure, the Supreme Court's decision in *West Virginia v. EPA* on 30 June 2022 (Supreme Court of the United States, 2022) and the SEC's adoption of its climate disclosure rule on 6 March 2024 (U.S. Securities and Exchange Commission, 2024), for eight days in the comparison with ratings and emissions and in the test of environmental management work. Section 6.6 examines two further days of news about the policy.

The days differ in how much news they carried. The 2016 election was a large surprise (Ramelli et al., 2021), and on the two days in July 2022 the probability that the legislation would pass first fell to close to zero and then rose to close to one (Bauer, Offner and Rudebusch, 2026). The withdrawal from the Paris Agreement had been promised during the 2016 campaign (Kruse, Mohnen and Sato, 2024), and Biden's victory was widely expected (Ramelli et al., 2021). Day 0 of the 2020 election, 9 November 2020, is also the day on which Pfizer and BioNTech announced the first results of their COVID-19 vaccine trial (Appendix D.3). On these two days, a small response would reflect a small surprise rather than a small weight of clean activity in firm value, so Section 6 reports each day separately. Elections also carry news unrelated to climate policy.

## 5.2 Earlier Evidence on Climate-Policy News

Earlier studies of these days measure firms' exposure in two ways, and their conclusions differ accordingly. Studies that rely on environmental ratings and emissions find that highly rated firms gained on days that moved climate policy in opposite directions. Ramelli et al. (2021) find that firms with strong climate-responsibility ratings gained on the 2016 election, which set back climate policy, as did carbon-intensive firms, and that ratings were rewarded again once Biden's victory was called in November 2020. Bauer, Offner and Rudebusch (2026) find that firms with higher LSEG environmental scores gained on the agreement on the IRA, and Martins et al. (2025) that energy firms with high environmental ratings gained after the 2024 election. Studies that rely on the clean products that firms sell, by contrast, find responses in the direction of the news. Kruse, Mohnen and Sato (2024) show that firms with a larger share of green revenue earned higher returns around the Paris Agreement, and Lu

et al. (2025*a*) that firms whose 10-K business descriptions mention more climate-solution products gained on the Paris Agreement, on the confirmation of Biden's election, and on the agreement on the IRA, and lost on the 2016 election. These studies compare firms within broad industry groups, if at all, and each examines one day or a few.

The two groups of measures record different objects, which may explain why their conclusions differ. Emissions record the outcome of past operations and are most relevant to carbon pricing and to the regulation of emissions (Bolton and Kacperczyk, 2021, 2023; Hsu, Li and Tsou, 2023). Environmental ratings are assessments based largely on disclosure, which diverge across providers (Berg, Kölbel and Rigobon, 2022). I use those of one provider, LSEG, and they load more on environmental management than on clean technology, so that the comparisons below concern this provider's score. Climate talk in earnings calls is forward-looking (Sautner et al., 2023*a*), but it records statements rather than resources. Clean products, green patents, and the revenue of clean business segments record activity specific to clean technology (Hege, Zhang and Pouget, 2025; Cohen, Gurun and Nguyen, 2026; Kruse, Mohnen and Sato, 2024), but they appear only once research produces output or a business generates revenue, and in the market sample green patents and clean segments exist for only 18 and 2 percent of firm-years. Measures built from hiring rules classify postings as green from their skills, titles, or occupations (Saussay, Sato and Vona, 2026; Curtis and Marinescu, 2023; Darendeli, Law and Shen, 2022; Ben Dor et al., 2022), and each yields a single indicator that does not distinguish clean technology from environmental management.

Most closely related to this paper is the work of Lu et al. (2025*a*), who measure the climate-solution products and services that firms describe in their 10-K filings with a language model (see also Lu et al., 2025*b*) and find that the stock prices of firms with more such products respond to climate-related regulatory events in the direction of the news. I diverge from their work in what is measured, in the comparison, and in scope. More specifically, their measure records the products that firms already sell, whereas the share of hiring records the work that firms are adding before it produces output, so that it can identify firms whose clean activity has yet to appear in their businesses. In addition, whereas they compare firms within broad industry groups, I compare firms within three-digit industries, separate clean technology from environmental management, and examine whether ratings and emissions are repriced once the orientation of hiring is held fixed. Finally, the procedure is not specific to climate, since it measures other capabilities with the same design.

## 5.3 Why the Orientation of Hiring Should Matter

To see how such news moves firm value, I write the value of firm $i$ on day $t$ as

$$V_{it} = V_{it}^{0} + \sum_{\tau > t} \delta_{\tau - t}\, \mathbb{E}_t\Big[\pi_\tau\Big]\, a_{i\tau}, \tag{7}$$

where $a_{i\tau}$ is the clean activity that the firm will be able to carry out in year $\tau$, $\pi_\tau$ is the margin per unit of that activity, $\delta_{\tau-t}$ is a discount factor, and $V_{it}^{0}$ is the value of the firm's other prospects. The expected margin depends on the path of climate policy through several channels: the payments that the policy makes, the demand for clean products, the regulations that favor them, and the competition they face. News about the policy changes this expected margin. If the news changes the expected margin in every year by the same proportion $\theta_t$, and leaves the firm's other prospects unchanged, it moves the firm's value by

$$\frac{\Delta V_{it}}{V_{it}} = \theta_t\, w_{it}, \qquad w_{it} = \frac{1}{V_{it}} \sum_{\tau > t} \delta_{\tau - t}\, \mathbb{E}_t\Big[\pi_\tau\Big]\, a_{i\tau}. \tag{8}$$

The weight $w_{it}$ is the share of the firm's value that comes from the clean activity it will be able to carry out, and news about the policy moves firms in proportion to this weight. The framework covers every day in the sample. The IRA changed the payments directly, whereas the Paris Agreement and the elections changed the expected path of payments, regulation, and demand. It abstracts from risk premia and from the effect of the news on the firm's other prospects, which comparisons within industries partly remove.

Because $a_{i\tau}$ includes activity that the firm does not yet carry out, the weight spans both parts of firm value that Myers (1977) distinguishes, the assets already in place and the growth opportunities, whose value depends on investments that the firm may choose to make later. The clean activity that a firm already conducts belongs to the first part, and the clean activity into which it could expand belongs to the second. Such an opportunity has value, however, only for a firm able to take it up. Expanding into an activity requires people who can design, build, and operate it, and these capabilities are what turn a possible investment into an option that the firm can exercise (Kogut and Kulatilaka, 2001). Because capabilities of this kind are embodied in the firm's employees (Eisfeldt and Papanikolaou, 2013) and take time to build (Pindyck, 1993), they must be in place before news about the policy arrives, so that the firms positioned to expand into clean activity when its return rises are those that developed these capabilities beforehand. News about the policy therefore reprices growth opportunities in a particular activity rather than growth opportunities in general. In the model of Kogan and Papanikolaou (2014), a shock to the productivity of new capital moves firm value in proportion to the share of that value which growth opportunities represent,

whatever activity they concern. A policy that pays for one activity instead moves firm value in proportion to the part that comes from that activity, so that two firms whose growth opportunities are equally large can be exposed very differently, and measures of growth in general, such as Tobin's $q$ or the market-to-book ratio, need not capture this exposure.

The weight $w_{it}$ is not observed, and the hiring share is not a measure of it. The share measures the direction of a firm's hiring, which should be related to $w_{it}$ in three ways. First, because hiring is how firms develop these capabilities, its direction indicates which growth opportunities a firm is positioned to take up, so a firm still building its capacity can carry a large weight before it has any clean output. Second, the policy pays mostly for clean activity rather than for environmental management, although the correspondence is not exact: not all clean-technology work is subsidized, and the policy also affects environmental management through regulation. Third, firms in the same industry differ in how well they are positioned for the policy, so comparisons within industries are informative. Policy shifts the relative return to clean and dirty technologies (Acemoglu et al., 2012), and because firms' own histories shape the direction of their innovation (Aghion et al., 2016), firms in the same industry meet this shift from different positions.

The measure lets me examine the second of these relations directly. Each posting that the labeller codes as environmental work is assigned one of fifteen classes, which I group into two parts (Section 3.1). Clean technology is the work that researches, builds, installs, or operates the technologies that generate, store, or use low-carbon energy, and environmental management is the work that measures, manages, reduces, or reports a firm's environmental impacts or provides environmental services. The classifier has one head for each part, so that the environmental share is the sum of a clean-technology share and an environmental management share (Section 3.2), and each part predicts the technologies that its own work produces (Section 4.3).

These considerations lead to five tests, which Sections 6 to 8 carry out in turn. First, if the orientation of hiring is related to the weight $w_{it}$, returns on days of news about the policy should move with the environmental share, in the direction of the news and relative to industry peers (Section 6). Second, I ask whether the share carries information that existing measures do not. The clean businesses that firms already report should not fully account for the repricing, since capacity under construction generates no revenue yet, and once the orientation of hiring is held fixed, environmental ratings need not move in the direction of the news, as they would if the news worked through tastes for green assets or concern about climate change (Pástor, Stambaugh and Taylor, 2022; Ardia et al., 2023), whereas emitters should respond the most if it worked through the prospect of a carbon price (Bolton and Kacperczyk, 2021, 2023) (Section 6.6). Third, if the market values the

growth opportunities in clean activity that these capabilities make it possible to take up, an interpretation that I call the activity channel, the firms whose hiring is oriented toward clean work should be those that later undertake the activity that the policy pays for, the repricing should follow clean-technology work more than environmental management, since the credits pay mostly for the former, and it should depend on the orientation of hiring rather than on how large a part of firm value growth opportunities represent in general. Analysts should revise their long-term growth forecasts for these firms in the direction of the news more than their near-term earnings forecasts, since the margins accrue over a decade, and because firms exercise growth opportunities when their value rises, these firms should invest more after favorable news and less after unfavorable news (Section 7). Fourth, if prices reflect the information in the share, it should predict returns when news about the policy arrives but not on average (Section 7.1). Fifth, because $w_{it}$ is a weight in firm value, the repricing should be largest where the capability is a large part of the firm, as in small firms and in firms with few tangible assets, and it may be larger where analysts follow the firm, since analysts have stronger incentives to follow firms whose intangible assets the financial statements do not recognize (Barth, Kasznik and McNichols, 2001) (Section 8).

# 6 Environmental Hiring and Firm Value on Policy Days

This section carries out the first two tests. I describe the design, report the main result and how it varies along the distribution of the share, show when the repricing occurs and that it persists, measure its size in dollars, and examine its robustness. Finally, I ask whether the result reflects other characteristics of firms that the news may reprice, among them the clean businesses that firms report and their ratings and emissions, and I state the limits of the design.

## 6.1 Empirical Design

For each firm $i$ and each day $e$, I compute the cumulative abnormal return over trading days $-1$ to $+1$, $CAR_{ie}$, in percentage points. Abnormal returns are measured against the five factors of Fama and French (2015) and momentum (Carhart, 1997), with betas estimated over trading days $-250$ to $-30$ before the event. I winsorize the returns at the 1st and 99th percentiles of each day. The sample comprises the firms in Compustat North America with a CRSP security and an environmental share in the year before the event, all of which trade on US exchanges, although about 12 percent of the firm-events belong to firms headquartered

elsewhere, mostly in Canada.[26] On the six days other than 14 July 2022, it contains 12,407 firm-events of about 3,900 firms, between 1,264 and 2,717 per day.

I estimate

$$d_e \, CAR_{ie} = \beta \, H^E_{ie} + \gamma' X_{ie} + \alpha_{j(i)e} + \varepsilon_{ie}, \tag{9}$$

where $d_e$ equals $+1$ on days favorable to climate policy and $-1$ on unfavorable days. The term $H^E_{ie}$ is the firm's environmental share in the calendar year before the event, standardized among the firms with returns around the event. The vector $X_{ie}$ contains the logarithms of market equity and book-to-market, momentum, market beta, idiosyncratic volatility, and the logarithms of the firm's openings, employees, and assets in the year before the event, so that neither firm size nor the coverage of the hiring data drives the estimate. The term $\alpha_{j(i)e}$ is a NAICS-3 industry-by-event effect, which absorbs what the news does to industries as a whole. If the share is related to the weight of clean activity in firm value, $\beta$ is positive. It measures the additional return, in the direction of the news, of a firm whose environmental share is one standard deviation higher than that of its industry peers.

I cluster standard errors by firm. Returns on the same day are correlated across firms, however, and conventional standard errors may then overstate the precision of event-day estimates. I therefore also compute placebo $p$-values. For each event, I repeat the regression on every trading day within 24 months of the event that lies more than seven days from any climate-policy day. For a single day, the $p$-value is the share of these placebo days on which the coefficient is at least as large in absolute value as the actual one. For the pooled regression, I draw one placebo day per event 20,000 times (Appendix D.2).

## 6.2 Main Results

Day by day, the environmental share enters with the sign of the news on five of the six main days and on 14 July 2022 (Figure 2, left panel, and Appendix Table E1); the exception is the 2020 election, to which I return below. The coefficient is large and significant on three days. A firm whose environmental share is one standard deviation higher lost 0.82 percentage points relative to its industry peers on the 2016 election and 1.15 points on the 2024 election, and it gained 0.83 points on the agreement on the IRA. The placebo $p$-values on these days are 0.003, 0.004, and 0.008, respectively. On the other four days, the coefficients are small and imprecise. Two of these days carried little news, as Section 5.1 explains. The 2020 election is also confounded by news about COVID-19 vaccines, since a firm's abnormal return on 16 November 2020, the day of the next vaccine announcement, is strongly related to its return

[26] Restricted to the 10,900 firm-events of firms headquartered in the United States, the pooled estimate of Table 11, column 1, is 0.55 ($t = 5.8$), against 0.59 in the full sample.

around 9 November (Appendix D.3). I therefore do not interpret the coefficient of −0.07 on that day.

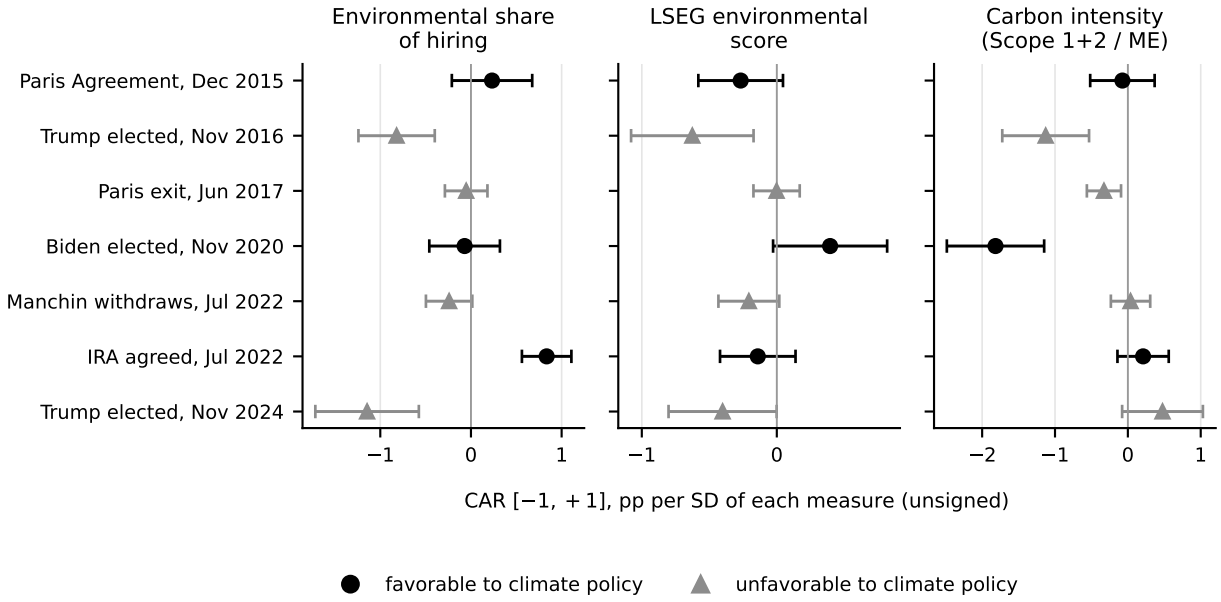



**Figure 2.** Hiring, Ratings, and Emissions on Policy Days. Each point is the coefficient of one firm attribute, standardized among the firms with returns around the event, in a regression of the unsigned CAR[−1,+1] with NAICS-3 effects and the controls of equation (9), with its 95 percent confidence interval. Each attribute enters alone, on the firms for which it exists.

Holding the event controls and industry effects fixed, only the orientation of hiring, among three firm attributes, sorts firms in the direction of the news (Figure 2). The other two panels show the same regression for two common measures of climate exposure, each estimated alone on the seven days. Neither lines up with the direction of the news. The environmental score of LSEG, the one rating provider I use, falls on six of these seven days, favorable and unfavorable alike, and never rises significantly at the 5 percent level. Carbon intensity has the sign that a carbon price would imply on four of them. Section 6.6 compares the three in the same regression.

Pooled over the six main days and signed by the direction of the news, the coefficient is 0.59, with a $t$-statistic of 6.6 (Table 11, column 1). A firm whose environmental share is one standard deviation higher thus earns 0.59 percentage points more in the direction of the news over the three days. Only one of the 20,000 placebo draws yields a coefficient as large, so the placebo $p$-value is 0.0001. Adding 14 July 2022 to the six days gives 0.53 ($t = 6.5$). The estimate holds across subsets of days. It is 0.39 on the three favorable days and 0.76 on the three unfavorable days, and 0.67 on the three elections and 0.50 on the other three days, with placebo $p$-values all below 0.03.

**Table 11.** Policy-Day Returns and the Environmental Share

| | Six days (1) | Seven days (2) | Favor-able (3) | Unfavor-able (4) | Elec-tions (5) | Other days (6) |
|---|---|---|---|---|---|---|
| Environmental share | 0.592*** | 0.527*** | 0.386*** | 0.762*** | 0.665*** | 0.498*** |
| | (0.090) | (0.081) | (0.109) | (0.136) | (0.145) | (0.092) |
| Placebo $p$ | <0.001 | <0.001 | 0.022 | <0.001 | 0.001 | 0.007 |
| Firm-events | 12,407 | 15,122 | 6,386 | 6,021 | 6,657 | 5,750 |

*Notes.* Signed CAR[−1,+1] (pp) on the standardized environmental share of the year before, with the event controls and NAICS-3-by-event effects. Columns: the six days without 14 July 2022, the seven days, and subsets of the six. Placebo $p$: share of 20,000 draws of placebo days with a coefficient at least as large (Appendix D.2). Standard errors clustered by firm in parentheses. *, **, ***: 10, 5, 1 percent.

The larger estimate on unfavorable days should not be read as an asymmetric response to the policy. Most of the gap comes from the 2020 election, whose outcome was widely expected (Ramelli et al., 2021) and which coincided with the news about vaccines. Without that day, the estimate on the two remaining favorable days is 0.67, close to the 0.76 on unfavorable days, and the two do not differ significantly at the 5 percent level. The days that carried the largest surprise in each direction, the agreement on the IRA and the 2016 election, give coefficients of almost equal size, 0.83 and −0.82. The evidence thus shows only that losses on unfavorable news are not smaller than gains on favorable news.

The repricing rises with the share across its distribution (Figure 3). Relative to firms in the lowest quintile of the environmental share within their industry, firms in the second to fifth quintiles earn 0.11, 0.15, 0.26, and 0.67 percentage points more in the direction of the news, and for the clean-technology share the differences are 0.31, 0.30, 0.53, and 0.66. The share is concentrated, since most firms devote little of their hiring to environmental work and a few devote much of it, but the estimate does not depend on the firms at the top. Without the top 1 or 5 percent of the share within industries, the slope per percentage point of the share remains significant, at 0.071 and 0.088 against 0.107 in the full sample, and the standardized coefficient is unchanged when the share is winsorized at its 95th percentile or taken in logarithms (Appendix Table E2).

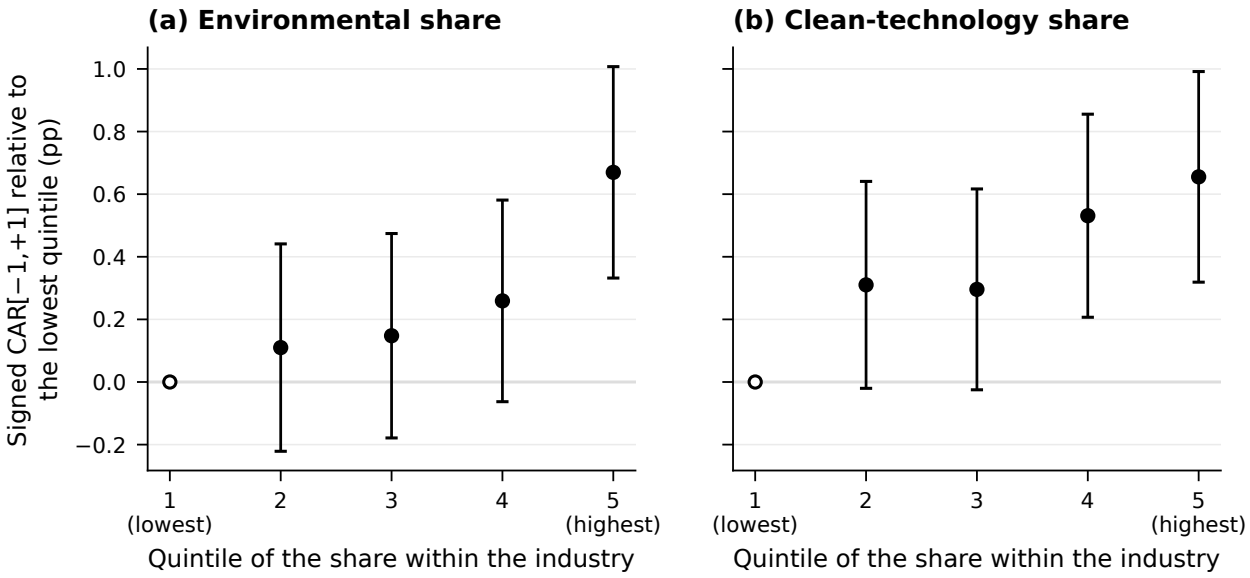



**Figure 3.** The Repricing along the Distribution of the Share. Coefficients of indicators of quintiles of each firm's share ranked within its NAICS-3 industry on each day (the lowest quintile omitted), in the signed regression of equation (9) on the six days, with 95 percent confidence intervals. Panel (b) holds the environmental management share fixed.

## 6.3 Timing and Persistence

If the repricing reflects the news, it should occur when the news reaches the market. Figure 4 plots the coefficient of the environmental share on each trading day around four days on which the timing of the news is precise. In each case, the largest coefficient falls on the first trading session after the news. On the 2016 and 2024 elections, it falls on the day after election day, at $-0.62$ and $-0.90$ ($t$-statistics of $-4.0$ and $-4.5$). Senator Manchin's withdrawal of support was reported in the evening of 14 July 2022, and the coefficient is $-0.28$ on 15 July ($t = -4.0$). The agreement on the IRA was announced after the close on 27 July 2022, and the coefficient is 0.65 on 28 July ($t = 6.0$).

The timing also separates the IRA from other news. On 27 July 2022, the Federal Open Market Committee raised its target range by 75 basis points. The coefficient on that day is 0.19, under a third of its value on 28 July, the first session after the IRA news. Some responses continue beyond the first session, as on 29 July 2022 and on 8 November 2024, and the response to Senator Manchin's statement partly reversed on 18 July 2022. Smaller coefficients also appear on some earlier days, such as 25 July 2022. The three-day window captures most of each response.

On the four days of Figure 4, the news arrived after the close, so the response can be located more precisely still. Election results arrive overnight, Senator Manchin's withdrawal was reported in the evening, and the agreement on the IRA was announced after the close, so

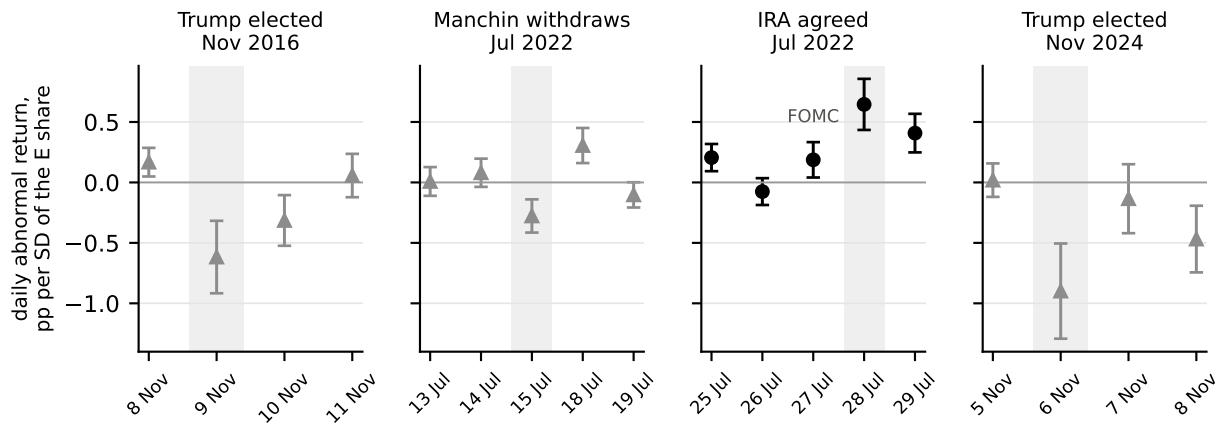



**Figure 4.** Timing of the Repricing. Each point is the coefficient of the environmental share in a regression of one day's abnormal return, as observed, around four days on which the timing of the news is precise, with 95 percent confidence intervals.

the return from the previous close to the open of the first session after the news should carry most of the repricing. It does. Pooled over the four days and signed by the direction of the news, the coefficient of the environmental share is 0.45 ($t = 6.3$) on the overnight return and 0.18 ($t = 3.4$) on the return from the open to the close, and the overnight return carries the larger part on each day (Appendix Table E4). On the agreement on the IRA, the overnight return of 28 July also leaves out the decision of the Federal Open Market Committee, which came in the afternoon of the day before.

Over a longer horizon, the repricing neither anticipates the news nor reverses (Figure 5). Over the ten trading days before the news, the cumulative coefficient of the environmental share stays close to zero, at 0.17 on the day before the window (standard error 0.11), and the window from day −10 to day −2 has a placebo $p$-value of 0.64. The coefficient then rises to 0.74 by day +1 and to 0.98 by day +2, and the further response over days +2 to +5, 0.29, suggests that part of the news is absorbed within the first week. Thereafter it neither drifts further nor reverses. At day +60 it is 0.78 (standard error 0.32), and the windows from day +2 to day +20 and to day +60 have placebo $p$-values of 0.63 and 0.93 (Appendix Table E3). These longer windows also contain later news about the policy, such as the passage of the IRA in August 2022 and the Georgia runoffs of January 2021, which accounts for part of their variation.

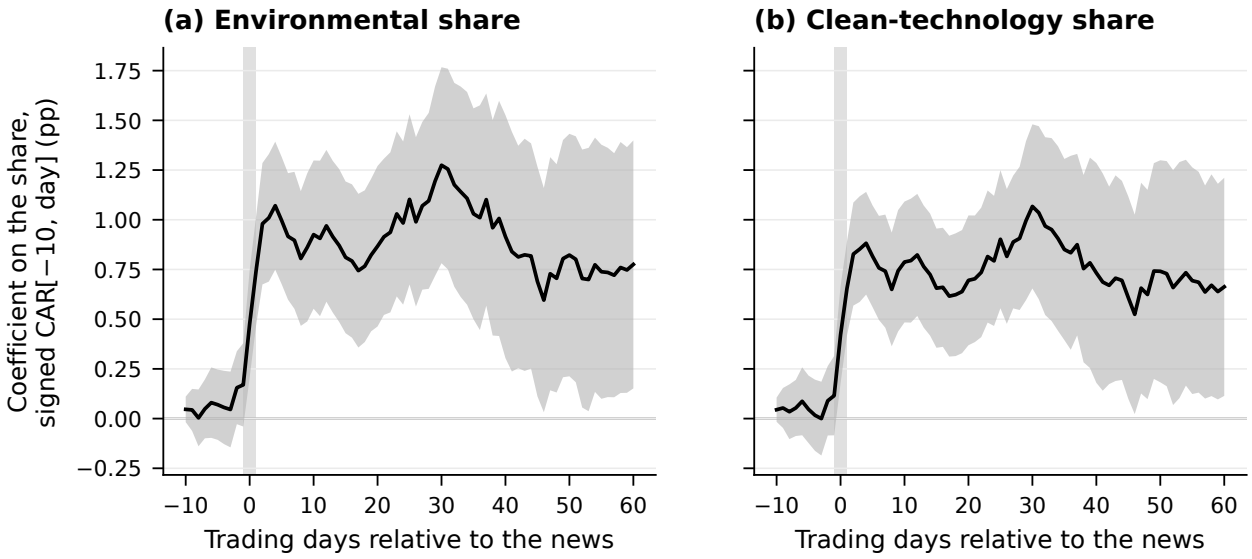



**Figure 5.** The Cumulative Response. Coefficient of the share in the signed regression of the CAR from day −10 to each day, on the six days, with 95 percent confidence bands; the shaded days are the event window. Firm-events with returns on every day from −10 to +60 (12,253).

## 6.4 Magnitudes

How large are these differences in dollars? For each day, I apply the coefficient of that day to each firm's environmental share minus the mean of its industry, which is the variation that the industry effects leave. I then multiply the implied return by the firm's market value at the end of the previous month. Table 12 reports the sum over the firms whose share exceeds their industry's mean. These amounts are regression-implied changes in market value relative to industry peers, the part of the repricing that the regression attributes to the share with all else held fixed, and not the observed returns of any firm. Because the industry effects absorb what the news does to entire industries, they are not estimates of the policy's total incidence, of the subsidies that firms receive, or of observed transfers of wealth.

The amounts are large on the days that carry the result, but they depend on how firms are weighted. With the coefficients of Table E1, which weight every firm equally, the agreement on the IRA is associated with a relative revaluation of $56 billion toward the firms above their industry's mean, the 2016 election with $20 billion away from them, and the 2024 election with $102 billion away from them (Table 12, columns 1 to 3). These amounts apply a single coefficient to firms of every size, and they are concentrated, since the tenth of firms furthest above their industry's mean accounts for 78 to 89 percent of the amount on every day (Internet Appendix Table IA.42). Because the response is weaker among the largest firms (Section 6.5), I also compute the amounts with coefficients that weight firms by their market value (columns 4 to 6). They are $48 billion on the IRA and $19 billion on the 2016 election,

**Table 12.** The Repricing in Dollars, Relative to Industry Peers

| | Equal-weighted coefficient | | | Value-weighted coefficient | | |
|---|---|---|---|---|---|---|
| | Coefficient (1) | $ billion (2) | 95% interval (3) | Coefficient (4) | $ billion (5) | 95% interval (6) |
| Paris Agreement, 2015 (+) | 0.232 | 5.6 | [−5.1, 16.2] | −0.124 | −3.0 | [−13.9, 7.9] |
| | (0.226) | | | (0.232) | | |
| Trump elected, 2016 (−) | −0.820*** | −19.7 | [−29.8, −9.6] | −0.771*** | −18.5 | [−30.6, −6.4] |
| | (0.215) | | | (0.257) | | |
| Paris withdrawal, 2017 (−) | −0.053 | −1.4 | [−7.7, 4.9] | −0.041 | −1.1 | [−5.1, 2.8] |
| | (0.120) | | | (0.075) | | |
| Biden elected, 2020 (+) | −0.070 | −3.4 | [−22.1, 15.4] | 0.275 | 13.3 | [−6.9, 33.4] |
| | (0.199) | | | (0.213) | | |
| Manchin withdraws†, 2022 (−) | −0.241* | −16.2 | [−33.3, 1.0] | −0.196 | −13.2 | [−29.1, 2.7] |
| | (0.130) | | | (0.121) | | |
| IRA agreement, 2022 (+) | 0.835*** | 56.0 | [37.7, 74.3] | 0.714*** | 47.9 | [24.6, 71.3] |
| | (0.139) | | | (0.178) | | |
| Trump re-elected, 2024 (−) | −1.146*** | −101.7 | [−152.4, −51.0] | −0.076 | −6.7 | [−87.7, 74.2] |
| | (0.291) | | | (0.465) | | |

*Notes.* Sum over the firms whose environmental share exceeds their NAICS-3 mean of the day's coefficient times the firm's share minus that mean, times its market value at the end of the previous month. Columns 1–3: unweighted coefficients; 4–6: coefficients weighting firms by market value. Intervals from the coefficient's standard error. †Not in the six main days.

both significant, and $7 billion on the 2024 election, which is not significantly different from zero. The amounts are thus best read as a range, from about $48 to $56 billion on the IRA and from $19 to $20 billion on the 2016 election. On the 2024 election, the value-weighted coefficient rests on a handful of the largest firms. Without the ten largest, it is −0.97 ($t = -4.2$), close to the equal-weighted −1.15.

## 6.5 Robustness

The estimate does not depend on the choice of days. Dropping one day at a time leaves a coefficient between 0.41, without the 2024 election, and 0.75, without the 2020 election, with $t$-statistics of 5.0 or more, so that three days carry most of the result but no single day drives it. The result also holds on two further days of news about the policy. On 6 January 2021, the Georgia Senate runoffs gave Democrats the Senate majority that later passed the IRA, and on 20 December 2021, Senator Manchin's rejection of the Build Back Better Act ended the prospects of its climate credits at the time. The environmental share has the sign of the news on both days, 0.89 ($t = 5.0$) and −0.53 ($t = -4.4$), and with these two days the pooled

estimate is 0.65 ($t = 7.6$; Appendix Table E7). Both days also carried broad news about fiscal policy.

Firms with a higher environmental share do not simply move more on days of large market news. On the 40 days with the largest market moves over 2015–2024, with returns signed by the direction of the market, the coefficient of the environmental share is 0.02 (standard error 0.02), and on the 20 largest of these days it is negative (Appendix Table E8). Moreover, the market rose on each of the six climate-policy days, by 0.29 to 2.91 percent, so that a higher sensitivity to market news would raise these firms' returns on unfavorable days as well, whereas they fall.

The estimate does not depend on the return benchmark or on the smallest stocks either (Appendix Table E6). Abnormal returns from the market model, and returns net of the market, give coefficients of 0.51 and 0.52, with placebo $p$-values below 0.001. Excluding the firm-events of stocks priced below \$5, or with market equity below the 20th percentile of NYSE firms, gives 0.59 and 0.55. Nor does the estimate rest on holding each control to the same relation with returns on every day. When each of the eight controls takes its own coefficient on each day, as in the regressions for single days, the coefficient is 0.57 ($t = 6.0$). Firm-specific news does not drive the estimate either. Several of the days fall in seasons of earnings announcements, and a fifth of the firm-events on the agreement on the IRA and a quarter of those on the 2024 election belong to firms that announced quarterly earnings within the window. Without them, the coefficient is 0.56 ($t = 7.1$). The estimate is not specific to the firms that the sample covers either. These firms are larger and older than the average listed firm, because they recruit through their own career sites, but when each firm-event is weighted by the inverse of its probability of inclusion, estimated from size, book-to-market, age, exchange, and industry among all listed firms, so that the sample resembles this population, the coefficient is 0.56 ($t = 7.1$). Weighting firms by their market equity, however, reduces the coefficient to 0.22 ($t = 2.1$). The 50 largest firms carry 40 to 50 percent of the weight in these regressions, and the response is weaker among the largest firms, as Section 8 shows in more detail. Weighting firms by the square root of their market equity gives 0.41 ($t = 4.5$). Portfolios sorted within industries, without controls, show the same contrast. With equal weights, the long–short portfolios sorted on the environmental share and on its clean-technology part each earn 0.53 percent per event, whereas with value weights they earn 0.31 percent (placebo $p = 0.10$) and 0.04 percent (Appendix Table E16). The repricing is thus concentrated among small and asset-light firms, as Section 8 shows.

Finally, the estimate is not an artifact of noise in the share. A firm-year's share is the mean of its postings' probabilities, so that it is measured more precisely when the firm posts more openings. Among firm-years with at least 100 or 250 openings, the coefficient rises

to 0.73 and 0.77, as attenuation from sampling noise would predict. Nor does the estimate depend on the rules for keeping postings. When the shares keep postings of any duration, the coefficient is 0.60, and it remains 0.60 when the postings still open at the end of the data are added (Appendix Table E9).

## 6.6 Is It the Capability?

The main threat to interpreting $\beta$ as reflecting the weight of clean activity in firm value is that the environmental share proxies for other characteristics that the news also reprices. Firms that hire toward environmental work may emit more, receive higher ratings, talk more about climate change, spend more on intangible capital, hold green patents, sell clean products, or differ in their exposure to taxes and trade, as Wagner, Zeckhauser and Ziegler (2018) show for the 2016 election. Figure 6 summarizes how the coefficient changes when measures of each of these characteristics enter the regression, alone and together, and in samples in which some of them cannot drive the result.

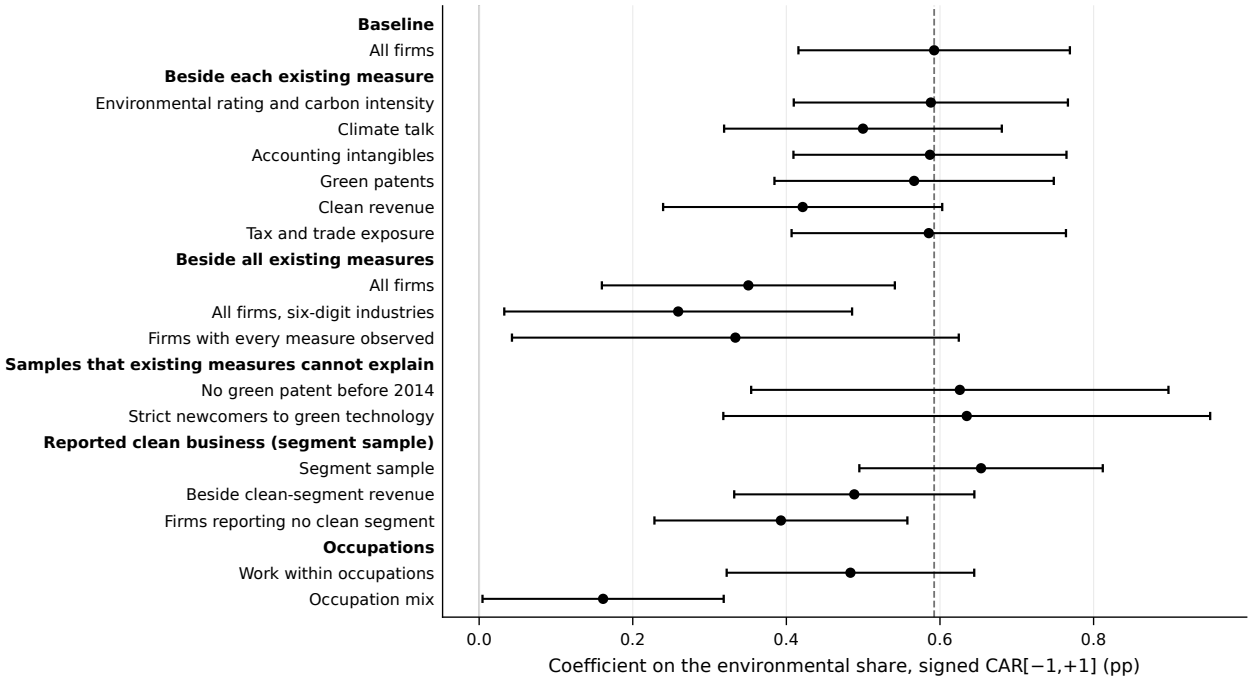



**Figure 6.** Is It the Capability? Coefficient of the environmental share in the signed regression of equation (9) on the six days, beside existing measures, in samples that existing measures cannot explain, beside reported clean business, and split into the parts within occupations and from the occupation mix, with 95 percent confidence intervals. The dashed line marks the baseline. Full regressions in Appendix Tables E11, E12, and E13.

The share does not simply reflect how a firm hires in general. The social, governance, and innovation shares are measured in the same way, and firms that hire toward one kind of work

may hire toward others, so that the social share correlates with the environmental share at 0.36 among the firm-events of the six days. When the three enter the regression beside the environmental share, its coefficient is 0.61 ($t = 6.7$), against 0.59 alone. Entered alone, none of the three has a significant coefficient ($-0.10$, $-0.10$, and 0.11; Appendix Table E10). I do not read these coefficients as placebo tests, since news about the policy may also bear on social or innovation work, and Part I establishes what each share measures (Section 4.5).

Each characteristic alone changes the estimate little. Climate talk and the revenue of clean-technology segments reduce it the most, to 0.50 and 0.42. With all characteristics together, the coefficient is 0.35 ($t = 3.6$), and within six-digit industries it is 0.26 ($t = 2.2$). More than two fifths of the baseline estimate therefore survive all the existing measures at once, within narrowly defined industries. This part is information that the orientation of hiring carries and the existing measures do not. Several of these measures are missing for many firms, and the baseline keeps every firm by setting missing values to zero with an indicator. On the 5,688 firm-events with every measure observed, the coefficient falls from 0.46 to 0.33 ($t = 2.2$) once all of them are added (Appendix Table E11).

In samples in which some of these characteristics cannot drive the result, the estimate is as large as in the full sample. Among firms with no green patent application in 2007–2013, which existing patent-based measures cannot see, the coefficient is 0.63 ($t = 4.5$). It is 0.63 ($t = 3.9$) among these firms when utilities and the tenth of six-digit industries with the highest 2007–2013 Y02 share of patent applications are also excluded. On the three elections, the tax and trade exposures that Wagner, Zeckhauser and Ziegler (2018) and Ramelli et al. (2021) control for change the coefficient from 0.67 to 0.64 (Appendix Table E12).

Nor does the share stand in for the green knowledge that a firm has accumulated. When the stocks of its Y02 and other patent applications since 2007 enter the regression, the coefficient of the share is 0.58 ($t = 6.3$), against 0.59 without them. The response is not confined either to the firms that supply the equipment that clean investment requires, whose sales the policy would raise directly. Among manufacturers of machinery, of electronic and electrical equipment, and of transportation equipment other than motor vehicles, and among construction firms, the clean-technology share has a coefficient of 0.55 ($t = 5.8$), and among all other firms it has a coefficient of 0.51 ($t = 4.6$; Appendix Table E25).

A firm may also be repriced because it already sells clean products, as Kruse, Mohnen and Sato (2024) show for the Paris Agreement. To examine this possibility, I identify the business segments in Compustat that sell clean-technology products, such as renewable power, solar and wind equipment, electric vehicles, batteries, and biofuels. Such segments are rare. They exist for 2 percent of firm-years, and across firm-years the clean-technology share and the revenue share of clean segments correlate at 0.50. In the sample of firms with segment

data, the coefficient of the environmental share is 0.65 ($t = 8.1$). With the revenue share of clean segments and an indicator of any clean segment, it remains 0.49 ($t = 6.1$). Reported clean revenue is also repriced (0.21, $t = 2.5$), as Kruse, Mohnen and Sato (2024) find, but it absorbs only a quarter of the coefficient. Among firms that report no clean segment, the environmental share has a coefficient of 0.39 ($t = 4.7$; Appendix Table E13). Hiring thus adds information beyond the clean businesses that firms report in their segments. Segments are aggregated, however, so a firm without a separately reported clean segment may still have clean-technology operations.

The repricing also follows the work that firms assign within occupations more than the occupations they hire. When the environmental share is split into the part that the firm's mix of occupations predicts and the part within occupations (equation (5)), the within-occupation part carries a coefficient of 0.48 ($t = 5.9$), against 0.16 ($t = 2.0$) for the occupation mix. This is the variation that occupation-based measures cannot capture.

If climate-policy news reprices firms through preferences for green assets, returns should follow environmental ratings in the direction of the news (Pástor, Stambaugh and Taylor, 2022; Ardia et al., 2023). Figure 2 shows that they do not. To compare the rating with the share directly, I estimate the response to each day on the firms with both an LSEG score and emissions data, with the environmental share, the score, and carbon intensity in the same regression. Holding the share and emissions fixed, the LSEG score falls on six of the eight climate-policy days I examine, significantly at the 10 percent level or better on five of them (Appendix Table E14). Three of these days are favorable to climate policy (the Paris Agreement, the IRA, and the SEC rule), and three are unfavorable (the two Trump elections and *West Virginia v. EPA*). The LSEG score is therefore not repriced in the direction of the news once the orientation of hiring is held fixed. Two limits qualify this conclusion. The score is the rating of a single provider, and environmental ratings diverge widely across providers (Berg, Kölbel and Rigobon, 2022), so that the result describes this score rather than ratings in general. And ratings are an imperfect proxy for investors' preferences, so that the result does not rule out a role for them, which Section 7.3 examines through the ownership of the stock.

Ratings nevertheless correlate with exposure. Sorted within NAICS-3 industries without the controls, firms with high ratings gain in the direction of the news on average, by 0.41 percent per event with equal weights, against 0.53 percent for firms with a high environmental share, and with value weights neither sort is significant at the 5 percent level (Appendix Table E16). Once the controls are held fixed, however, firms with high LSEG scores fell on the 2016 election, and holding the share fixed as well, they also fell on the IRA day. These estimates bear on earlier findings. Ramelli et al. (2021) find that firms with strong climate

ratings from MSCI KLD gained on the 2016 election, and Bauer, Offner and Rudebusch (2026) find that firms with higher LSEG scores gained on the day of the IRA agreement, so that highly rated firms gained on two days that moved the prospects of climate policy in opposite directions. Both studies compare firms within broad industry groups, twelve Fama–French industries in the first and scores constructed relative to industry peers in the second, whereas the regressions here compare firms within three-digit NAICS industries and hold the orientation of hiring fixed. Pástor, Stambaugh and Taylor (2022), moreover, find that the returns associated with ratings arise across industries rather than within them.

Emissions do not follow the news systematically either. If the news repriced firms through the prospect of carbon pricing or of stricter regulation of emissions, emitters would gain on unfavorable days and lose on favorable ones. Carbon intensity has this sign on only four of the seven days (Figure 2). Its two large responses are on the 2016 election, when emitters fell, against the prediction, and on 9 November 2020, the day of the vaccine news. On the day of *West Virginia v. EPA*, which limited the regulation of emitters, carbon intensity does not move (0.08, $t = 0.5$). A single ruling is a weak test, since its outcome may have been anticipated. These results do not rule out carbon-related channels, but they give no sign that the policy news of this period reprices firms mainly through their emissions.

The same holds for the measures with which Bolton and Kacperczyk (2021, 2023) find that investors price carbon risk. On the five days that Trucost covers, which exclude the Paris Agreement, adding the logarithms of scope 1 and scope 2 emissions, the intensity of scope 1 emissions, their growth, and an indicator of disclosure leaves the coefficient of the environmental share at 0.68, against 0.66 on the same firm-events without them, and none of these measures is significant at the 5 percent level (Appendix Table E15). Nor is the repricing a reward for low emissions. Within the lowest tercile of scope 1 emissions in each industry and day, the clean-technology share has a coefficient of 0.58 ($t = 3.5$), against 0.77 and 0.70 in the other two terciles, and among the firms with the least clean-technology hiring, those with low emissions do not gain relative to those with high emissions ($-0.02$, $t = -0.1$).

The repricing is not exposure to interest rates either. Firms that hire toward environmental work are younger and less profitable, so their cash flows may lie further in the future, and several of the days moved long-term yields. When each firm's sensitivity to the 10-year Treasury yield, estimated over the year before each day, is allowed its own price on each day, the coefficient remains 0.58. Oil prices and the market as a whole also moved on several of the days, and within industries, firms with higher environmental shares are somewhat more sensitive to yields and to the price of oil ($-0.03$ and 0.10 per standard deviation of the share, with $t$-statistics of $-2.4$ and 6.3). When the sensitivities to oil and to the market also receive their own prices on each day, together with that to yields, the coefficient is 0.55 ($t = 6.2$),

although the market beta is priced significantly on four of the six days and the oil beta on one (Appendix Table E17).

The share might also reflect the political alignment of firms, if firms that hire toward environmental work gained on any news favorable to Democrats rather than on news about climate policy. The midterm elections of 2018 serve as a placebo. Their results changed the majority in the House but carried no identifiable news about climate policy, since a Republican Senate and President could still block any climate legislation. The share does not move in the first session after the results (0.02, $t = 0.2$), and none of the windows around that election is significant at the 5 percent level (Appendix Table E18). The placebo thus gives no indication that the repricing on the policy days reflects political alignment.

The midterm elections of 2022 are not a placebo, since their results bore on the risk to the IRA as well as on the partisan balance of Congress, and because Republican gains fell short of expectations, the predicted sign is positive. Their timing, however, is uninformative. As on the policy days, the results arrived after the close, so that the first session after them is the window that identifies the election, and in that session the coefficient is 0.09 ($t = 0.9$). The sessions around it carry other news. On election day itself, before any result was known, the coefficient is 0.26 ($t = 2.7$), which may reflect polls and prediction markets as much as expectations about inflation, and on the following day the consumer price index came in below expectations and long-term yields fell sharply. The three-day sum of 0.51 ($t = 2.4$) is therefore not attributable to the election result. It remains 0.47 when the firm's rate and oil betas are held fixed, so that these exposures do not account for it, although the betas are a coarse control on a day on which the whole yield curve shifted. As with the 2020 election and the news about vaccines (Appendix D.3), the day cannot be isolated, and I draw no inference from it.

Industry codes can also group firms that sell different products. Within the product markets that Hoberg and Phillips (2010, 2016) define from the product descriptions in 10-K filings, the firm's own share keeps a coefficient of 0.44 ($t = 3.9$) when the mean share of its rivals is held fixed, against 0.62 on the same firms without it (Appendix Table E19). The rivals' share enters positively (0.47, $t = 3.5$), whereas it would enter negatively if firms with the capability gained at the expense of rivals without it.

The design has limits. Three of the six main days carry most of the evidence, so the number of independent shocks is small, although no single day drives the result (Section 6.5) and the two further days point the same way. Elections carry news about many policies besides climate policy. The result on the 2016 election weakens in the smaller sample of firms with both a rating and emissions data.[27] The response is weaker among the largest firms, so

[27] On the 2016 election, the 835 firms with both an LSEG score and Trucost emissions are much larger than

that value-weighted estimates are about two fifths of the equal-weighted ones. The share used for the 2024 election is that of 2023, the last year of hiring data. Finally, the placebo $p$-values are conditional on the chosen set of days and on the placebo procedure, and they are not equivalent to the inference of a randomized experiment.

Within these limits, the orientation of hiring shows which firms within an industry gain or lose value when the prospects of the policy change, and more than two fifths of this information is not in ratings, emissions, climate talk, patents, or reported segments.

# 7 What Explains the Repricing

Section 6 shows that firm value moves with the environmental share on days of news about the policy and that the characteristics that existing measures record do not explain this association. The evidence cannot identify a single mechanism behind it, but it can distinguish expected cash flows from a change in the returns that investors require, and it can rule some channels out. Under the activity channel of Section 5.3, the repricing reflects the value of the growth opportunities in clean activity that this work makes it possible to take up, and therefore expected cash flows. This section asks, in turn, whether the repricing reflects expected cash flows or required returns, which cash flows it follows, who prices them, and which predictions the data do not support.

## 7.1 Cash Flows or Discount Rates

If the news changed the returns that investors require on these firms rather than the cash flows they expect, their expected returns would move on the policy days, they would earn different returns outside them, and the response would be largest where firm value rests most on distant growth. None of these appears in the data.

I measure each firm's expected return from its option prices, following Martin and Wagner (2019), for whom the expected excess return of a stock differs across firms on the same day by one half of its SVIX index, and I compare the index over the four trading days after each event with its level over the four trading days before. Expected returns measured in this way do not change with the environmental share in the direction of the news ($-0.21$ percentage points a year per standard deviation of the share, $t = -1.0$, over 30 days, and $-0.04$, $t = -0.2$, over 91 days; Appendix Table E28). Nor does the share predict returns outside the days

the average firm. Among them, the coefficient of the environmental share is $-0.40$ ($t = -1.1$) before any control and $-0.14$ once the rating and emissions are held fixed, because in these firms the environmental share correlates with emissions (0.44). On these firms, the clean-technology part of the share keeps a coefficient of $-0.64$ ($t = -2.5$) beside both measures (Appendix Table E20).

of policy news. Returns around the quarterly earnings announcements of the two years after the share is measured are unrelated to it (0.00, standard error 0.05), although these firms report earnings below the analysts' consensus, and in monthly Fama–MacBeth regressions over 2015–2024 its coefficient is −0.02 (standard error 0.08), while a portfolio long the top and short the bottom quintile of the share earns no significant alpha (Appendix Tables F1 and F2). Together with the absence of drift after the policy days (Section 6.3), these results indicate that prices reflect the information in the share when the news arrives and that the share carried no detectable premium over this decade (Appendix F). They have limited power against one alternative, however. Because the capability is valued for margins that accrue over a decade, a change in the discount rate of a few hundredths of a percentage point a year would suffice to explain the repricing, well within the precision of the option-implied estimates, so that the data rule out a detectable change in short-horizon required returns rather than any role for discount rates.

A change in the returns required on long-duration equity would also reprice firms in proportion to how much of their value rests on growth, whatever the orientation of their hiring. The repricing does not scale in this way. When the share is interacted with the firm's rank, within its industry and day, on Tobin's $q$, on the total $q$ of Peters and Taylor (2017), on the market-to-book ratio, or on a low forward earnings yield, none of the interactions is significant at the 5 percent level, and the only one significant at 10 percent, with the market-to-book ratio, is negative. Across terciles of an index that combines Tobin's $q$, the market-to-book ratio, the forward earnings yield, and the ratio of property, plant, and equipment to assets, the coefficient of the share is 0.68, 0.64, and 0.47, and its interaction with the index is close to zero (−0.04; Appendix Table E25). Nor is the response significantly weaker where organization capital, which the key employees in whom it is embodied may capture (Eisfeldt and Papanikolaou, 2013), is larger. The policy days were therefore not simply days on which growth stocks outperformed value stocks.

The revisions of analysts' forecasts point instead to cash flows. Because the credits pay over a decade, news about them should move long-term forecasts more than forecasts of the next few years' earnings, whereas a change in tastes or discount rates predicts no revision. Pooled over the six days and signed by the direction of the news, the consensus forecast of long-term growth moves 0.18 percentage points more for a firm whose environmental share is one standard deviation higher ($t = 2.1$), a revision that comes from the clean-technology share (0.17, $t = 2.0$) rather than from environmental management (0.06), while forecasts of earnings per share for the next three fiscal years move by less than 0.05 percent of the price per standard deviation, none significantly at 5 percent (Table 13). The revisions rest on two days. Around the IRA, analysts raise the long-term growth forecasts of firms with a higher

clean-technology share by 0.54 percentage points ($t = 2.3$), and around the 2024 election they lower those of firms with a higher environmental share by 0.53 points ($t = -2.3$), whereas on the other four days the revisions are small and insignificant (Appendix Table E26). Long-term forecasts exist for about 800 firms per day, and the monthly windows of the consensus contain other news, so that this evidence is less precise than the price responses.

**Table 13.** Revisions of Analysts' Forecasts on Policy Days

| | Environmental share (1) | Clean technology (2) | Environmental management (3) | Firm-events (4) |
|---|---|---|---|---|
| EPS, fiscal year 1 (% of price) | 0.042 | 0.046* | −0.002 | 11,407 |
| | (0.034) | (0.028) | (0.027) | |
| EPS, fiscal year 2 (% of price) | −0.012 | −0.008 | −0.004 | 11,387 |
| | (0.027) | (0.024) | (0.019) | |
| EPS, fiscal year 3 (% of price) | 0.046 | 0.042 | 0.011 | 9,969 |
| | (0.029) | (0.026) | (0.024) | |
| Long-term growth (pp) | 0.184** | 0.168** | 0.057 | 5,943 |
| | (0.088) | (0.084) | (0.057) | |
| Net share of analysts revising up, year 2 | 0.001 | 0.002 | −0.001 | 11,387 |
| | (0.006) | (0.005) | (0.005) | |

*Notes.* Change in the IBES consensus from the last monthly forecast before each day to the first after it, pooled over the six days and signed, on the standardized environmental share (column 1) or its two parts (columns 2 and 3), with the controls of Table 11 except beta and idiosyncratic volatility, and NAICS-3-by-event effects. Standard errors clustered by firm in parentheses. *, **, ***: 10, 5, 1 percent.

The repricing thus reflects expected cash flows at least in part. The data reject a detectable change in short-horizon required returns, a premium, and a rotation between growth and value stocks, although they cannot exclude a small change in the rates at which investors discount distant cash flows.

## 7.2 Which Cash Flows

The cash flows that the repricing follows are those of the work that the policy pays for. The credits pay mostly for clean-technology work, and the clean-technology part of the share carries the repricing (Table 14). On the six days, a one-standard-deviation higher clean-technology share is associated with a return 0.56 percentage points higher in the direction of the news ($t = 8.1$), with a placebo $p$-value below 0.001, against 0.09 ($t = 1.4$) for the environmental management share. The difference between the two, 0.47 ($t = 5.0$), is positive in every subset of days, and with all the existing measures and six-digit industry-by-event effects, the clean-

technology share keeps a coefficient of 0.32 ($t = 3.2$), whereas the environmental management share falls to −0.12 (Appendix Table E23). Nor does environmental management respond to news about the regulation of emissions, as a rival view would predict. Its share does not move significantly when the Supreme Court limited the authority of the Environmental Protection Agency to regulate emissions from power plants on 30 June 2022 (*West Virginia v. EPA*; Supreme Court of the United States, 2022) or when the Securities and Exchange Commission adopted rules on climate-related disclosure on 6 March 2024 (U.S. Securities and Exchange Commission, 2024) (−0.13, $t = -1.4$, and 0.01).

**Table 14.** Clean Technology and Environmental Management on Policy Days

| | Six days (1) | Seven days (2) | Favor-able (3) | Unfavor-able (4) | Elec-tions (5) | Other days (6) |
|---|---|---|---|---|---|---|
| Clean-technology share | 0.564*** | 0.511*** | 0.409*** | 0.700*** | 0.688*** | 0.435*** |
| | (0.070) | (0.063) | (0.089) | (0.122) | (0.119) | (0.083) |
| Placebo $p$ | <0.001 | <0.001 | 0.022 | <0.001 | <0.001 | 0.012 |
| Environmental management share | 0.090 | 0.082 | 0.046 | 0.123 | 0.028 | 0.158** |
| | (0.066) | (0.059) | (0.088) | (0.106) | (0.115) | (0.067) |
| Placebo $p$ | 0.117 | 0.101 | 0.376 | 0.164 | 0.499 | 0.082 |
| Difference | 0.474*** | 0.429*** | 0.363*** | 0.577*** | 0.661*** | 0.277*** |
| | (0.094) | (0.085) | (0.122) | (0.159) | (0.162) | (0.102) |
| Firm-events | 12,407 | 15,122 | 6,386 | 6,021 | 6,657 | 5,750 |

*Notes.* Signed CAR[−1,+1] (pp) on the two parts of the environmental share, entered together and standardized, with the controls and effects of Table 11. Difference: clean technology minus environmental management. Placebo $p$ as in Table 11. Standard errors clustered by firm in parentheses. *, **, ***: 10, 5, 1 percent.

The share also identifies the firms that the policy later paid. I measure the activity that the credits pay for from two sources that do not come from hiring, the mentions of nine of the credits in the firm's 10-K filings from January 2023 to October 2026, found with the full-text search of SEC EDGAR, and the clean facilities that the Clean Investment Monitor records as announced from August 2022 to June 2026, matched to their owners by name. The share is measured on hiring in 2019–2021, before any of these outcomes, and each regression holds fixed the firm's earlier activity. Among 2,370 US firms, a firm whose environmental share was one standard deviation higher is 6.8 percentage points more likely to mention one of the credits in a later 10-K ($t = 5.8$), against a mean of 2.9 percent, 1.9 points more likely to announce a clean facility ($t = 2.1$), against 3.3 percent, and 3.1 points more likely to announce a clean manufacturing plant ($t = 3.5$), against 1.1 percent (Table 15). Among firms with no earlier mention, the share also predicts a later one (4.7 points, $t = 3.2$). The

clean-technology part carries these associations, whereas the environmental management part predicts neither the credits nor manufacturing plants, and the social, governance, and innovation shares are not positively associated with either outcome (Appendix Table E22).

These results describe the policy rather than the market. They show that the share picks out the firms that later claim the credits and announce facilities, but they establish no causal effect, since every outcome follows the act and no comparison group was left untouched by it, and they test no mechanism. A mention in a 10-K is not a receipt, since a filing may discuss eligibility, expectations, or credits bought from others, and an announced facility may never be built. Internet Appendix IA.III.7 describes the search terms, the match of facilities to firms, and an audit of both, and Appendix Table E21 shows that the results hold for all North American firms.

**Table 15.** Environmental Hiring and the Activity the Credits Pay For

| | Tax credits in 10-K filings, 2023–2026 | | | Clean facilities announced, Aug. 2022–June 2026 | | |
|---|---|---|---|---|---|---|
| | Any clean credit (1) | Manufacturing credit (45X) (2) | Number of credits (3) | Any facility (4) | Any manufacturing facility (5) | Announced capex (6) |
| *A. The environmental share* | | | | | | |
| Environmental share | 6.81*** | 3.27*** | 0.157*** | 1.93** | 3.13*** | 0.202** |
| | (1.17) | (0.96) | (0.035) | (0.90) | (0.90) | (0.080) |
| *B. Its two parts, entered together* | | | | | | |
| Clean technology | 6.92*** | 3.73*** | 0.161*** | 2.84*** | 3.22*** | 0.192*** |
| | (1.05) | (0.91) | (0.033) | (0.94) | (0.93) | (0.072) |
| Environmental management | 0.70 | −0.41 | 0.013 | −1.07*** | 0.48 | −0.210 |
| | (0.77) | (0.37) | (0.011) | (0.38) | (0.31) | (0.248) |
| Activity before August 2022 | Yes | Yes | Yes | Yes | Yes | Yes |
| Firm controls | Yes | Yes | Yes | Yes | Yes | Yes |
| NAICS-3 effects | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of dependent variable | 2.95 | 1.09 | 0.044 | 3.28 | 1.05 | 33.7 |
| Firms | 2,370 | 2,370 | 2,370 | 2,370 | 2,370 | 974 |

*Notes.* Cross-section of US-located firms with an environmental share in 2019–2021 and Compustat data for fiscal 2023 or later. The shares are the means over the firm's years of the estimation sample in 2019–2021, standardized across North American firms. Columns (1) to (3): 10-K filings, including exhibits, filed from January 2023 to October 2026 and found by the full-text search of SEC EDGAR. The dependent variable equals 100 if a filing mentions any of nine credits that the Inflation Reduction Act created or extended (sections 45X, 45Y, 48E, 45V, 45Q, 45Z, 48C, 45W, and 45U; Internet Appendix Table IA.36), equals 100 if a filing mentions the advanced manufacturing production credit (section 45X), or counts the credits mentioned. Activity before August 2022: an indicator for a 10-K filed from January 2019 to July 2022 that mentions production or investment tax credits or section 45Q. Columns (4) to (6): facilities of the Clean Investment Monitor announced from August 2022 to June 2026 and matched to their owners by name. The dependent variable equals 100 if the firm announces any facility or a manufacturing facility, or is the announced capital expenditure in millions of 2024 dollars, estimated by Poisson pseudo-maximum likelihood, which drops the industries without a facility. Activity before August 2022: an indicator for a facility announced earlier and the logarithm of one plus their number. Firm controls: the logarithm of assets and the ratio of property, plant, and equipment to assets in 2021, and the innovation share, return on assets, logarithm of age, Tobin's $q$, and leverage in 2019–2021, and the logarithm of the firm's mean annual openings and the mean logarithm of its employees in 2019–2021. Heteroskedasticity-robust standard errors in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

The repricing on the agreement on the IRA is nonetheless not concentrated among the firms that later acted (Table 16). Among the 95 percent of these firms that neither mention a credit nor announce a facility later, the clean-technology share has a coefficient of 0.89 ($t = 4.4$) on that day, and its interaction with later activity is small and insignificant ($-0.26$, $t = -1.0$), with similar results on the other days of 2022 and 2024. The repricing is concentrated instead among the firms already active before August 2022, among which it is larger by 0.78 ($t = 2.8$), and once this earlier activity is held fixed, the interaction with later activity turns negative ($-0.60$, $t = -2.2$), although it rests on few firms. Prices thus capitalized capability in place. Where a firm already had clean operations, the people whom its hiring revealed were positioned to claim the credits at once, since the credits pay for activity already under way as well as for its expansion. Whether a firm without such operations would later act depended instead on news that arrived after the day, and the market did not identify these firms beyond what their hiring revealed. That the repricing also reaches firms whose clean activity had not appeared in filings or facility announcements by 2026 is consistent with hiring revealing a capability before outcomes record it.

**Table 16.** The Repricing on the IRA, by Earlier and Later Activity

| | IRA agreement (1) | IRA days (2) | Manchin withdraws (3) | 2024 election (4) | IRA agreement (5) |
|---|---|---|---|---|---|
| Clean-technology share | 0.885*** | 0.502*** | 0.120 | 1.067*** | 0.597*** |
| | (0.200) | (0.153) | (0.169) | (0.398) | (0.214) |
| × later activity | −0.257 | −0.087 | 0.080 | 0.391 | −0.602** |
| | (0.267) | (0.215) | (0.239) | (0.487) | (0.269) |
| Environmental management share × later activity | 0.026 | 0.008 | −0.011 | −0.132 | −0.013 |
| | (0.150) | (0.085) | (0.137) | (0.488) | (0.149) |
| Clean-technology share × activity before Aug 2022 | | | | | 0.775*** |
| | | | | | (0.277) |
| Firm-events | 2,184 | 4,366 | 2,182 | 1,988 | 2,184 |

*Notes.* US firms of Table 15. Later activity: a 10-K filed from 2023 that mentions one of the IRA credits, or a clean facility announced from August 2022 to June 2026; activity before August 2022 as in Table 15. One day: NAICS-3 effects, as observed; pooled days: signed, NAICS-3-by-event effects. Standard errors in parentheses. *, **, ***: 10, 5, 1 percent.

## 7.3 Who Prices the Repricing

Investors' tastes for green assets, as the portfolios of funds that invest by environmental, social, and governance criteria reveal them, do not account for the repricing, although these

funds amplify it in recent years. I identify the funds by the words in their names in the CRSP Mutual Fund database and measure, at their last report before each day, the share of each firm's stock that they hold and the weight of the stock in their combined portfolio relative to its weight in the market, the overweight through which tastes for green assets appear in the model of Pástor, Stambaugh and Taylor (2021) (Internet Appendix IA.V.7). The stocks that these funds overweight do not gain in the direction of the news, and beside the overweight the coefficient of the environmental share is 0.61, against 0.59 without it (Appendix Table E30). On the days of 2015–2017, when the funds' holdings of listed US stocks were worth between \$6 and \$10 billion, the coefficient of the share is 0.40 ($t = 3.6$), and it does not vary with their ownership. From 2020, when these holdings had grown to between \$61 and \$140 billion, the repricing is larger where the funds hold more of the stock, by 0.09 percentage points per standard deviation of their ownership ($t = 2.9$), against a coefficient of the share of 0.41 at average ownership (Appendix Table E29), and larger where they overweight it (0.67, $t = 2.1$). The funds thus amplify a repricing that their tastes do not produce, and a standard deviation of their ownership changes it by less than a quarter.

The response also depends on whether investors attend to the firm. Section 8 shows that the repricing is more than twice as large among firms that analysts follow, even when size and tangible intensity are held fixed, and largest among small and asset-light firms, in which the capability is a larger part of firm value. Together with the role of these funds since 2020, this pattern is consistent with a repricing that is largest where investors who attend to the capability hold or follow the stock and where the capability weighs most in its value.

## 7.4 What the Evidence Does Not Show

Two further predictions of the cash-flow channel find no independent support. The first uses the IRA's own design. The act let firms sell most of its credits for cash, which raised their value most for firms without the taxable income to use them (U.S. Congress, 2022). Clean-technology firms with losses or tax loss carryforwards respond more on the days of 2022, but they respond as much on the earlier days, on which the transfer of credits was not at stake, and the difference between the two periods is not significant (Appendix Table E27). The test thus finds no response specific to the transfer of the credits.

The second concerns investment. If the repricing reflects the value of growth opportunities in clean activity, firms with a higher clean-technology share should invest more after favorable news and less after unfavorable news. In an annual panel with firm and industry-by-year effects, their capital expenditure in the following year moves by 0.09 percent of assets per standard deviation of the share in the direction of the news ($t = 3.0$), against a mean of

3.3 percent, but this response rests on the year of the IRA (0.24, $t = 3.0$). After the 2024 election it is 0.05 ($t = 1.0$), and without the year of the IRA it is 0.05 ($t = 1.9$; Appendix Table E24). With so few years of news, each shaped by other events, the data do not show that investment responds to news about the policy outside the year of the IRA.

Taken together, the evidence points to expected cash flows from the work that the policy pays for. The repricing follows clean-technology work rather than environmental management, and revisions of long-term growth forecasts move with it, whereas expected returns, returns outside the policy days, and the weight of growth in firm value do not, although the data cannot exclude a small change in long-horizon discount rates. The share identifies the firms that the policy later paid, the repricing is largest where the capability was already in place, and environmentally minded funds amplify it in recent years without accounting for most of it. The design of the credits and the investment of these firms add no independent confirmation beyond the year of the IRA. The hiring data end in September 2023, so the paper does not study how firms' hiring responds to the policy.

# 8 Where the Repricing Is Largest

Every estimate compares a firm with the other firms of its industry on the same day. The news therefore moves value among firms of the same industry, according to the orientation of their hiring, and within product markets defined from 10-K text the repricing is shared with rivals rather than redistributed among them (Section 6.6). This section carries out the fifth test, asking where, within industries, the repricing is largest.

In equation (8), clean activity enters firm value as a weight. For a given orientation of hiring, this weight should be larger where the capability is a large part of the firm, as in small firms and in firms with few tangible assets. The estimates in Figure 7 are consistent with this prediction. The coefficient of the environmental share is 1.51 ($t = 3.7$) among young firms in the lowest tercile of property, plant, and equipment over assets, and between 0.38 and 0.61 in the other groups. It is 0.70 ($t = 5.2$) among small firms and 0.41 ($t = 3.7$) among large firms. In one regression with all interactions, the share's coefficient falls by 0.20 per standard deviation of tangible intensity ($t = -2.6$) and rises by 0.52 among small firms ($t = 3.3$), and once tangible intensity, size, and coverage are held fixed, firm age adds nothing (Appendix Table E31). The repricing is thus largest in asset-light and small firms.[28] These are also the

[28]The gradient in tangible intensity uses the current ratio of property, plant, and equipment to assets. For the clean-technology part, the interaction with tangible intensity is smaller and insignificant when tangible intensity is measured before the sample period, in 2011–2013, partly because many young firms have no pre-sample data. Firm size, which needs no pre-sample data, gives the same conclusion (Appendix Tables E32 and E31).

firms whose balance sheets record the least of the capabilities they develop, which is where a measure from hiring can add the most. The concentration in asset-light firms does not mean that the capability substitutes for tangible capital, since across firms clean-technology hiring goes with higher capital expenditures (Appendix B.3).

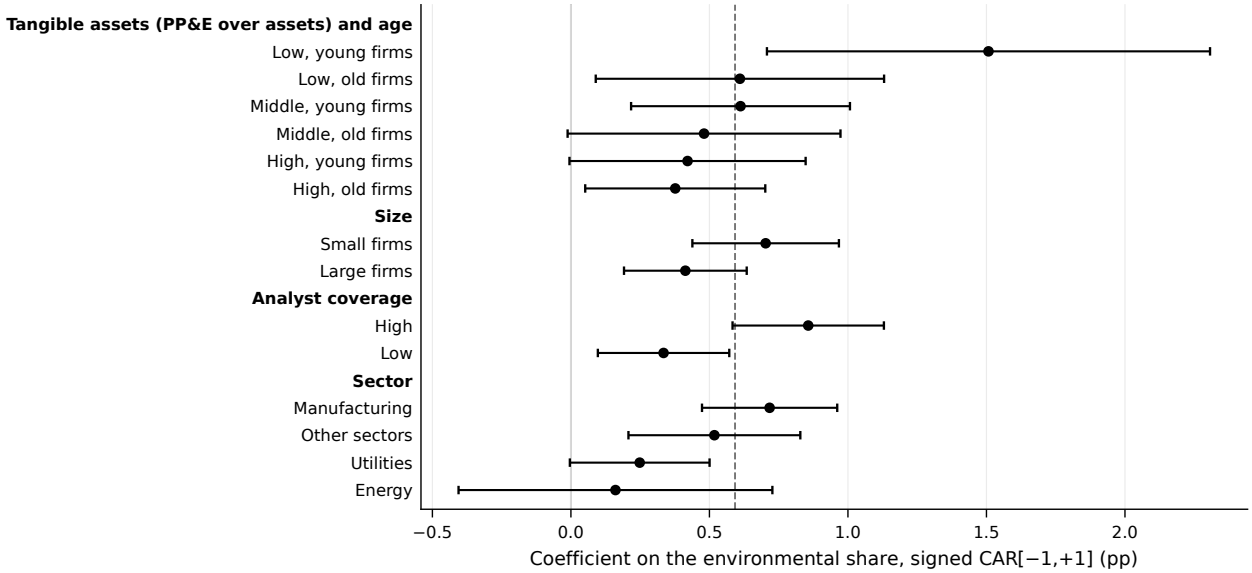



**Figure 7.** Where the Repricing Is Largest. Coefficient of the environmental share in the signed regression of equation (9) on the six days, estimated separately in each group, with 95 percent confidence intervals; the dashed line marks the full-sample estimate. Terciles of PP&E over assets and median splits of age, size, and analyst coverage are formed within each event; sectors are NAICS 31–33 except 324 (manufacturing), 22 (utilities), and 21 and 324 (energy).

The repricing is also larger among firms that analysts follow. Among firms with above-median analyst coverage, the coefficient of the environmental share is 0.86 ($t = 6.1$), against 0.33 ($t = 2.8$) among the other firms. In the joint regression, the coefficient is lower by 0.80 among firms with low coverage ($t = -4.3$). One possible interpretation is information production. No account records the capability, and analysts have stronger incentives to follow firms whose intangible assets the accounts do not recognize (Barth, Kasznik and McNichols, 2001). Other explanations, such as slower incorporation of news into the prices of thinly followed stocks within a three-day window, could produce the same pattern.

Finally, the repricing is not confined to the sectors in which clean activity is most visible. It is 0.72 ($t = 5.8$) within manufacturing and 0.52 ($t = 3.3$) in the sectors outside manufacturing, utilities, and energy, which together hold nine in ten of the firm-events, whereas within utilities and within energy, where the firm-events are few and the shares are high throughout, the estimates are small and imprecise. Without utilities or without energy, the pooled estimate is 0.65 and 0.61. The orientation of hiring thus distinguishes firms within the broad range of industries in which clean activity is a choice rather than the business itself.

# 9 Conclusion

Firms differ in the activities they are capable of undertaking, yet conventional data reveal little about the activities they are preparing to undertake before those activities generate outcomes. This paper asks whether the specific activities toward which firms direct their hiring can be identified systematically, and whether this information reveals economically relevant differences between firms. The approach reads the responsibilities described in job postings with a calibrated classifier, trained on the text embeddings of a small sample labelled by a language model, and measures, for each firm and year, the share of new openings devoted to a given kind of work. The share captures the orientation of hiring, not the size of a firm's capacity or the value of its capabilities.

The validation indicates that the share records the work it is meant to record. The labels agree with blind human coding, and the calibrated probabilities match the rate of labelled work in postings that were not used for training, with errors unrelated to the year of the posting or to the firm that posted it. Where an expense line records spending on the same broad activity, the innovation share recovers R&D, and where none exists, each share predicts the records of its own work more strongly than the others do, from green patents to the injuries recorded by a firm's establishments and securities litigation. The environmental share is related to existing measures of environmental activity but remains distinct from them, and a firm's orientation is persistent, as the gradual building of a capability implies, so that the measure is most informative in comparisons between firms.

The application asks whether such capability-oriented hiring helps explain firms' exposure to climate industrial policy beyond conventional measures. On days of news about US climate industrial policy, firm value moves with the environmental orientation of hiring relative to industry peers and in the direction of the news, by about 0.6 percentage points over three days for a firm whose share is one standard deviation higher than those of its peers. The repricing follows clean-technology work, the part of environmental work that the policy pays for, rather than environmental management. It persists when ratings, emissions, climate talk, patents, and reported clean segments are held fixed, with more than two fifths of it surviving all of them together within six-digit industries, and it remains among firms without green patents. It is largest in small and asset-light firms and among firms that analysts follow, yet it does not depend on how large a part of firm value growth opportunities represent in general. Firms with a higher clean-technology share also invested more in the year after the agreement on the IRA, although their investment shows no reliable response to the news of other years.

These results support a reading in which the market values the growth opportunities in

the activity that the orientation of hiring makes possible, a channel distinct from investors' tastes for green assets and from the risk of carbon pricing. The repricing is strongest where the subsidized activity was already under way, yet it reaches firms whose clean activity had not appeared in filings or announcements of facilities, so that hiring can reveal exposure before output or patents record it. The evidence is less decisive about why this exposure is valued. Revisions of long-term growth forecasts point to the cash flows that the activity is expected to generate, but they are imprecise, and the data cannot exclude that part of the repricing reflects the rates at which investors discount distant cash flows or, in recent years, the demand of funds that invest by environmental criteria. Prices, for their part, appear to incorporate the information in the share when the news arrives, since the repricing is neither anticipated nor reversed and the share predicts no returns outside the policy days. This leaves little room for mispricing over the period, although a decade of returns cannot exclude a modest premium for the exposure, nor adjustments over horizons that later news about the policy would confound.

These findings bear on how exposure to industrial policy can be observed. Policies that pay for specific activities create exposure that differs among firms of the same industry and that resides in firms' plans before it appears in their assets or revenue. Because postings reveal these plans as they are formed, they make it possible to identify the firms whose value depends on the prospects of such a policy at a stage at which conventional measures still show little difference between them, which matters for investors who bear the risk of its reversal and for policymakers who seek to understand whom it reaches. The same reasoning suggests why the repricing is largest where financial statements record least, since the capabilities of small and asset-light firms are the least visible in their accounts.

The broader implication is that job postings can measure, firm by firm, activities that financial statements do not record. Financial statements record what firms spend and what they have built, and only a few broad activities have lines of their own, whereas postings describe the work that firms are adding, activity by activity, at the moment they choose to add it. Because the measure is built from public postings, it makes observable differences in firms' positioning that investors appear to price already but that researchers and others outside the firm cannot easily see, and it does so within industries, before these differences appear in output, and where financial statements record no expense line for the activity.

The results have limits. A posting records a hire that a firm intends to make, not one it has made, and the measure exists only for firms that recruit through their own career sites, which are larger and older than the average listed firm. Because the orientation of a firm's hiring changes slowly, the measure distinguishes firms more reliably than it tracks a single firm over time. The labels come from one language model, checked against a second

model and against the blind codes of a single coder, and comparisons of social shares within industries are attenuated. In the application, three of the six main days carry most of the evidence, although two further days point the same way, elections carry news about many policies, the 2020 election cannot be separated from news about vaccines, and the placebo tests are conditional on the chosen days. The response is weaker among the largest firms, so that aggregate amounts in dollars depend on how firms are weighted. The comparisons with ratings rest on the score of a single provider, and environmental ratings diverge across providers. All results are associations between firm value and an orientation that firms choose, identified from dated news.

More broadly, the procedure applies to any activity that can be defined by the work it requires. Quality assurance, worker safety, cybersecurity, and artificial intelligence are examples of activities that firms staff with their own employees and that financial statements do not record separately, and measuring a new one requires only a written definition and a labelled sample. The same approach can relate the activities toward which firms direct their hiring to the policies, technologies, or events that change their returns, and, with hiring data that extend beyond the enactment of a policy, it can show how firms redirect their hiring once those returns have changed.

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# Appendix A. Variable Definitions

This appendix defines every variable used in the paper (Table A1). Internet Appendix IA.III describes each source and how it is linked to issuers.

**Table A1.** Variable Definitions

| Variable | Definition | Source |
|---|---|---|
| *A. Hiring* | | |
| Environmental share | Expected share of a firm-year's new openings whose responsibilities involve environmental work, the mean of the calibrated probabilities of its postings (equation (3)), in percent. Standardized within year over all firm-years with openings (panel regressions), within event (event studies), or across firms when it enters as a firm's mean over 2019–2021 (Table 15). | LinkUp; this paper |
| Clean-technology share | The part of the environmental share credited to postings whose most likely part of environmental work is clean technology (Section 3.1); standardized like the share. | LinkUp; this paper |
| Environmental management share | The remainder of the environmental share: environmental management systems, permitting, and services; waste; water; pollution and remediation; air emissions; greenhouse-gas accounting; sustainability reporting; climate risk; and land. | LinkUp; this paper |
| Social, governance, and innovation shares | Measured in the same way from their own definitions; in the regressions of the environmental share on policy days, they enter only to check that it does not reflect the overall composition of hiring. | LinkUp; this paper |
| Sub-function shares | The part of a share credited to each part of its definition, each posting credited to its most likely part: worker safety, product and patient safety, and people and rights (social); quality assurance, risk and financial controls, information security and privacy, and compliance and integrity (governance); software and research (innovation). | LinkUp; this paper |
| Occupation mix, work within occupations | The two terms of equation (5), with six-digit SOC occupations and $\bar{s}_{ot}$ the share among all openings of consolidated firm-years; standardized like the share. | LinkUp; this paper |
| Rivals' share | Similarity-weighted mean environmental (clean-technology) share of the firm's TNIC-3 rivals in the year before the event, in standard deviations of the firm's own share in the event sample. | LinkUp; Hoberg and Phillips (2016) |
| *B. Returns, forecasts, and options* | | |
| CAR$[-1,+1]$ | Sum of daily abnormal returns over trading days $-1$ to $+1$, in percentage points; the abnormal return is the excess return minus the loadings on the five factors of Fama and French (2015) and momentum, estimated over days $-250$ to $-30$ with at least 120 returns, without the intercept. Winsorized at the 1st and 99th percentiles within event. Figure 4 reports the daily components. | CRSP |
| Other windows and benchmarks | The same sum over days $-10$ to $-2$, $-5$ to $-2$, $+2$ to $+5$, $+2$ to $+20$, and $+2$ to $+60$, and from day $-10$ to each day up to $+60$ (cumulative path), every day required; abnormal returns from the market model with an intercept, estimated over the same days, and market-adjusted returns, the return minus the Fama–French market return. Daily returns after 2024 from the CRSP daily file in its new format. | CRSP |

*Continued on the next page*

*Table A1, continued*

| Variable | Definition | Source |
|---|---|---|
| Price and size screens | CRSP price of at least \$5, and market equity at or above the 20th percentile of ordinary common shares listed on the NYSE, both at the end of the month before the event. | CRSP |
| Market equity, book-to-market | Log CRSP market equity at the end of the month before the event; log of Compustat common equity of the latest fiscal year ended at least four months before the event over that market equity. | CRSP; Compustat |
| Momentum | Cumulative return over months $-12$ to $-2$ before the event. | CRSP |
| Beta, idiosyncratic volatility | Market beta and standard deviation of the daily residuals, in decimals, of the factor regression over trading days $-250$ to $-30$. | CRSP |
| Rate beta | Coefficient on the daily change in the 10-year Treasury yield, in basis points, added to that regression; winsorized at 1/99 and standardized within event. | CRSP; Federal Reserve |
| Oil beta | Coefficient on the daily return of Brent crude oil, in percent (winsorized at 1/99), added to that regression with the yield change; winsorized at 1/99 and standardized within event. | CRSP; FRED |
| Overnight and intraday returns | On the first session after news that arrived after the close, the return from the previous close to the open, with distributions on the day, and from the open to the close, in percent, winsorized at 1/99 within day. | CRSP |
| Earnings announcement in the window | Indicator of a quarterly earnings announcement (Compustat report date) within trading days $-1$ to $+1$ of the event. | Compustat |
| Population weights | Inverse of the probability that a listed firm is in the event sample, from a logit, within each event, of inclusion on log market equity and its square, log book-to-market, log age, an NYSE indicator, and NAICS-2 indicators, among CRSP common stocks and ADRs on the NYSE, AMEX, and Nasdaq with a Compustat record; trimmed at the 1st percentile of the probability and normalized to mean one within event. | CRSP; Compustat |
| Implied change in value | The day's coefficient of the environmental share, estimated with equal weights or with weights proportional to market equity, times the firm's share minus its NAICS-3 mean, times CRSP market equity at the end of the month before the event. | CRSP; this paper |
| Forecast revisions | Change in the IBES mean forecast between the last monthly consensus before the event and the first after it, for the same fiscal period: earnings per share for fiscal years 1 to 3, in percent of the CRSP price at the end of the month before the first consensus, put on the share basis of the IBES forecasts (traded price above \$1); long-term growth, in percentage points; analysts revising up minus down over estimates, fiscal year 2. Winsorized at 1/99 within event. | IBES; CRSP |
| SUE, consensus revision | Mean over the quarters of fiscal year $t+1$ ($t+2$) of IBES standardized unexpected earnings, winsorized at 1/99 within year; last minus first monthly fiscal-year-1 consensus before the announcement (at least three), over the absolute price at the first, for fiscal year $t+1$. | IBES; CRSP |
| Announcement return | Market-adjusted return over days $-1$ to $+1$ around each quarterly earnings date in calendar year $t+1$ ($t+2$), in percent, winsorized at 1/99 within year; annual sum scaled to four announcements. | CRSP; Compustat |
| Monthly excess return | Return minus the risk-free rate from July of $Y+1$ to June of $Y+2$, on characteristics of year $Y$. | CRSP |

*Continued on the next page*

*Table A1, continued*

| Variable | Definition | Source |
|---|---|---|
| Option-implied expected return | One half of the stock's risk-neutral variance $\text{SVIX}^2_{i,t}$ (Martin and Wagner, 2019), annualized, from out-of-the-money puts and calls with deltas of 0.10 to 0.50 on the standardized volatility surface at 30 (91) days, integrated over strikes; the outcome is its mean over trading days +2 to +5 minus its mean over days −5 to −2, with at least three days in each. | OptionMetrics; CRSP |
| *C. Existing measures and controls* | | |
| Openings, employees, assets | Logarithms of new openings, employees, and total assets of year $t$ (panel regressions) or of the year before the event; in Table 15, log assets of 2021, log mean openings, and mean log employees of 2019–2021. | LinkUp; Compustat |
| Environmental rating | LSEG environmental pillar score (0–100), with its history restated under LSEG's current methodology, standardized. | LSEG |
| Carbon intensity | Log of Trucost Scope 1 (including estimates) plus location-based Scope 2 emissions over Compustat fiscal-year-end market value, year before the event (Ramelli et al., 2021). | Trucost; Compustat |
| Emissions intensity | As carbon intensity, over sales (Table B4). | Trucost; Compustat |
| Climate talk | Share of earnings-call content about climate change, overall and by topic, among them the opportunity and the regulatory exposure (Appendix Table C6), standardized within year (Sautner et al., 2023*a*). | Sautner et al. (2023*a*) |
| Occupation-based measures | Opening-weighted mean green task intensity of the firm's occupations, the share of an occupation's task statements that O*NET classifies as green (Vona et al., 2018); share of openings in occupations that O*NET classifies as green occupations with new or enhanced skills; share of openings in environmental occupations, the tenth of six-digit occupations with at least 1,000 openings that has the highest mean environmental probability. The O*NET measures use the openings whose occupation has an O*NET value, carried to 2018 SOC codes through O*NET's crosswalk; all are standardized within year. | O*NET; LinkUp |
| Intangible capital | SG&A and R&D over lagged assets (zero where missing); knowledge and organization capital (Peters and Taylor, 2017) over assets, winsorized at 1/99 within year. | Compustat; WRDS (Peters–Taylor) |
| Green patents | Indicator of a Y02 or IRA-class application in 2007–2013; log of one plus mean annual Y02 and IRA-class applications in 2007–2013; Y02 share of 2007–2013 applications; log of one plus Y02, IRA-class, and all applications in $t-2$ to $t$. | Google Patents |
| Clean revenue share | Revenue of clean-technology segments over that of all segments, and an indicator of any clean segment; set to zero with an indicator where segment data are missing. | Compustat Segments |
| Tax and trade | Income taxes paid over pretax income, if positive, bounded to $[0, 1]$; foreign share of pretax income, bounded to $[-1, 2]$; foreign share of geographic-segment sales. | Compustat |
| Sales, employees, market value | Compustat sales and employees; market value is the fiscal year-end price times shares outstanding. Amounts in Canadian dollars are converted at the exchange rate of the fiscal year-end month. | Compustat |
| Openings per 100 employees | Openings in the firm-year over Compustat employees, times 100, winsorized at 1/99 within year. | LinkUp; Compustat |
| Tobin's $q$, leverage, cash | Assets plus market value minus common equity and deferred taxes, over assets; long-term plus current debt over assets; cash and short-term investments over assets. | Compustat |

*Continued on the next page*

*Table A1, continued*

| Variable | Definition | Source |
|---|---|---|
| Return on assets, sales growth | Operating income before depreciation over lagged assets; log change in sales. | Compustat |
| Capital expenditure, R&D, SG&A | Over lagged assets; R&D and SG&A zero where missing, with an indicator of missing R&D where stated; SG&A includes R&D. | Compustat |
| PP&E over assets | Net property, plant, and equipment over assets of the year, or its 2011–2013 mean; terciles within event in Table E31. | Compustat |
| Age, dividend payer | Years since the firm's first fiscal year in the Compustat files used (North America from 1975, Global from 2005), as the log of one plus; young: below the event's median; indicator of positive common dividends. | Compustat |
| Analyst coverage | Number of analysts in the December IBES consensus of fiscal-year-1 earnings of year $t$, zero if not covered; low coverage: below the year's or event's median. | IBES |
| Small firm | Market equity at the end of the month before the event below the event's median. | CRSP |
| Patent-position samples | No green patent: no Y02 or IRA-class application in 2007–2013; strict newcomers: in addition, outside utilities (NAICS 221) and outside the tenth of six-digit industries with the highest 2007–2013 Y02 share of applications (industries with at least 20 applications). | Google Patents |
| Activity before August 2022 | Indicator of a 10-K filed from 2019 to July 2022 that mentions production or investment tax credits or section 45Q; indicator and log of one plus the number of facilities announced before August 2022. | SEC EDGAR; Clean Investment Monitor |
| ESG-fund ownership | Shares held by mutual funds whose names, as in force at the report date, identify investment by environmental, social, and governance criteria, at each fund's last report in the 183 days before the event, over shares outstanding at the end of the month before; standardized within event, or above the event's median. | CRSP Mutual Funds; CRSP |
| Institutional ownership | Shares held by institutions filing Form 13F at the last quarter-end before the event, each manager's largest position in the quarter summed over managers, over shares outstanding; standardized within event, or above the event's median. | Thomson Reuters 13F; CRSP |
| Low tax capacity | Indicator of a tax loss carryforward (positive, missing treated as none) or negative pretax income in the fiscal year before the event. | Compustat |
| *D. Validation outcomes* | | |
| Y02, IRA-class, ENV-TECH, other applications | Applications filed in $t+1$ to $t+2$ in the Y02 scheme, in the IRA classes, in ENV-TECH, in ENV-TECH but not Y02, in Y02 or ENV-TECH, or outside Y02 (Section 2.4). | Google Patents |
| Green share of applications | Y02 applications over all applications filed in $t+1$ to $t+2$, for firm-years with at least one, as a fraction or in percent as each table states. | Google Patents |
| Scope 1 intensity | Log of one plus Scope 1 tonnes per $ million of revenue in $t+1$, reported over Compustat sales or Trucost's revenue-scaled value including estimates. | Trucost; Compustat |
| Environmental incidents | Environmental stories in $t+1$ to $t+2$, each counted once; severe if rated two or more on RepRisk's three-point scale; zero within coverage. | RepRisk |
| Social and governance incidents | Social (governance) stories that RepRisk records about the firm in $t+1$ to $t+2$, each counted once; severe if rated two or more on its three-point scale; zero within coverage. | RepRisk |
| EPA cases, environmental suits | Formal enforcement cases at matched TRI facilities (firms with one); federal environmental suits; both in $t+1$ to $t+2$. | EPA; Audit Analytics |
| Change in toxic releases | Log of one plus TRI releases in $t+2$ minus that in $t$, at matched facilities. | EPA TRI |

*Continued on the next page*

*Table A1, continued*

| Variable | Definition | Source |
|---|---|---|
| R&D, SG&A net of R&D, capital expenditure, cost of goods sold | In year $t$ (R&D also in $t+1$) over lagged assets, in percentage points, winsorized at 1/99 within year; SG&A net of R&D subtracts R&D where SG&A exceeds it. | Compustat |
| Intangible investment | Following Peters and Taylor (2017), R&D, counted as zero where not reported, plus organization capital investment, 0.3 times SG&A net of R&D (SG&A whole where R&D exceeds it), in $t$ or $t+1$, in percent of lagged assets, winsorized at 1/99 within year. | Compustat |
| First green application | 100 if the firm's first Y02 (IRA-class) application is filed in $t+1$ to $t+2$, among firm-years with none from 2007 through $t$. | Google Patents |
| Tax-credit mentions | 100 if a 10-K filed from January 2023 to October 2026 mentions any of nine credits that the IRA created or extended, or the section 45X credit; number of credits mentioned. | SEC EDGAR |
| Clean facilities | 100 if any facility (any manufacturing facility) is announced from August 2022 to June 2026; announced capital expenditure, in millions of 2024 dollars. | Clean Investment Monitor |
| Injury rate | Cases with days away, restricted, or transferred per 200,000 hours worked in $t+1$, winsorized at 1/99; state version over $t+1$ to $t+2$ or $t-2$ to $t-1$, in the states of the firm's openings (cells with at least 10 openings); injury cases per hour worked: cases in $t+1$ to $t+2$, with the log of hours worked as a control. | OSHA |
| Inspections, violations, suits | Programmed inspections, which the agency schedules from lists of high-hazard establishments; serious violations in programmed (triggered by complaints, referrals, and accidents) inspections, with the log of the number of such inspections as a control; federal securities suits, class actions, derivative and fraud suits, and employment suits; all in $t+1$ to $t+2$. | OSHA; Audit Analytics |
| Audit opinions | Non-standard (qualified, adverse, or no) opinion in $t+1$; material weakness in internal control in $t+1$. | Compustat |
| Innovation outcomes | All applications in $t+1$ to $t+2$; log of one plus R&D in $t+1$; value per granted patent in $t+1$ (Kogan et al., 2017), in millions of 1982 dollars, cohorts to 2020; firm R&D over sales, capped at 2. | Compustat; Google Patents; Kogan et al. (2017) |
| *E. Investment after the news and further characteristics (Sections 6.6 and 7)* | | |
| Investment outcomes | In year $t+1$, in percent: capital expenditure and R&D (zero where not reported) over lagged assets, the log growth of net property, plant, and equipment and of employees, and the sale of common and preferred stock over lagged assets, each winsorized at 1/99 within year; 100 if the firm announces a facility in the Clean Investment Monitor in calendar year $t+1$, observed from 2018. | Compustat; Clean Investment Monitor |
| News of the year | $+1$ in 2015, 2020, and 2022; $-1$ in 2016, 2017, and 2024; zero in 2018, 2019, 2021, and 2023. | This paper |
| Measures of growth | Tobin's $q$; total $q$ (Peters and Taylor, 2017); minus the logarithm of book-to-market; minus the IBES consensus for the first fiscal year over the price, on the same share basis; minus PP&E over assets; each ranked within NAICS-3 industry and event and centred, and the index of Table E25. | Compustat; IBES; CRSP; WRDS (Peters–Taylor) |
| Organization capital | SG&A capitalized by perpetual inventory since 1975, with a depreciation rate of 15 percent and an initial stock of the first SG&A over 0.25, in nominal terms (Eisfeldt and Papanikolaou, 2013); the organization capital of Peters and Taylor (2017); both over assets. | Compustat; WRDS (Peters–Taylor) |

*Continued on the next page*

*Table A1, continued*

| Variable | Definition | Source |
|---|---|---|
| Emission measures | Logarithms of scope 1 and of one plus scope 2 emissions; logarithm of one plus scope 1 emissions over revenue; log change of scope 1 emissions from the previous fiscal year; an indicator of reported rather than estimated scope 1 emissions; standardized within event (Bolton and Kacperczyk, 2021). | Trucost |
| ESG-fund overweight | The stock's weight in the combined holdings of ESG funds minus its weight in the market, over the latter, at the end of the month before the event. | CRSP Mutual Funds; CRSP |
| Patent stocks | Logarithms of one plus the firm's Y02 and other patent applications from 2007 to the year before the share. | Google Patents |

*Notes.* Ratios of accounting variables are winsorized at the 1st and 99th percentiles within year; event controls and logarithms are not. Missing values of the existing measures are set to zero with an indicator in the regressions that include them, unless a table states that it uses complete cases.

# Appendix B. Measurement

This appendix reports the performance of the classifiers for the four kinds of work, their comparison with simpler measures, and the agreement between labellers (Appendix B.1), the four shares over time (Appendix B.2), and which firms orient their hiring toward environmental work (Appendix B.3). Internet Appendix IA.I reproduces the labelling instructions, and Internet Appendix IA.II documents how each classifier was trained, selected, and calibrated.

## B.1 The Performance of the Classifiers

Table B1 reports, for each kind of work, how closely the classifier's probabilities match the labelled rates, for postings and for firm-year shares, how well its scores rank postings, and how often it agrees with the language model and with my blind codes. Table B2 compares the classifier with three simpler measures, scored on the same random sample: the average probability of the posting's occupation, a dictionary of the terms most associated with positives in the training labels, and a hand-curated dictionary of ESG terms, both listed in Internet Appendix IA.II.7. Table B3 compares the labels of the language model with those of a second language model.

**Table B1.** Calibration, Ranking, and Agreement

| | Environmental | Social | Governance | Innovation |
|---|---|---|---|---|
| *A. Calibration of postings (random sample and score-band supplements, cross-fitted)* | | | | |
| Labelled positive rate (%) | 2.0 | 4.4 | 10.5 | 8.5 |
| Mean calibrated probability (%) | 2.0 | 4.4 | 10.5 | 8.6 |
| Expected calibration error (points) | 0.43 | 0.79 | 0.96 | 0.71 |
| *B. Calibration of firm-year shares (random sample)* | | | | |
| Slope on the firm-year's expected share | 1.16 (0.14) | 0.98 (0.19) | 1.03 (0.10) | 1.08 (0.06) |
| with industry-by-year effects | 1.11 (0.24) | 0.54 (0.21) | 1.03 (0.15) | 0.88 (0.11) |
| Postings (positive) | 2,949 (52) | 2,949 (141) | 2,949 (325) | 2,949 (288) |
| *C. Ranking* | | | | |
| ROC-AUC, random sample | 0.955 | 0.889 | 0.896 | 0.969 |
| PR-AUC, random sample | 0.592 | 0.432 | 0.618 | 0.773 |
| PR-AUC relative to the base rate, random sample | 36.5 | 9.6 | 5.9 | 8.8 |
| ROC-AUC, validation sample, blind human codes | 0.966 | 0.966 | 0.927 | 0.928 |
| | [0.94, 0.99] | [0.95, 0.98] | [0.90, 0.95] | [0.90, 0.95] |
| PR-AUC, validation sample, blind human codes | 0.902 | 0.821 | 0.798 | 0.814 |
| | [0.85, 0.95] | [0.72, 0.90] | [0.68, 0.87] | [0.72, 0.86] |
| *D. Agreement (Cohen's $\kappa$)* | | | | |
| Classifier and labeller, random sample | 0.57 | 0.39 | 0.52 | 0.71 |
| Classifier and blind human codes, validation sample | 0.84 | 0.74 | 0.72 | 0.79 |
| Labeller and blind human codes, validation sample | 0.91 | 0.83 | 0.81 | 0.84 |
| *E. Training* | | | | |
| Labelled training postings | 31,995 | 28,953 | 27,063 | 15,390 |
| of which positive | 4,768 | 2,908 | 4,415 | 4,332 |
| Classifier heads | 2 | 3 | 4 | 1 |

*Notes.* Each kind of work has its own classifier. Panel A: the random sample of 4,000 postings and its supplement of postings drawn within a band of the classifier's score, weighted by their probability of selection, are split into two halves; the calibration of equation (1) is fitted on one half and evaluated on the other, and the table reports the mean over the two directions. Expected calibration error: mean absolute difference, in percentage points, between the probability and the labelled rate across bins of the probability, weighted by bin size. Panel B: the labels of the postings of the random sample that fall in firm-years of the estimation sample, regressed on the expected share of their firm-year computed without the posting itself; a slope of one indicates calibration across firm-years. Industry-by-year effects: NAICS-3 by year. Standard errors clustered by firm in parentheses. Panel C: ROC-AUC is the probability that a random positive outranks a random negative; PR-AUC is the area under the precision–recall curve, whose chance level is the base rate. The last rows score the validation sample against the blind human codes, with 95 percent intervals in brackets; because that sample over-represents postings likely to involve the work, its PR-AUC exceeds that of the random sample and is not comparable with it. Panel D: agreement of binary codes, with the classifier flagging postings whose score exceeds the threshold that maximizes agreement with the labeller; the validation sample of 800 postings coded blind by a human coder. Panel E: postings in the training samples labelled by the language model, and the number of logistic regressions (heads) selected by cross-validation. Internet Appendix IA.II describes the samples, the selection of the model, its calibration, and the threshold.

**Table B2.** The Classifier and Simpler Measures

| | Environmental | Social | Governance | Innovation |
|---|---|---|---|---|
| *A. Area under the precision–recall curve (random sample)* | | | | |
| Classifier (calibrated probability) | 0.592 | 0.432 | 0.618 | 0.773 |
| Occupation average | 0.301 | 0.285 | 0.427 | 0.652 |
| Dictionary learned from the training labels | 0.216 | 0.373 | 0.450 | 0.629 |
| ESG keyword dictionary[a] | 0.267 | 0.097 | 0.143 | 0.193 |
| Base rate (chance level) | 0.016 | 0.045 | 0.105 | 0.088 |
| *B. Area under the ROC curve (random sample)* | | | | |
| Classifier (calibrated probability) | 0.955 | 0.889 | 0.896 | 0.969 |
| Occupation average | 0.898 | 0.829 | 0.824 | 0.952 |
| Dictionary learned from the training labels | 0.857 | 0.836 | 0.808 | 0.907 |
| ESG keyword dictionary[a] | 0.849 | 0.660 | 0.564 | 0.635 |
| *C. Agreement with the labels at a binary rule (Cohen's $\kappa$)* | | | | |
| Classifier at the threshold $\tau_q$ | 0.57 | 0.39 | 0.52 | 0.71 |
| Any dictionary term present[a] | 0.18 | 0.09 | 0.13 | 0.11 |
| Share flagged by the dictionary (%) | 10.3 | 26.2 | 9.3 | 30.2 |
| Precision of the dictionary | 0.12 | 0.09 | 0.23 | 0.15 |
| *D. Firm-year shares (random-sample postings in estimation firm-years)* | | | | |
| $R^2$, label on the expected share | 0.221 | 0.031 | 0.089 | 0.157 |
| $R^2$, label on the occupation-based share | 0.124 | 0.025 | 0.065 | 0.163 |
| Jointly: coefficient on the expected share | 1.13 (0.16) | 0.72 (0.26) | 1.08 (0.21) | 0.36 (0.19) |
| on the occupation-based share | 0.10 (0.30) | 0.72 (0.46) | -0.10 (0.28) | 0.91 (0.22) |

*Notes.* Panels A to C: the 4,000 postings of the simple random sample, with the language model's labels as reference; no measure was built on them. Classifier: the calibrated probability, cross-fitted as in Table B1. Occupation average: the mean calibrated probability of the posting's six-digit SOC occupation over the openings of consolidated firm-years. Dictionary learned from the training labels: the 100 words and two-word phrases most associated with positives in the classifier's training samples (smoothed log-odds, terms in at least 20 positive postings), scored by the number present. [a]ESG keyword dictionary: 118 environmental, 112 social, and 80 governance terms, curated by hand for an earlier version of this project, which play no part in the measure; for innovation, the pattern of research and development terms that drew part of the innovation training sample. Scored by the number of terms present; the rule flags a posting with any term. Internet Appendix Tables IA.21 and IA.22 list the terms. Panel D: the 2,949 postings of the random sample in firm-years of the estimation sample, their labels regressed on the firm-year's expected share (computed without the posting itself), on its occupation-based share (the occupation averages weighted by the firm-year's openings by occupation), and on both; standard errors clustered by firm in parentheses.

**Table B3.** Agreement between Labellers

| | Environmental | Social | Governance | Innovation |
|---|---|---|---|---|
| Postings | 791 | 787 | 782 | 692 |
| Positive rate, labeller (%) | 15.9 | 17.8 | 31.6 | 21.7 |
| Positive rate, second labeller (%) | 13.5 | 13.1 | 22.1 | 18.8 |
| Observed agreement (%) | 96.6 | 93.0 | 81.6 | 93.9 |
| Cohen's $\kappa$ | 0.86 | 0.73 | 0.72 | 0.81 |
| 95% confidence interval | [0.80, 0.92] | [0.65, 0.80] | [0.68, 0.84] | [0.76, 0.86] |

*Notes.* The labels of the language model in use, Claude Sonnet 5.5, against those of a second language model, GPT-5.4 mini (`gpt-5.4-mini-2026-03-17`), on the validation samples of 800 postings for environmental, social, and governance work and 700 postings for innovation work, less the postings that the labeller marks as insufficient to judge. A posting is positive for the labeller when it scores it 1 or 2, and for the second labeller when it labels it 1. Observed agreement is the share of postings on which the two agree. With $p_o$ that share and $p_e$ the agreement expected if each labelled postings positive at its own rate independently of the other, Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$. Confidence intervals are from a bootstrap that resamples whole firms (2,000 draws).

## B.2 The Four Shares over Time

Figure B1 shows the mean of each of the four shares by year. All four rise over the decade, the environmental share most in relative terms, from 2.1 to 4.2 percent of a firm-year's openings, and the social share from 4.0 to 6.4 percent, whereas the governance and innovation shares move within narrower ranges, from 13.4 to 16.5 percent and from 10.5 to 13.3 percent.

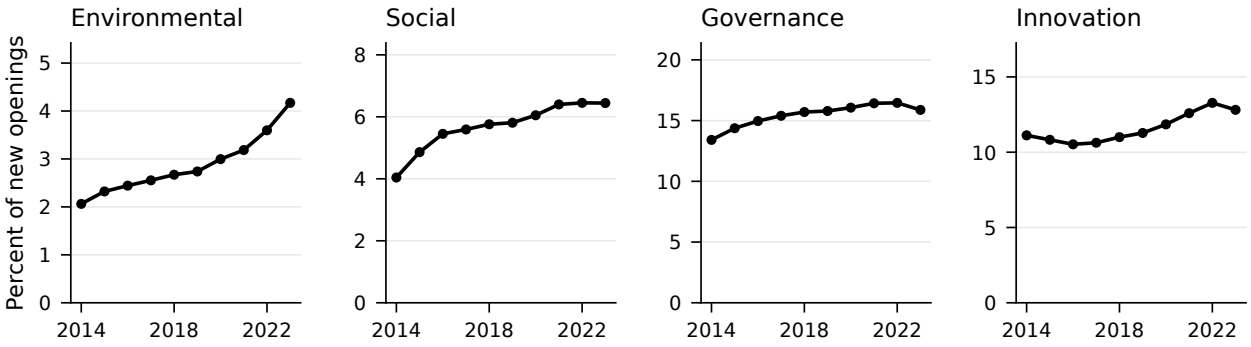



**Figure B1.** The Four Capability Shares over Time. The mean over the firm-years of the estimation sample of each capability's share, in percent of a firm-year's new openings, by year of removal; 2023 ends on 30 September. The environmental share and its two parts are shown in Figure 1.

## B.3 Which Firms Orient Their Hiring toward Environmental Work

Within industries, firms that hire toward environmental work talk more about climate change, have higher environmental ratings, and are younger and less profitable (Table B4). The regressions are descriptive. Within six-digit industries and years (column 3), firms with a higher environmental share talk more about climate change in their earnings calls (2.1 percentage points per standard deviation of talk, $t = 6.9$) and have higher LSEG environmental scores (0.32, $t = 3.5$). They are also less profitable ($-0.31$ per standard deviation of return on assets, $t = -3.3$), younger ($-0.34$ per standard deviation of log age, $t = -4.3$), and more likely to have filed a green patent before 2014. Size and Tobin's $q$ do not matter within industries.

The two parts differ. Clean-technology hiring goes with climate talk (2.0, $t = 6.7$), capital expenditures (0.25, $t = 2.7$), and young, less profitable firms that do not pay dividends (column 4). Environmental management hiring goes with tangible assets (0.34 per standard deviation of property, plant, and equipment over assets, $t = 3.2$) and with higher LSEG scores (0.20, $t = 3.5$), but not with climate talk (0.08). Environmental management work thus sits with plants and with ratings, whereas clean-technology work sits with new investment and with young firms that talk about climate opportunities. The accounts show the same pattern. Across firms, a one-standard-deviation higher clean-technology share goes with capital expenditures higher by 0.38 percentage points of lagged assets (Appendix Table C7). Clean technology work thus accompanies investment in physical capacity rather than replacing it, consistent with the capability complementing tangible capacity (Internet Appendix IA.VI.1). These patterns anticipate the results of Sections 6.6 and 8: ratings load on environmental management, and the repricing is largest in asset-light and small firms.

**Table B4.** Which Firms Orient Their Hiring toward Environmental Work

| Share (pp of new openings) | (1) Environ. | (2) Environ. | (3) Environ. | (4) Clean technology | (5) Environ. management |
|---|---|---|---|---|---|
| Log assets | −0.560*** | −0.261** | −0.035 | −0.031 | −0.004 |
| | (0.143) | (0.124) | (0.113) | (0.097) | (0.063) |
| Tobin's q | 0.074 | 0.049 | 0.056 | 0.033 | 0.023 |
| | (0.059) | (0.056) | (0.053) | (0.047) | (0.030) |
| Leverage | −0.122** | −0.006 | 0.071 | 0.056 | 0.015 |
| | (0.059) | (0.053) | (0.058) | (0.048) | (0.036) |
| Cash / assets | −0.243*** | −0.070 | 0.001 | 0.051 | −0.049 |
| | (0.071) | (0.063) | (0.063) | (0.057) | (0.031) |
| ROA | −0.368*** | −0.344*** | −0.310*** | −0.301*** | −0.009 |
| | (0.076) | (0.079) | (0.095) | (0.092) | (0.029) |
| Sales growth | 0.076** | 0.041 | −0.006 | −0.005 | −0.001 |
| | (0.039) | (0.037) | (0.036) | (0.035) | (0.013) |
| Capex / assets | 0.341*** | 0.306*** | 0.190* | 0.252*** | −0.061 |
| | (0.114) | (0.110) | (0.106) | (0.094) | (0.059) |
| PP&E / assets | 0.273* | 0.404** | 0.275* | −0.063 | 0.338*** |
| | (0.154) | (0.172) | (0.166) | (0.129) | (0.105) |
| R&D / assets | −0.038 | −0.281*** | −0.204** | −0.194** | −0.010 |
| | (0.086) | (0.087) | (0.104) | (0.099) | (0.034) |
| SG&A / assets | −0.419*** | −0.150*** | −0.049 | −0.050 | 0.001 |
| | (0.055) | (0.057) | (0.045) | (0.035) | (0.031) |
| Log firm age | −0.473*** | −0.427*** | −0.336*** | −0.253*** | −0.083* |
| | (0.089) | (0.084) | (0.078) | (0.067) | (0.047) |
| Log (1 + analysts) | 0.086 | 0.035 | 0.059 | 0.066 | −0.007 |
| | (0.097) | (0.097) | (0.081) | (0.068) | (0.044) |
| Emissions intensity | 0.985*** | 0.301* | 0.216 | 0.070 | 0.146 |
| | (0.176) | (0.158) | (0.146) | (0.120) | (0.089) |
| LSEG environmental score | 0.789*** | 0.519*** | 0.324*** | 0.122 | 0.202*** |
| | (0.121) | (0.103) | (0.093) | (0.078) | (0.058) |
| Climate talk | 2.743*** | 2.441*** | 2.115*** | 2.039*** | 0.076 |
| | (0.226) | (0.292) | (0.308) | (0.306) | (0.075) |
| Dividend payer | −0.164 | −0.132 | −0.294* | −0.307*** | 0.013 |
| | (0.188) | (0.169) | (0.153) | (0.116) | (0.104) |
| R&D missing | 1.130*** | 0.581*** | 0.250 | −0.047 | 0.297** |
| | (0.227) | (0.188) | (0.168) | (0.106) | (0.138) |
| Green patent before 2014 | 1.286*** | 0.733*** | 0.518** | 0.451** | 0.068 |
| | (0.291) | (0.271) | (0.263) | (0.217) | (0.164) |
| Fixed effects | Year | NAICS-3 × year | NAICS-6 × year | NAICS-6 × year | NAICS-6 × year |
| Firm-years | 23,443 | 23,388 | 21,174 | 21,174 | 21,174 |
| $R^2$ | 0.37 | 0.54 | 0.68 | 0.57 | 0.73 |

*Notes.* Market sample: North American firms with a CRSP security, 2014–2023. The dependent variable is the environmental share of the firm-year's new openings (columns 1 to 3), its clean-technology share (column 4), or its environmental management share (column 5), in percentage points. Characteristics are of the same fiscal year. Continuous ones are winsorized at the 1st and 99th percentiles and standardized within year, so that coefficients are per standard deviation; the dividend payer, missing R&D, and green patent indicators are 0 or 1, the last equal to one for firms with a Y02 or IRA-class application filed in 2007–2013. Emissions intensity is the logarithm of Scope 1 and 2 emissions over sales (Trucost); the LSEG environmental score and climate talk, the overall exposure of Sautner et al. (2023*a*), enter set to zero where missing, with indicators for missing values that are not shown. Firm age is the number of years since the firm's first Compustat record; analysts are those with an IBES forecast. Fixed effects: year (column 1), NAICS-3-by-year (column 2), and NAICS-6-by-year (columns 3 to 5). The coefficients describe which firms hire toward environmental work, not what causes it. Standard errors clustered by firm in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

# Appendix C. Validation

This appendix first shows where each share is highest, and then reports the tests that validate the four shares of Section 4, one subsection for each kind of work, followed by the exhibits that compare them. Each kind of work is measured from its own written definition, and the instructions from which the labeller codes it, together with its classes, are reproduced in Internet Appendix IA.I. Unless a table states otherwise, each regression relates an outcome to the four standardized shares of year $t$, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects, with standard errors clustered by firm, and the outcomes are defined in Appendix Table A1.

## C.1 Where Each Share Is Highest

Table C1 lists, for each share, the five industries, firms, occupations, and job titles where it is highest. Industries are the NAICS-3 industries with at least 20 firms in the estimation sample, ranked by the mean share of their firm-years. Firms are those observed in at least five years with at least 1,000 openings a year, ranked by their mean share. Occupations are the six-digit SOC occupations with at least 1,000 consolidated openings, ranked by the mean probability of their openings. Titles are those carried by at least 1,000 postings of 2014 to 2023, from at least 50 firms, whose postings involve the work more often than not, with a mean probability above one half, ranked by the expected number of their postings that involve the work, the sum of their probabilities. The condition on firms keeps a title that a single employer posts thousands of times from leading the list.

## C.2 Environmental Work

The tests of environmental work use two kinds of outcomes, both measured outside the postings. What the work produces is measured by the patent applications that the firm files in the two years after the year of hiring, classified by their codes into the Y02 scheme, the IRA classes, and the OECD's ENV-TECH list (Section 2.4). Where the work is needed is measured by the firm's footprint: its Scope 1 emissions per dollar of revenue, reported by the firm or estimated by Trucost, the environmental incidents that RepRisk compiles, EPA enforcement cases, environmental suits, and the change in its toxic releases. Each regression enters the environmental share beside the other three, and the tables also report its two parts, entered together in place of the share, and the specification with firm effects, which asks whether a firm's share moves with its own outcomes over time.

Table C2 relates the environmental share and its two parts to the records of what

**Table C1.** Where Each Share Is Highest

| Industries | Firms | Occupations | Titles |
|---|---|---|---|
| *A. Environmental share* | | | |
| Waste management and remediation services (37.7) | Waste Management (56.7) | Wind Turbine Service Technicians (74.5) | Helper (57) |
| Utilities (16.7) | Waste Connections (51.2) | Zoologists and Wildlife Biologists (74.1) | Environmental Technician (95) |
| Pipeline transportation (11.1) | Vestas (51.2) | Solar Photovoltaic Installers (71.3) | Environmental Specialist (93) |
| Oil and gas extraction (9.8) | GFL Environmental (49.9) | Refuse and Recyclable Material Collectors (63.5) | Environmental Engineer (88) |
| Electrical equipment, appliances and components (9.4) | Covanta (46.5) | Water and Wastewater Treatment Plant and System Operators (59.8) | Sorter (74) |
| *B. Social share* | | | |
| Mining (except oil and gas) (19.0) | BHP (32.3) | Health and Safety Engineers, Except Mining Safety Engineers and Inspectors (68.2) | Human Resources Manager (51) |
| Oil and gas extraction (17.1) | Freeport-McMoRan (32.0) | Occupational Health and Safety Specialists (65.1) | Human Resources Business Partner (50) |
| Pipeline transportation (14.6) | Dow (27.6) | Labor Relations Specialists (60.8) | EHS Specialist (71) |
| Support activities for mining (13.3) | Covanta (27.5) | Occupational Health and Safety Technicians (47.4) | EHS Manager (80) |
| Petroleum and coal products (13.0) | Holcim (23.6) | First-Line Supervisors of Firefighting and Prevention Workers (43.2) | Safety Manager (91) |
| *C. Governance share* | | | |
| Chemical manufacturing (29.1) | Sojitz (81.6) | Financial Examiners (78.4) | Quality Engineer (87) |
| Securities and investments (23.8) | Catalent (53.8) | Life, Physical, and Social Science Technicians, All Other (76.6) | Quality Manager (89) |
| Credit intermediation (22.4) | Mitsubishi UFJ Financial Group (46.2) | Information Security Analysts (70.0) | Quality Inspector (98) |
| Miscellaneous manufacturing (22.0) | Shire (45.1) | Financial Risk Specialists (68.9) | Supplier Quality Engineer (91) |
| Pipeline transportation (21.9) | Moderna (44.8) | Compliance Officers (66.6) | Quality Technician (92) |
| *D. Innovation share* | | | |
| Computer and electronic products (27.0) | Advanced Micro Devices (74.6) | Computer and Information Research Scientists (74.3) | Software Engineer (89) |
| Publishing, including software (2022 codes) (25.6) | Nvidia (68.0) | Computer Hardware Engineers (73.9) | Senior Software Engineer (83) |
| Publishing, including software (2017 codes) (24.7) | Cadence Design Systems (67.0) | Software Developers (73.1) | Software Development Engineer (92) |
| Other information services (21.1) | CS Communication & Systèmes (65.0) | Biochemists and Biophysicists (69.4) | Sr. Software Engineer (84) |
| Transportation equipment (19.8) | Intel (64.6) | Materials Scientists (65.4) | Data Engineer (85) |

*Notes.* In parentheses: for industries and firms, the mean share of their firm-years in the estimation sample; for occupations and titles, the mean probability of their openings or postings, in percent. Industries: NAICS-3 industries with at least 20 firms. Firms: observed in at least five years, with at least 1,000 openings a year. Occupations: six-digit SOC occupations with at least 1,000 consolidated openings in 2014–2023, with the titles of O*NET-SOC 2019. Titles: lowercased and stripped of punctuation, among titles carried by at least 1,000 postings of 2014–2023 from at least 50 firms with a mean probability above one half, ranked by the sum of their probabilities, and shown in one of their spellings.

environmental work produces and of the firm's footprint. The clean-technology part predicts Y02 and IRA-class applications, whereas the environmental management part predicts the technologies that the OECD classifies outside Y02, those of pollution abatement, waste, water, and environmental monitoring. With firm effects, the associations with green applications keep a little over half their size, whereas those with emissions and incidents vanish, since a footprint is a lasting attribute that distinguishes firms rather than one that moves with a firm's hiring. Table C3 restricts the sample to firms that have not yet filed an application of the class. Among firms that have filed some patent, for which a first green application marks a change in the direction of their technology, the share predicts a first IRA-class application and, less precisely, a first Y02 application, and the clean-technology part carries both associations. Among all firms without a green application, most of which never patent, it predicts only a first IRA-class application, and it predicts neither once utilities and the industries whose patenting was already most green are excluded.

Table C4 varies the sample, the timing, and the controls of the patent regressions. The coefficients change little with region-by-year effects, with the shares of the years through 2019, or among firms located in North America or filing with the SEC, whereas among North American firms alone the Y02 coefficient falls to 0.169, significant at the 10 percent level, while the IRA-class coefficient remains close to 0.3. When three lags of the share enter beside it, the lags take over the coefficient of the current share, so that across firms it is the lasting orientation of a firm's hiring that predicts its green patenting, whereas with firm effects the current share keeps its coefficient beside its lags. In the smaller fixed sample of North American firms with accounting data (Panel C), the Y02 coefficient falls to 0.099 and the IRA-class coefficient to 0.214, neither significant, and adding the four accounting controls changes them by at most 0.02, so that the fall reflects the sample rather than the controls.

Table C5 places the environmental share beside measures built from occupations and beside the environmental pillar of the LSEG score. Beside the task intensity and the green occupations of O*NET, and beside the share of environmental occupations, the share keeps most of its association with green applications and all of its association with emissions and incidents. Beside the occupation-mix term of equation (5), which predicts green applications from the occupations a firm hires alone, the share keeps its association with incidents (0.144) but loses significance for green applications (0.111 for Y02, $t = 1.1$), so that across firms the occupations a firm hires carry much of the information about its green patenting, whereas the responsibilities within them carry the information about its footprint. The two terms are not independent measures, since the occupation-mix term is built from the same probabilities as the share and accounts for 43 percent of its variance (Table 8), so that beside it the share is identified only from the part of its variation that the occupation mix does not predict. That

part, the variation within occupations, nonetheless carries the repricing on the policy days, with a coefficient of 0.48 ($t = 5.9$) against 0.16 for the occupation mix (Appendix Table E12). The share and the LSEG pillar each predict Y02 applications and incidents beside the other, and within industry-years they correlate at only 0.11. Table C6 relates the two parts to the exposure to climate change that firms discuss on their earnings calls (Sautner et al., 2023*a*). The clean-technology share predicts next year's opportunity and regulatory exposure beyond the current exposure, and opportunity exposure predicts next year's clean-technology share beyond the current share, so that hiring and talk move together rather than one leading the other.

Table C7 relates environmental hiring to the lines of the accounts, whose treatment of labor costs Internet Appendix IA.IV describes. Across firms, both parts go with higher capital expenditures, by 0.38 percentage points of assets per standard deviation of the clean-technology share and by 0.18 points for the environmental management share, and with lower SG&A and R&D, consistent with the finding of Appendix B.3 that firms that hire for environmental work invest more in tangible assets than their industry peers. Within firms, no line moves with either part at the 5 percent level. Environmental hiring is a small part of a firm's costs, spread across several lines, so that the accounts cannot reveal how a firm's capability changes, whereas its hiring can.

**Table C2.** The Environmental Share and Environmental Records

| | Environmental share (1) | Clean technology (2) | Environmental management (3) | Difference per point (4) | Environmental, firm FE (5) | Firm-years (6) |
|---|---|---|---|---|---|---|
| *A. What environmental work produces, $t+1$ to $t+2$* | | | | | | |
| Y02 applications | 0.205*** | 0.144*** | 0.044 | 0.154 | 0.128** | 20,349 |
| | (0.049) | (0.033) | (0.140) | (0.205) | (0.055) | |
| IRA-class applications | 0.296*** | 0.203*** | 0.089 | 0.178 | 0.162** | 18,461 |
| | (0.062) | (0.042) | (0.197) | (0.278) | (0.067) | |
| ENV-TECH outside Y02 | 0.214*** | 0.030 | 0.461*** | −0.601*** | 0.072 | 16,997 |
| | (0.082) | (0.084) | (0.087) | (0.181) | (0.069) | |
| Green share of applications[a] | 0.076*** | 0.067*** | 0.015** | 0.078*** | 0.008 | 11,266 |
| | (0.007) | (0.005) | (0.007) | (0.012) | (0.006) | |
| Other applications | −0.218* | −0.111 | −0.343** | 0.327 | −0.049 | 22,400 |
| | (0.127) | (0.095) | (0.148) | (0.264) | (0.041) | |
| *B. Where the footprint is* | | | | | | |
| Scope 1 intensity, reported, $t+1$[a] | 0.137*** | 0.064** | 0.137** | −0.115 | −0.008 | 5,699 |
| | (0.041) | (0.032) | (0.057) | (0.104) | (0.019) | |
| Scope 1 intensity, estimated, $t+1$[a] | 0.120*** | 0.065*** | 0.107*** | −0.060 | −0.015 | 24,443 |
| | (0.028) | (0.023) | (0.039) | (0.070) | (0.021) | |
| Environmental incidents, $t+1$ to $t+2$ | 0.142*** | 0.091*** | 0.106*** | −0.017 | −0.009 | 21,769 |
| | (0.030) | (0.031) | (0.023) | (0.047) | (0.015) | |
| Severe incidents, $t+1$ to $t+2$ | 0.132*** | 0.082*** | 0.099*** | −0.022 | −0.035* | 20,805 |
| | (0.032) | (0.031) | (0.026) | (0.051) | (0.019) | |
| EPA enforcement cases, $t+1$ to $t+2$ | 0.179* | 0.169 | 0.105 | 0.100 | 0.035 | 3,896 |
| | (0.103) | (0.105) | (0.075) | (0.173) | (0.072) | |
| Environmental suits, $t+1$ to $t+2$ | −0.020 | −0.239* | 0.080 | −0.467** | −0.143 | 6,682 |
| | (0.104) | (0.140) | (0.084) | (0.234) | (0.277) | |
| Change in log toxic releases, $t$ to $t+2$[a] | 0.002 | −0.002 | 0.004 | −0.008 | −0.102 | 4,327 |
| | (0.032) | (0.038) | (0.029) | (0.073) | (0.092) | |

*Notes.* The sample is the estimation sample of firm-years from 2014 to 2023. Each row reports regressions of the outcome on the shares of year $t$, entered together with the social, governance, and innovation shares, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects. Column (1) reports the coefficient of the standardized environmental share. Columns (2) and (3) report the coefficients of its clean-technology and environmental management parts, each standardized and entered together in place of the share. Column (4) reports the difference between the coefficients of a percentage point of clean-technology work and of a percentage point of environmental management work, per standard deviation of the environmental share; it is the coefficient of the clean-technology share, scaled by the standard deviation of the environmental share and entered beside the environmental share. Column (5) adds firm effects to the specification of column (1), and column (6) gives the firm-years of column (1). Counts are estimated by Poisson pseudo-maximum likelihood, so that coefficients approximate proportional changes, and outcomes marked [a] by OLS. Outcomes are defined in Table A1. Standard errors clustered by firm are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C3.** The First Green Application

| | First Y02 application | | | First IRA-class application | | |
|---|---|---|---|---|---|---|
| | No green patent (1) | Strict newcomers (2) | Patenting firms (3) | No green patent (4) | Strict newcomers (5) | Patenting firms (6) |
| *A. The environmental share* | | | | | | |
| Environmental share | 0.18 | −0.17 | 1.64* | 0.53** | 0.46 | 1.85** |
| | (0.33) | (0.35) | (0.90) | (0.25) | (0.29) | (0.80) |
| *B. Its two parts, entered together* | | | | | | |
| Clean technology | 0.22 | 0.02 | 3.19** | 0.82** | 0.55 | 5.21*** |
| | (0.32) | (0.43) | (1.44) | (0.38) | (0.44) | (1.58) |
| Environmental management | 0.05 | −0.21 | 0.58 | 0.09 | 0.19 | 0.46 |
| | (0.31) | (0.30) | (0.61) | (0.17) | (0.23) | (0.43) |
| Base rate (percent) | 3.2 | 3.1 | 6.8 | 1.6 | 1.5 | 3.2 |
| Firms | 5,654 | 5,408 | 2,132 | 5,930 | 5,677 | 2,441 |
| Firm-years | 15,471 | 14,768 | 6,212 | 16,676 | 15,931 | 7,423 |

*Notes.* The sample is the firm-years in which the firm has filed no application of the class from 2007 through year $t$. The dependent variable equals 100 if the firm files an application of the class in $t+1$ to $t+2$, so that coefficients are in percentage points per standard deviation of the share. Columns (1) and (4) keep the firms without a Y02 or IRA-class application in 2007–2013. Columns (2) and (5) further exclude utilities and the tenth of six-digit industries whose applications in 2007–2013 were most often in Y02, among industries with at least 20 applications. Columns (3) and (6) keep the firms without a green patent that have filed at least one application of any class by year $t$. The regressions are estimated by OLS with the controls and effects of Table C2. The base rate is the mean of the dependent variable. Standard errors clustered by firm are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C4.** Green Patents: Samples, Timing, and Accounting Controls

| | Y02 applications (1) | IRA-class applications (2) | All applications (3) | Firm-years Y02 (4) | Firm-years IRA-class (5) |
|---|---|---|---|---|---|
| *A. Samples and fixed effects* | | | | | |
| Baseline (Table C2, column 1) | 0.205*** | 0.296*** | −0.124 | 20,349 | 18,461 |
| | (0.049) | (0.062) | (0.097) | | |
| Region-by-year effects | 0.216*** | 0.305*** | −0.119 | 20,349 | 18,461 |
| | (0.050) | (0.062) | (0.100) | | |
| North America and US filers | 0.204*** | 0.296*** | −0.124 | 18,910 | 17,160 |
| | (0.049) | (0.062) | (0.096) | | |
| North America only | 0.169* | 0.299** | −0.249** | 16,572 | 14,865 |
| | (0.096) | (0.118) | (0.116) | | |
| Shares through 2019 | 0.194*** | 0.291*** | −0.090 | 13,615 | 12,328 |
| | (0.053) | (0.065) | (0.108) | | |
| *B. Timing: firm-years with the share observed in $t-1$, $t-2$, and $t-3$* | | | | | |
| *NAICS-3-by-year effects* | | | | | |
| Share of $t$ alone | 0.198*** | 0.261*** | | | |
| | (0.076) | (0.093) | | | |
| Share of $t$, with its three lags | −0.017 | −0.032 | | 8,316 | 7,639 |
| | (0.174) | (0.200) | | | |
| Sum of the coefficients of the three lags | 0.244* | 0.329** | | 8,316 | 7,639 |
| | (0.142) | (0.151) | | | |
| *Firm effects added* | | | | | |
| Share of $t$ alone | 0.139*** | 0.134** | | | |
| | (0.053) | (0.054) | | | |
| Share of $t$, with its three lags | 0.141*** | 0.143*** | | 2,930 | 1,954 |
| | (0.052) | (0.054) | | | |
| Sum of the coefficients of the three lags | 0.155 | 0.163 | | 2,930 | 1,954 |
| | (0.102) | (0.100) | | | |
| *C. North American firms with accounting data, one fixed sample* | | | | | |
| Controls of Panel A | 0.099 | 0.214 | | 13,933 | 12,356 |
| | (0.123) | (0.151) | | | |
| Adding the four accounting controls | 0.097 | 0.193 | | 13,933 | 12,356 |
| | (0.121) | (0.153) | | | |

*Notes.* Each cell reports the coefficient of the standardized environmental share in a Poisson pseudo-maximum-likelihood regression of the applications filed in $t+1$ to $t+2$ on the four shares of year $t$, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects, as in column (1) of Table C2. In Panel A, the second row adds effects for each year of Compustat North America and of Compustat Global; the third row keeps the North American firms and the firms of Compustat Global that file patent applications in the United States; the fourth row keeps the North American firms only; and the fifth row keeps the firm-years through 2019, whose outcome window closes before the publication lag of applications truncates it. Column (3) relates the share to all applications. Panel B keeps the firm-years from 2017 to 2023 whose firm is in the estimation sample in each of the three preceding years, in which the share's correlation with its first lag is 0.95. The first row of each block enters the share of year $t$ alone, the second enters it together with the shares of $t-1$, $t-2$, and $t-3$, and the third reports the sum of the coefficients of the three lags; the other three shares enter for year $t$ only, and the second block adds firm effects. Panel C keeps the North American firms with R&D, SG&A net of R&D, and capital expenditures over lagged assets and with intangible capital over assets (Peters and Taylor, 2017), 16,110 firm-years before estimation, so that both rows use the same sample; the second row adds these four variables as controls. Firm-years are those that remain after the Poisson estimator drops firms whose outcome is always zero; in Panel B they refer to the regressions with lags. Standard errors clustered by firm are in parentheses; with firm effects they count the industry-by-year effects in the degrees of freedom, as in Table C2. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C5.** The Environmental Share beside Occupation-Based Measures and the LSEG Score

| | Y02 applications (1) | ENV-TECH outside Y02 (2) | IRA-class applications (3) | Green share of applications[a] (4) | Scope 1 intensity, $t+1$[a] (5) | Environmental incidents (6) |
|---|---|---|---|---|---|---|
| Environmental share alone | 0.205*** | 0.214*** | 0.296*** | 0.076*** | 0.137*** | 0.142*** |
| | (0.049) | (0.082) | (0.062) | (0.007) | (0.041) | (0.030) |
| Firm-years | 20,349 | 16,997 | 18,461 | 11,266 | 5,699 | 21,769 |
| *A. O*NET green task intensity of the firm's occupations* | | | | | | |
| O*NET measure alone | 0.217*** | 0.109 | 0.380*** | 0.021*** | −0.043 | 0.026 |
| | (0.072) | (0.092) | (0.079) | (0.005) | (0.033) | (0.045) |
| Together: environmental share | 0.161** | 0.202** | 0.210** | 0.075*** | 0.167*** | 0.147*** |
| | (0.066) | (0.099) | (0.093) | (0.007) | (0.044) | (0.033) |
| Together: O*NET measure | 0.120 | 0.041 | 0.217* | 0.007* | −0.103*** | −0.030 |
| | (0.086) | (0.110) | (0.120) | (0.003) | (0.036) | (0.050) |
| *B. Share of openings in O*NET green occupations* | | | | | | |
| O*NET measure alone | 0.240*** | 0.195** | 0.479*** | 0.016*** | 0.098*** | 0.008 |
| | (0.070) | (0.095) | (0.104) | (0.004) | (0.035) | (0.045) |
| Together: environmental share | 0.167*** | 0.187** | 0.224*** | 0.076*** | 0.125*** | 0.144*** |
| | (0.055) | (0.087) | (0.071) | (0.007) | (0.042) | (0.031) |
| Together: O*NET measure | 0.175** | 0.148 | 0.359*** | 0.002 | 0.072* | −0.023 |
| | (0.077) | (0.101) | (0.121) | (0.003) | (0.037) | (0.047) |
| *C. Share of openings in environmental occupations* | | | | | | |
| Occupation share alone | 0.171 | | 0.329** | | | 0.065 |
| | (0.108) | | (0.157) | | | (0.048) |
| Together: environmental share | 0.197*** | | 0.255*** | | | 0.139*** |
| | (0.069) | | (0.092) | | | (0.034) |
| Together: occupation share | 0.020 | | 0.117 | | | 0.013 |
| | (0.127) | | (0.184) | | | (0.051) |
| *D. Occupation-mix term of equation (5)* | | | | | | |
| Occupation-mix term alone | 0.270*** | | 0.380*** | | | 0.086* |
| | (0.045) | | (0.066) | | | (0.045) |
| Together: environmental share | 0.111 | | 0.157 | | | 0.144*** |
| | (0.099) | | (0.137) | | | (0.038) |
| Together: occupation-mix term | 0.160 | | 0.217 | | | −0.006 |
| | (0.109) | | (0.150) | | | (0.055) |
| *E. LSEG environmental pillar score* | | | | | | |
| Environmental share alone | 0.197*** | | | | | 0.142*** |
| | (0.054) | | | | | (0.030) |
| Together: environmental share | 0.174*** | | | | | 0.118*** |
| | (0.053) | | | | | (0.029) |
| Together: pillar score | 0.362*** | | | | | 0.375*** |
| | (0.110) | | | | | (0.061) |
| Firm-years | 17,190 | | | | | 18,517 |

*Notes.* The first row repeats column (1) of Table C2 and gives its firm-years, which Panels C and D share; Panels A and B have at most three fewer, and Panel E gives its own. Each panel relates the outcome to an alternative measure, standardized within year, first alone and then together with the environmental share; the social, governance, and innovation shares, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects enter throughout. Panel A uses the opening-weighted mean green task intensity of the firm's occupations, the share of an occupation's task statements that O*NET classifies as green (Vona et al., 2018), carried to 2018 SOC codes through O*NET's crosswalk. Panel B uses the share of the firm's openings in occupations that O*NET classifies as green occupations with new or enhanced skills. Both are computed over the openings whose occupation has an O*NET value. Panel C uses the share of openings in environmental occupations, the tenth of six-digit occupations with at least 1,000 openings that has the highest mean environmental probability. Panel D uses the occupation-mix term of equation (5), which credits each opening with the mean environmental probability of its six-digit occupation in the year. Panels C and D are estimated for Y02 and IRA-class applications and environmental incidents only. Panel E uses the LSEG environmental pillar score (LSEG Data & Analytics, 2024) in the firm-years that have one, and its first row re-estimates the environmental share alone in these firm-years. Counts in $t+1$ to $t+2$ are estimated by Poisson pseudo-maximum likelihood, and outcomes marked [a] by OLS; the green share of applications is a fraction, as in Table C2. Standard errors clustered by firm are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C6.** Climate Talk and Environmental Hiring, Lead and Lag

| | Opportunity talk, $t+1$ (1) | Regulatory talk, $t+1$ (2) | Clean-tech. share, $t+1$ (3) | Environ. man. share, $t+1$ (4) |
|---|---|---|---|---|
| Clean-technology share, $t$ | 0.045*** | 0.052*** | 0.884*** | 0.000 |
| | (0.016) | (0.019) | (0.014) | (0.003) |
| Environmental management share, $t$ | −0.013 | 0.016 | 0.000 | 0.899*** |
| | (0.009) | (0.013) | (0.004) | (0.020) |
| Opportunity talk, $t$ | 0.804*** | | 0.044*** | −0.008* |
| | (0.034) | | (0.010) | (0.005) |
| Regulatory talk, $t$ | | 0.578*** | 0.003 | 0.007 |
| | | (0.049) | (0.007) | (0.006) |
| Firm-years | 17,669 | 17,669 | 17,784 | 17,784 |

*Notes.* The sample is the firm-years of the market sample, the North American firms of the estimation sample with a CRSP security, whose earnings-call measures of climate-change exposure (Sautner et al., 2023*a*) are available in years $t$ and $t+1$. Columns (1) and (2) regress the opportunity and the regulatory exposure of year $t+1$ on the clean-technology and environmental management shares of year $t$ and on the same exposure in year $t$. Columns (3) and (4) regress each part of year $t+1$ on both exposures and both parts of year $t$. The exposures are standardized within year, and the parts as in Table C2. All regressions are estimated by OLS with the social, governance, and innovation shares, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects. Standard errors clustered by firm are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C7.** Environmental Hiring and the Lines of the Accounts

| | Across firms | | Within firms | | Firm-years | |
|---|---|---|---|---|---|---|
| | Clean technology (1) | Environmental management (2) | Clean technology (3) | Environmental management (4) | Across (5) | Within (6) |
| SG&A net of R&D | −0.416** | −0.979*** | −0.099 | −0.021 | 22,801 | 22,173 |
| | (0.173) | (0.349) | (0.226) | (0.111) | | |
| R&D | −0.218** | −0.865*** | 0.184* | −0.001 | 26,542 | 25,876 |
| | (0.103) | (0.143) | (0.110) | (0.033) | | |
| Capital expenditures | 0.379*** | 0.178*** | −0.036 | −0.058 | 26,469 | 25,805 |
| | (0.079) | (0.064) | (0.077) | (0.066) | | |
| Cost of goods sold | −0.621 | 0.115 | −0.698 | −0.241 | 26,542 | 25,876 |
| | (0.680) | (0.737) | (0.658) | (0.356) | | |

*Notes.* The sample is the North American firms of the estimation sample. Each line of the accounts in year $t$, over lagged total assets in percentage points, is regressed on the clean-technology and environmental management shares of year $t$, each standardized and entered together with the social, governance, and innovation shares and the logarithms of assets, employees, and openings. Columns (1) and (2) include NAICS-3-by-year effects, and columns (3) and (4) add firm effects. SG&A net of R&D is Compustat SG&A minus R&D. Standard errors clustered by firm are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

## C.3 Social Work

What social work achieves is an absence, fewer violations and injuries at the firm's establishments, which the Occupational Safety and Health Administration documents through its inspections and the injury reports it collects. Where the work is needed is measured by the violations found in inspections that complaints and accidents trigger, by the social incidents that RepRisk compiles, and by employment suits. Because the share can relate to injuries in either direction, with more safety work where hazards are greater and fewer injuries where it succeeds, and because worker safety varies mostly across industries, the main test compares the same firm's hiring across the US states in which it operates, with firm-by-year effects that hold fixed everything about the firm in a given year.

Table C8 relates the social share and its three parts to the inspections, violations, and injury rates recorded by the Occupational Safety and Health Administration, to employment suits recorded by Audit Analytics, and to the social incidents recorded by RepRisk. Firms whose social share is one standard deviation higher incur about 10 percent fewer serious violations in the inspections that the agency schedules for high-hazard establishments, an imprecise estimate carried by worker safety, and their injury rates in the following year are lower by 7 percent of the mean. They are not inspected more often, so that selection into inspection does not drive the comparison. The share also predicts more serious violations in inspections triggered by complaints and accidents and more social incidents in the news, which mark where the work is needed, whereas employment suits are too rare for a precise estimate. With firm effects, the associations with violations and injury rates vanish. Table C9 compares the same firm's hiring across the US states in which it operates. Within a firm and year, the social share of a state's openings is higher where the firm's establishments record more injuries, both in the following year and in the two preceding years, so that it reflects a stable placement of safety work in the firm's more hazardous operations, and once firm-by-state effects remove these permanent differences, a rise in a state's social hiring is not followed by fewer injuries.

**Table C8.** The Social Share and Social Records

| | Social share | Worker safety | Product and patient safety | People and rights | Social, firm FE | Firm-years |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *A. What social work produces, $t+1$ to $t+2$* | | | | | | |
| Serious violations, programmed inspections | −0.107* | −0.117* | −0.013 | 0.038 | 0.162* | 3,358 |
| | (0.065) | (0.063) | (0.081) | (0.067) | (0.094) | |
| Injury rate (DART), $t+1$[a] | −0.116*** | −0.092*** | −0.071** | −0.055* | 0.040 | 10,160 |
| | (0.034) | (0.032) | (0.033) | (0.033) | (0.037) | |
| *B. Design check, $t+1$ to $t+2$* | | | | | | |
| Programmed inspections | 0.075 | 0.092* | −0.412*** | −0.040 | −0.002 | 16,252 |
| | (0.058) | (0.051) | (0.106) | (0.063) | (0.047) | |
| *C. Where social work is needed, $t+1$ to $t+2$* | | | | | | |
| Serious violations, triggered inspections | 0.085* | 0.086** | −0.041 | −0.013 | 0.043 | 6,624 |
| | (0.044) | (0.039) | (0.069) | (0.030) | (0.043) | |
| Social incidents | 0.129*** | 0.063** | −0.052 | 0.161*** | −0.045** | 22,360 |
| | (0.039) | (0.028) | (0.062) | (0.039) | (0.019) | |
| Severe social incidents | 0.102*** | 0.064** | −0.210*** | 0.133*** | −0.046* | 21,846 |
| | (0.036) | (0.031) | (0.057) | (0.031) | (0.026) | |
| Employment suits | −0.546 | 0.149 | −1.962 | −0.282 | n.e.[b] | 1,665 |
| | (0.709) | (0.574) | (1.331) | (0.313) | | |

*Notes.* The sample is the estimation sample of firm-years from 2014 to 2023 with each outcome. Each row reports regressions of the outcome on the four shares of year $t$, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects. Column (1) reports the coefficient of the standardized social share. Columns (2) to (4) replace it with its worker-safety, product and patient safety, and people and rights parts, each standardized and entered together. Column (5) adds firm effects to the specification of column (1), and column (6) gives the firm-years of column (1). Serious violations are those cited in the inspections that the Occupational Safety and Health Administration programs from lists of high-hazard establishments or in the inspections that complaints, referrals, and accidents trigger, with the logarithm of the number of such inspections as a control. The injury rate is the number of cases with days away, restricted, or transferred per 200,000 hours worked at the firm's establishments, as reported to OSHA. Social incidents are the social stories that RepRisk records about the firm, each counted once, and severe incidents those it rates two or more on its three-point scale. Employment suits are federal employment cases against the firm recorded by Audit Analytics. Counts in $t+1$ to $t+2$ are estimated by Poisson pseudo-maximum likelihood, so that coefficients approximate proportional changes, and outcomes marked [a] by OLS. [b]Not estimable: with firm effects, only 27 firm-years remain for employment suits. Standard errors clustered by firm are in parentheses; with firm effects they count the industry-by-year effects in the degrees of freedom, as in Table C2. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C9.** Social Hiring across a Firm's US States

| | Social share (1) | Worker safety (2) | Environ-mental (3) | Govern-ance (4) | Inno-vation (5) | Cells (6) | Firms (7) |
|---|---|---|---|---|---|---|---|
| *A. Placement across a firm's states: firm-by-year effects* | | | | | | | |
| Injury rate in the state, $t+1$ to $t+2$ | 0.089*** | 0.128*** | 0.042 | −0.103*** | −0.178*** | 68,098 | 1,632 |
| | (0.018) | (0.018) | (0.034) | (0.021) | (0.025) | | |
| Injury rate in the state, $t-2$ to $t-1$ | 0.107*** | 0.141*** | −0.015 | −0.099*** | −0.205*** | 60,330 | 1,610 |
| | (0.020) | (0.021) | (0.028) | (0.021) | (0.023) | | |
| *B. Changes within a firm's states: firm-by-year and firm-by-state effects* | | | | | | | |
| Injury rate in the state, $t+1$ to $t+2$ | 0.019 | 0.026 | −0.018 | 0.002 | 0.032* | 65,665 | 1,632 |
| | (0.018) | (0.018) | (0.032) | (0.019) | (0.019) | | |
| Injury cases per hour worked[a] | −0.009 | −0.001 | 0.019 | −0.015 | 0.029*** | 61,628 | 1,632 |
| | (0.015) | (0.014) | (0.018) | (0.011) | (0.011) | | |
| Injury rate, other effects[b] | 0.022 | 0.020 | −0.026 | 0.005 | 0.043** | 66,922 | 1,632 |
| | (0.028) | (0.029) | (0.033) | (0.028) | (0.019) | | |

*Notes.* Each observation is a firm, US state, and year with at least 10 openings, in a state where the firm's establishments were inspected by the Occupational Safety and Health Administration from 2007 to 2013 or file injury records, for firm-years with at least two such states. The shares of each cell's openings are standardized within year across cells. Each row regresses an outcome of the firm's establishments in the state on the social share of the cell's openings, with the environmental, governance, and innovation shares of the cell and the logarithm of its openings. Column (2) replaces the social share with its worker-safety part, and columns (3) to (5) report the other three shares of the regression of column (1). The injury rate is the number of cases with days away, restricted, or transferred per 200,000 hours worked at the firm's establishments in the state over $t+1$ to $t+2$, or over $t-2$ to $t-1$, winsorized at the 1st and 99th percentiles within year and estimated by OLS. Panel A includes firm-by-year effects, which hold fixed everything about the firm in a given year. Panel B adds firm-by-state effects, which absorb the permanent hazard of the firm's operations in each state and leave 16 percent of the variance of the social share. [a]Poisson pseudo-maximum likelihood on the number of cases in $t+1$ to $t+2$, with the logarithm of hours worked as a control. [b]Firm-by-state and state-by-year effects instead of firm-by-year and firm-by-state effects. Standard errors clustered by firm are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

## C.4 Governance Work

Governance work, as measured here, operates the firm's controls and its compliance with the law. What it achieves is measured by the absence of failures of control, material weaknesses in internal control and non-standard audit opinions in the following year. Where it is needed is measured by the firm's legal exposure: the federal securities suits, class actions, and derivative and fraud suits against SEC registrants that Audit Analytics collects, and the governance incidents that RepRisk compiles. Each part of the work is also entered separately, since the risk and financial controls part should relate to weaknesses in internal control and the compliance and integrity part to litigation.

Table C10 relates the governance share and its four parts to the federal legal cases against SEC registrants recorded by Audit Analytics, to auditors' opinions from Compustat, and to the governance incidents recorded by RepRisk. Firms whose governance share is one standard deviation higher face 21 percent more securities suits over the following two years, 26 percent more class actions, and 36 percent more derivative and fraud suits, and they are involved in 16 percent more governance incidents in the news. The compliance and integrity part, whose work responds to legal and regulatory exposure, carries most of the association with litigation. With firm effects, only the association with derivative and fraud suits remains, since legal exposure, like an environmental footprint, is a lasting attribute of a firm. Material weaknesses in internal control are not related to the share as a whole (0.41 percentage points, $t = 1.3$), but they are related to its risk and financial controls part (0.68 points, $t = 2.3$), and non-standard audit opinions are not related to it.

**Table C10.** The Governance Share and Governance Records

| | Governance share (1) | Quality (2) | Risk and financial controls (3) | Information security and privacy (4) | Compliance and integrity (5) | Governance, firm FE (6) | Firm-years (7) |
|---|---|---|---|---|---|---|---|
| *A. What governance work produces, $t+1$* | | | | | | | |
| Non-standard audit opinion (pp)[a] | 0.042 | 0.049 | −0.005 | −0.002 | 0.026 | 0.037 | 21,870 |
| | (0.033) | (0.046) | (0.019) | (0.014) | (0.026) | (0.042) | |
| Material weakness (pp)[a] | 0.412 | 0.284 | 0.677** | 0.162 | −0.173 | −0.025 | 18,401 |
| | (0.311) | (0.339) | (0.299) | (0.296) | (0.341) | (0.464) | |
| *B. Where governance work is needed, $t+1$ to $t+2$* | | | | | | | |
| Securities suits | 0.195*** | 0.098 | 0.085* | 0.029 | 0.187*** | −0.001 | 16,682 |
| | (0.055) | (0.063) | (0.051) | (0.036) | (0.041) | (0.091) | |
| Class actions | 0.232*** | 0.110* | 0.110** | −0.012 | 0.213*** | 0.103 | 17,226 |
| | (0.054) | (0.062) | (0.050) | (0.040) | (0.037) | (0.073) | |
| Derivative and fraud suits | 0.305*** | 0.333*** | 0.008 | −0.001 | 0.140* | 0.426** | 10,678 |
| | (0.095) | (0.104) | (0.100) | (0.108) | (0.076) | (0.209) | |
| Governance incidents | 0.147** | −0.069 | 0.158*** | −0.051 | 0.121*** | 0.012 | 22,208 |
| | (0.060) | (0.054) | (0.051) | (0.043) | (0.034) | (0.032) | |
| Severe governance incidents | 0.184*** | −0.018 | 0.180*** | −0.004 | 0.121*** | −0.009 | 21,848 |
| | (0.056) | (0.059) | (0.048) | (0.041) | (0.037) | (0.040) | |

*Notes.* The sample is the estimation sample of firm-years from 2014 to 2023 with each outcome. Each row reports regressions of the outcome on the four shares of year $t$, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects. Column (1) reports the coefficient of the standardized governance share. Columns (2) to (5) replace it with its quality, risk and financial controls, information security and privacy, and compliance and integrity parts, each standardized and entered together. Column (6) adds firm effects to the specification of column (1), and column (7) gives the firm-years of column (1). A non-standard audit opinion is a qualified, adverse, or no opinion in $t+1$, and a material weakness is one reported in internal control over financial reporting in $t+1$, from Compustat; both are indicators times 100, so that coefficients are in percentage points, against means of 0.07 and 4.4 percent. Securities suits, class actions, and derivative and fraud suits are federal cases against the firm recorded by Audit Analytics, and governance incidents are the governance stories that RepRisk records, each counted once, severe if rated two or more on its three-point scale. Counts in $t+1$ to $t+2$ are estimated by Poisson pseudo-maximum likelihood, so that coefficients approximate proportional changes, and outcomes marked [a] by OLS. Standard errors clustered by firm are in parentheses; with firm effects they count the industry-by-year effects in the degrees of freedom, as in Table C2. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

## C.5 Innovation Work

Innovation work is the benchmark because R&D, which the accounts report, pays mostly for the labor of the people who do the work (Section 4.2). Its tests use R&D, intangible investment as measured by Peters and Taylor (2017), patent applications and the value of granted patents, and Tobin's $q$, each across firms within industry-years and within firms.

Table C13 relates the innovation share to R&D and patents. Across firms, a firm's mean innovation share correlates at 0.42 with its R&D over sales and at 0.43 with its patent applications, with rank correlations of 0.67 and 0.50, and it keeps its association with R&D when the occupation-mix term of equation (5) enters beside it, whose coefficient then turns negative, so that the responsibilities described within occupations carry the information about R&D. Across firm-years, the share predicts patent applications and R&D in the following years, also within firms. Table C14 relates the four shares to R&D and to intangible investment as measured by Peters and Taylor (2017). A firm whose innovation share is one standard deviation higher spends 3.96 percentage points more of its assets on R&D in the same year, against a mean of 4.63, and 3.86 points more in the following year, and within firms 0.29 and 0.32 points more. The share also predicts intangible investment across firms, but not within them. Figure C1 shows that the relation with R&D is close to linear across the distribution of the share, both across firms within industry-years and within firms.

Table C11 shows where the innovation share is highest. Among the NAICS-3 industries with at least 20 firms, it is highest in computer and electronic products, at 27.0 percent of openings, in software publishing and information services, and in research and technical services, and it is lowest in nursing care, accommodation, general merchandise stores, hospitals, and restaurants, below 2 percent. Among the firms that post at least 1,000 openings a year, the highest shares belong to semiconductor makers and the developers of chip-design software, such as Advanced Micro Devices, Nvidia, Intel, Qualcomm, Cadence, and Synopsys, which devote 60 to 75 percent of their openings to innovation work. A firm's share is also persistent. Measured relative to its industry and year, it correlates at 0.88 from one year to the next, against 0.89 for the environmental share.

Table C12 relates the innovation share to Tobin's $q$. A firm whose innovation share is one standard deviation higher than those of other firms in its industry and year has a $q$ higher by 0.31, about a seventh of the average $q$ of 2.15. The association is larger across firms' long-run averages, at 0.39, and it is positive in each of the ten yearly cross-sections. Controls for scale, the balance sheet, performance, age, and dividends leave it unchanged, but it falls to 0.06, and is no longer significant, once R&D and SG&A expenses enter. Panel C shows which of the two absorbs it. R&D spending alone accounts for all of it, leaving −0.02, whereas SG&A spending leaves it at 0.33. The knowledge capital that Peters and Taylor (2017) accumulate

from past R&D accounts for about two-fifths of it, leaving 0.18, and the organization capital that they accumulate from SG&A accounts for none. Markets thus value the innovation orientation of hiring as they value current R&D spending, as expected if the share records the labor that R&D pays for, which accounts for more than two-thirds of business R&D spending in the United States (Moris and Shackelford, 2023).

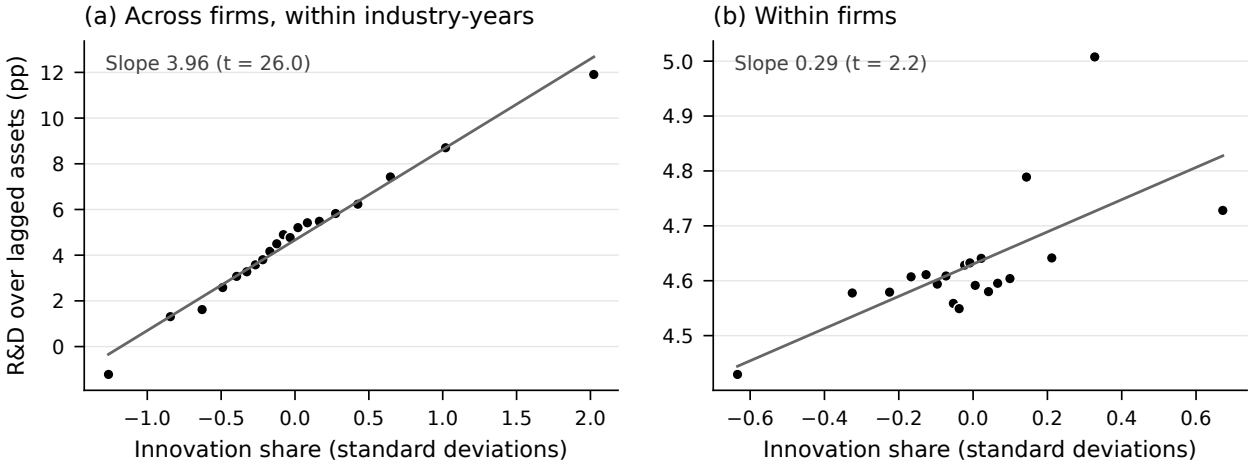



**Figure C1.** Innovation Hiring and R&D. Binned scatter plots of R&D over lagged assets, in percentage points, on the standardized innovation share, for the North American firms of the estimation sample. Both variables are first residualized on the social, governance, and environmental shares, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects in panel (a), with firm effects added in panel (b); the residuals of the share are sorted into 20 groups of equal size, and each marker plots the group means, with the sample means added back. The line has the slope of the corresponding regression in Table C14, whose $t$-statistic uses standard errors clustered by firm.

**Table C11.** Where the Innovation Share Is Highest

| | Firm-years | Firms | Innovation share (%) |
|---|---|---|---|
| *A. NAICS-3 industries with at least 20 firms* | | | |
| *The ten with the highest share* | | | |
| Computer and electronic products (334) | 2,782 | 570 | 27.0 |
| Publishing industries, including software (2022 codes) (513) | 487 | 178 | 25.6 |
| Publishing industries, including software (2017 codes) (511) | 745 | 176 | 24.7 |
| Other information services (519) | 1,307 | 336 | 21.1 |
| Transportation equipment (336) | 1,223 | 235 | 19.8 |
| Data processing, hosting and related services (518) | 952 | 260 | 17.8 |
| Professional, scientific and technical services (541) | 1,799 | 403 | 17.8 |
| Chemical manufacturing (325) | 3,439 | 815 | 17.2 |
| Electrical equipment, appliances and components (335) | 545 | 118 | 16.1 |
| Machinery (333) | 1,243 | 232 | 15.2 |
| *The five with the lowest share* | | | |
| Food services and drinking places (722) | 473 | 93 | 1.7 |
| Hospitals (622) | 145 | 31 | 1.7 |
| General merchandise stores (452) | 181 | 35 | 1.6 |
| Accommodation (721) | 242 | 43 | 1.6 |
| Nursing and residential care facilities (623) | 118 | 21 | 0.7 |
| All firms | 35,772 | 7,128 | 11.8 |
| *B. The ten firms with the highest share among large recruiters* | | | |
| | Years | Openings a year | Innovation share (%) |
| Advanced Micro Devices (electronics) | 10 | 3,297 | 74.6 |
| Nvidia (electronics) | 10 | 3,211 | 68.0 |
| Cadence Design Systems (software) | 10 | 1,388 | 67.0 |
| CS Communication & Systèmes (technical services) | 5 | 1,285 | 65.0 |
| Intel (electronics) | 10 | 14,663 | 64.6 |
| Qualcomm (electronics) | 10 | 3,729 | 64.6 |
| Synopsys (software) | 10 | 2,106 | 63.7 |
| Altair Engineering (software) | 7 | 1,795 | 62.0 |
| Marvell Technology (electronics) | 10 | 1,061 | 60.7 |
| SoftBank Group (telecommunications) | 9 | 1,247 | 60.3 |
| *C. Persistence within firms* | | | |
| Year-on-year correlation, innovation share | | | 0.88 |
| Year-on-year correlation, environmental share | | | 0.89 |
| Pairs of consecutive firm-years | | | 27,883 |

*Notes.* Estimation sample (35,772 firm-years of 7,128 firms, 2014–2023). Panel A: the NAICS-3 industries with at least 20 firms, by the mean innovation share of their firm-years; industries are classified by their historical NAICS codes, so that software publishers appear under the 2017 and 2022 codes of publishing. Panel B: firms with at least five firm-years and at least 1,000 openings a year on average, by the mean innovation share of their firm-years. Panel C: correlation between a firm's share in year $t$ and in year $t-1$, each measured as the deviation from the mean of its NAICS-3 industry and year, over pairs of consecutive firm-years.

**Table C12.** The Innovation Share and Tobin's $q$

| *A. Three comparisons* | Firm-years (1) | Firm means (2) | Fama–MacBeth (3) |
|---|---|---|---|
| Innovation share | 0.307*** | 0.394*** | 0.390*** |
| | (0.038) | (0.036) | [$t = 7.3$] |
| Observations | 24,264 | 3,325 | 10 years, all positive |
| *B. Controls added in turn, on a fixed sample* | Coefficient | (s.e.) | |
| Industry-by-year effects only | 0.296*** | (0.039) | |
| + Scale | 0.326*** | (0.039) | |
| + Balance sheet | 0.292*** | (0.039) | |
| + Performance | 0.314*** | (0.039) | |
| + Age and dividends | 0.311*** | (0.040) | |
| + R&D and SG&A | 0.055 | (0.037) | |
| + Intangible capital | 0.063* | (0.037) | |
| Firm-years | 23,197 | | |
| *C. Each expense alone, added to the controls through age and dividends* | Coefficient | (s.e.) | |
| + R&D | −0.018 | (0.038) | |
| + SG&A | 0.329*** | (0.038) | |
| + Knowledge capital (capitalized R&D) | 0.180*** | (0.039) | |
| + Organization capital (capitalized SG&A) | 0.325*** | (0.040) | |

*Notes.* Market sample: firms of Compustat North America with a CRSP security; Tobin's $q$ is assets plus market value minus common equity and deferred taxes, over assets, winsorized at 1 and 99 percent within year. The four standardized shares enter together and only the innovation share is reported. Panel A: (1) firm-years with the logarithms of assets, employees, and openings and NAICS-3-by-year effects; (2) firm means of $q$ on firm means of the shares and controls, with NAICS-2 effects; (3) Fama–MacBeth means of yearly cross-sections with NAICS-2 effects, with the $t$-statistic of the yearly coefficients. Panel B: on a fixed sample, controls added in turn to NAICS-3-by-year effects: the logarithms of assets, employees, and openings; leverage, property, plant, and equipment, and capital expenditure over assets; return on assets and sales growth; the logarithm of age and a dividend indicator; R&D over assets, an indicator of unreported R&D, and SG&A over assets; and the knowledge and organization capital of Peters and Taylor (2017) over assets. Panel C: on the same sample, each expense added alone to the controls through age and dividends. Standard errors clustered by firm in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C13.** The Innovation Benchmark

| | R&D over sales | | Log patent | Log value |
|---|---|---|---|---|
| | All firms | R&D reporters | applications | per patent |
| | (1) | (2) | (3) | (4) |
| *A. Across firms: firm means* | | | | |
| Correlation with the innovation share | 0.42 | 0.39 | 0.43 | -0.10 |
| Rank correlation | 0.67 | 0.70 | 0.50 | -0.09 |
| Innovation share | 0.146*** | 0.161*** | 0.431*** | 0.007 |
| | (0.008) | (0.010) | (0.029) | (0.031) |
| *Entered together:* | | | | |
| Innovation share | 0.229*** | 0.236*** | 0.101* | -0.189** |
| | (0.020) | (0.023) | (0.054) | (0.079) |
| Occupation-mix term | -0.096*** | -0.093*** | 0.388*** | 0.229*** |
| | (0.017) | (0.021) | (0.058) | (0.084) |
| Firms | 5,041 | 2,922 | 5,062 | 1,252 |
| | Patent applications $t+1$ to $t+2$ | Log R&D $t+1$ | Value per patent, $t+1$ | |
| *B. Across firm-years: the shares in year t* | | | | |
| Innovation share | 0.378*** | 0.017*** | 0.755 | |
| | (0.064) | (0.004) | (2.188) | |
| Environmental share | -0.124 | -0.008* | 6.563* | |
| | (0.097) | (0.005) | (3.863) | |
| Social share | 0.100 | 0.006 | -1.167 | |
| | (0.096) | (0.004) | (2.459) | |
| Governance share | -0.250*** | -0.001 | -1.866 | |
| | (0.076) | (0.004) | (2.238) | |
| *Innovation share beside the occupation-mix term:* | | | | |
| Innovation share | 0.178* | 0.012 | | |
| | (0.105) | (0.008) | | |
| Occupation-mix term | 0.203* | 0.005 | | |
| | (0.113) | (0.008) | | |
| Within firm (firm effects) | 0.082** | 0.024*** | -2.217 | |
| | (0.040) | (0.007) | (3.197) | |
| Firm-years | 22,404 | 14,819 | 4,125 | |

*Notes.* The innovation share is measured exactly as the environmental share (Section 3), from its own definition of innovation work. Panel A uses one observation per firm with at least three firm-years in the estimation sample. The firm's mean innovation share, standardized across firms, is related to its total R&D over its total sales across its years, capped at 2, with unreported R&D counted as zero in column (1) and the firms that never report R&D excluded in column (2); to the logarithm of one plus its mean number of patent applications per year; and to the logarithm of one plus the mean value of its granted patents (Kogan et al., 2017), for firms with a CRSP security and filing cohorts through 2020. The occupation-mix term is the firm's mean of the first term of equation (5), standardized. These regressions include NAICS-2 sector effects, with heteroskedasticity-robust standard errors. Panel B uses the firm-years from 2014 to 2023. The four standardized shares of year $t$ enter together, with NAICS-3-by-year effects and the logarithms of assets, employees, and openings; applications over $t+1$ to $t+2$ are estimated by Poisson pseudo-maximum likelihood, log R&D in $t+1$ by OLS with log R&D in $t$ as a control, and the value per patent in $t+1$ by OLS. The occupation-mix term, standardized within year, enters beside the innovation share in the rows that include it, and firm effects are added in the row so labelled. Standard errors clustered by firm are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C14.** Innovation Hiring, R&D, and Intangible Investment

| | NAICS-3-by-year effects | | | | Firm | Flow of $t$ | Firm- | Mean |
|---|---|---|---|---|---|---|---|---|
| | Inno-vation | Environ-mental | Social | Govern-ance | effects | as control | years | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *A. Year $t$, percent of assets at the end of $t-1$* | | | | | | | | |
| R&D | 3.962*** | −0.687*** | 0.361*** | 0.647*** | 0.294** | | 26,542 | 4.63 |
| | (0.153) | (0.125) | (0.124) | (0.147) | (0.134) | | | |
| Organization capital investment | −0.188** | −0.269*** | −0.325*** | −0.319*** | −0.048 | | 22,801 | 6.04 |
| | (0.095) | (0.076) | (0.085) | (0.100) | (0.093) | | | |
| Intangible investment | 3.420*** | −0.928*** | −0.078 | 0.019 | 0.252 | | 22,801 | 10.30 |
| | (0.205) | (0.188) | (0.167) | (0.198) | (0.183) | | | |
| *B. Year $t+1$, percent of assets at the end of $t$* | | | | | | | | |
| R&D | 3.860*** | −0.676*** | 0.415*** | 0.592*** | 0.319*** | 0.755*** | 25,393 | 4.29 |
| | (0.145) | (0.111) | (0.118) | (0.137) | (0.114) | (0.053) | | |
| Organization capital investment | −0.174* | −0.256*** | −0.321*** | −0.240** | 0.014 | −0.046* | 21,807 | 5.71 |
| | (0.089) | (0.064) | (0.081) | (0.095) | (0.075) | (0.026) | | |
| Intangible investment | 3.225*** | −0.902*** | −0.032 | 0.093 | 0.207 | 0.670*** | 21,807 | 9.69 |
| | (0.189) | (0.162) | (0.160) | (0.188) | (0.147) | (0.065) | | |

*Notes.* The sample is the North American firms of the estimation sample, for which Compustat's definitions of SG&A and R&D hold. Each row regresses a flow, in percent of assets at the end of the previous year, on the four standardized shares of year $t$, the logarithms of assets, employees, and openings, and NAICS-3-by-year effects, and columns (1) to (4) report the four shares of this regression. Column (5) adds firm effects and reports the innovation share. Column (6), for the flows of year $t+1$, adds the same flow of year $t$ as a control and reports the innovation share. Following Peters and Taylor (2017), R&D is Compustat R&D, counted as zero where it is not reported; organization capital investment is 0.3 times SG&A net of R&D, with SG&A used whole where R&D exceeds it; and intangible investment is their sum. Each flow is winsorized at the 1st and 99th percentiles within year. Column (7) gives the firm-years of column (1), and column (8) the mean of the dependent variable. Standard errors clustered by firm are in parentheses; with firm effects they count the industry-by-year effects in the degrees of freedom, as in Table C2. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

## C.6 Across the Four Kinds of Work

Table C15 shows where the social and governance shares are highest. Among the NAICS-3 industries with at least 20 firms, the social share is highest in mining, oil and gas extraction, and pipeline transportation, at 15 to 19 percent of openings, and lowest in car dealers, banking, and appliance retailers, below 2 percent. The governance share is highest in chemical manufacturing, which includes pharmaceuticals, at 29.1 percent, in securities, banking, and the manufacture of medical and other instruments, and lowest in nursing care and retail. Each of the four shares is also persistent within firms. Measured relative to its industry and year, a firm's share correlates from one year to the next at 0.89 for environmental work, 0.82 for social work, 0.86 for governance work, and 0.88 for innovation work.

Table C16 describes the four shares and the parts into which each splits, and Table C17 their correlations. Governance and innovation work are the most common, with means of 15.7 and 11.8 percent of openings, and environmental work the least common and the most right-skewed. The environmental share correlates with the social share, at 0.31 between firms, and only weakly with the others. Tables C18 and C19 report the regressions of every validation outcome on the four shares, which Section 4.5 summarizes. Table C20 decomposes the largest rises in each share into the change within the career sites that posted openings in both years and the contribution of the sites that entered or left (Section 4.6).

**Table C15.** Where the Social and Governance Shares Are Highest

| | Firm-years | Firms | Share (%) |
|---|---|---|---|
| *A. Social share: NAICS-3 industries with at least 20 firms* | | | |
| *The five with the highest share* | | | |
| Mining (except oil and gas) (212) | 495 | 120 | 19.0 |
| Oil and gas extraction (211) | 420 | 86 | 17.1 |
| Pipeline transportation (486) | 112 | 29 | 14.6 |
| Support activities for mining (213) | 353 | 67 | 13.3 |
| Petroleum and coal products (324) | 269 | 50 | 13.0 |
| *The three with the lowest share* | | | |
| Motor vehicle and parts dealers (441) | 222 | 39 | 1.9 |
| Credit intermediation (522) | 2,848 | 495 | 1.6 |
| Furniture, electronics and appliance retailers (2022 codes) (449) | 89 | 31 | 1.6 |
| All firms | 35,772 | 7,128 | 5.9 |
| *B. Governance share: NAICS-3 industries with at least 20 firms* | | | |
| *The five with the highest share* | | | |
| Chemical manufacturing (325) | 3,439 | 815 | 29.1 |
| Securities and investments (523) | 578 | 101 | 23.8 |
| Credit intermediation (522) | 2,848 | 495 | 22.4 |
| Miscellaneous manufacturing (339) | 771 | 152 | 22.0 |
| Pipeline transportation (486) | 112 | 29 | 21.9 |
| *The three with the lowest share* | | | |
| Nursing and residential care facilities (623) | 118 | 21 | 4.3 |
| Motor vehicle and parts dealers (441) | 222 | 39 | 4.2 |
| Furniture, electronics and appliance retailers (2022 codes) (449) | 89 | 31 | 3.7 |
| All firms | 35,772 | 7,128 | 15.7 |
| *C. Persistence within firms: year-on-year correlation* | | | |
| Environmental share | | | 0.89 |
| Social share | | | 0.82 |
| Governance share | | | 0.86 |
| Innovation share | | | 0.88 |
| Pairs of consecutive firm-years | | | 27,883 |

*Notes.* Estimation sample (35,772 firm-years of 7,128 firms, 2014–2023). Panels A and B: the NAICS-3 industries with at least 20 firms, by the mean share of their firm-years, classified by their historical NAICS codes. Panel C: correlation between a firm's share in year $t$ and in year $t-1$, each measured as the deviation from the mean of its NAICS-3 industry and year, over pairs of consecutive firm-years. The industries with the highest environmental share are in Table 6, and those with the highest innovation share in Table C11.

**Table C16.** The Four Shares and Their Parts

| | Mean | SD | P25 | Median | P75 | P90 | Firm-years |
|---|---|---|---|---|---|---|---|
| *A. The four shares (percent of openings)* | | | | | | | |
| Environmental share | 3.03 | 6.36 | 0.29 | 0.82 | 2.81 | 7.79 | 35,772 |
| Social share | 5.89 | 5.67 | 2.01 | 4.04 | 7.88 | 13.11 | 35,772 |
| Governance share | 15.70 | 11.14 | 7.30 | 13.34 | 21.37 | 31.01 | 35,772 |
| Innovation share | 11.82 | 12.79 | 2.42 | 7.27 | 16.95 | 29.48 | 35,772 |
| Openings (thousands) | 2.57 | 10.13 | 0.10 | 0.33 | 1.29 | 4.86 | 35,772 |
| *B. The parts of each share (percent of openings)* | | | | | | | |
| *Environmental* | | | | | | | |
| Clean technology | 1.44 | 4.60 | 0.05 | 0.20 | 0.85 | 2.89 | 35,772 |
| Environmental management | 1.59 | 4.10 | 0.14 | 0.41 | 1.35 | 3.80 | 35,772 |
| *Social* | | | | | | | |
| Worker safety | 3.37 | 5.05 | 0.35 | 1.27 | 4.30 | 9.42 | 35,772 |
| Product and patient safety | 1.21 | 2.59 | 0.06 | 0.30 | 0.99 | 3.27 | 35,772 |
| People and rights | 1.31 | 1.61 | 0.35 | 0.84 | 1.69 | 2.96 | 35,772 |
| *Governance* | | | | | | | |
| Quality | 6.50 | 9.06 | 0.45 | 2.55 | 9.19 | 18.60 | 35,772 |
| Risk and financial controls | 3.09 | 4.01 | 0.80 | 1.80 | 3.71 | 7.27 | 35,772 |
| Information security and privacy | 1.24 | 2.55 | 0.09 | 0.46 | 1.38 | 3.05 | 35,772 |
| Compliance and integrity | 4.88 | 5.41 | 1.41 | 3.01 | 6.29 | 11.70 | 35,772 |

*Notes.* The sample is the estimation sample of 35,772 firm-years of 7,128 firms from 2014 to 2023. Each share is the expected share of the firm-year's new openings whose responsibilities involve the work, in percent, measured as in Section 3. Panel B splits each share into its parts by crediting each opening's probability to the part whose classifier head predicts it most strongly, so that the parts of a share sum to the share. Innovation work has a single part.

**Table C17.** Correlations across the Four Shares

| | Environmental | Social | Governance | Innovation |
|---|---|---|---|---|
| Environmental | 1 | *0.21* | *0.06* | *0.10* |
| Social | 0.31 | 1 | *0.38* | *−0.01* |
| Governance | 0.05 | 0.40 | 1 | *0.00* |
| Innovation | 0.05 | −0.09 | 0.08 | 1 |
| Firms (firm-years) | 7,128 (35,772) | | | |

*Notes.* The sample is the estimation sample. Below the diagonal, the entries are correlations between firms, of the firms' mean shares over their years. Above the diagonal, in italics, they are correlations within firms, of the deviations of each firm-year's shares from the firm's means.

**Table C18.** Every Validation Outcome on the Four Shares: Environmental Records

| | Environmental share (1) | Social share (2) | Governance share (3) | Innovation share (4) | Firm-years (5) |
|---|---|---|---|---|---|
| *A. Records of environmental work* | | | | | |
| Y02 applications | 0.205*** | 0.093 | −0.162 | 0.392*** | 20,349 |
| | (0.049) | (0.110) | (0.115) | (0.071) | |
| ENV-TECH applications | 0.213*** | 0.107 | −0.238* | 0.373*** | 20,216 |
| | (0.050) | (0.112) | (0.126) | (0.072) | |
| Y02 or ENV-TECH applications | 0.203*** | 0.104 | −0.171 | 0.382*** | 20,584 |
| | (0.049) | (0.102) | (0.112) | (0.070) | |
| ENV-TECH outside Y02 | 0.214*** | 0.175* | −0.208 | 0.216* | 16,997 |
| | (0.082) | (0.105) | (0.166) | (0.126) | |
| IRA-class applications | 0.296*** | 0.099 | −0.137 | 0.177* | 18,461 |
| | (0.062) | (0.150) | (0.184) | (0.108) | |
| Green share of applications[a] | 0.076*** | −0.010** | 0.003 | −0.001 | 11,266 |
| | (0.007) | (0.005) | (0.004) | (0.003) | |
| Other applications | −0.218* | 0.111 | −0.256*** | 0.382*** | 22,400 |
| | (0.127) | (0.099) | (0.077) | (0.066) | |
| Change in log toxic releases, $t$ to $t+2$[a] | 0.002 | −0.049 | 0.059 | 0.088 | 4,327 |
| | (0.032) | (0.034) | (0.048) | (0.057) | |
| Scope 1 intensity, reported, $t+1$[a] | 0.137*** | 0.290*** | −0.092** | −0.159*** | 5,699 |
| | (0.041) | (0.047) | (0.042) | (0.036) | |
| Scope 1 intensity, estimated, $t+1$[a] | 0.120*** | 0.214*** | −0.079*** | −0.075*** | 24,443 |
| | (0.028) | (0.027) | (0.021) | (0.019) | |
| Environmental incidents | 0.142*** | 0.039 | −0.096* | 0.086 | 21,769 |
| | (0.030) | (0.034) | (0.050) | (0.065) | |
| Severe environmental incidents | 0.132*** | 0.065* | −0.083 | 0.113 | 20,805 |
| | (0.032) | (0.039) | (0.055) | (0.071) | |
| EPA enforcement cases | 0.179* | 0.303*** | −0.414*** | 0.018 | 3,896 |
| | (0.103) | (0.092) | (0.131) | (0.151) | |
| Environmental suits | −0.020 | 0.307*** | −0.270* | 0.356** | 6,682 |
| | (0.104) | (0.094) | (0.149) | (0.163) | |

*Notes.* The sample is the estimation sample of firm-years from 2014 to 2023 with each outcome. Each row reports one regression of the outcome on the four standardized shares of year $t$, entered together with the logarithms of assets, employees, and openings and NAICS-3-by-year effects, and columns (1) to (4) report the four coefficients. Counts are summed over $t+1$ to $t+2$ and estimated by Poisson pseudo-maximum likelihood, so that coefficients approximate proportional changes; outcomes marked [a] are estimated by OLS. The green share of applications is a fraction. Column (1) repeats column (1) of Table C2, and Table C19 reports the records of the other three kinds of work. Outcomes are defined in Table A1. Standard errors clustered by firm are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C19.** Every Validation Outcome on the Four Shares: Social, Governance, and Innovation Records

| | Environmental share (1) | Social share (2) | Governance share (3) | Innovation share (4) | Firm-years (5) |
|---|---|---|---|---|---|
| *B. Records of social work* | | | | | |
| Serious violations, programmed inspections | −0.061 | −0.107* | 0.197** | −0.126 | 3,358 |
| | (0.083) | (0.065) | (0.083) | (0.168) | |
| Injury rate (DART), $t+1$[a] | 0.043 | −0.116*** | −0.041 | −0.231*** | 10,160 |
| | (0.027) | (0.034) | (0.036) | (0.036) | |
| Programmed inspections | −0.022 | 0.075 | −0.020 | −0.747*** | 16,252 |
| | (0.062) | (0.058) | (0.073) | (0.120) | |
| Serious violations, triggered inspections | −0.104*** | 0.085* | −0.104** | −0.129** | 6,624 |
| | (0.033) | (0.044) | (0.047) | (0.065) | |
| Social incidents | 0.115*** | 0.129*** | −0.119** | 0.160*** | 22,360 |
| | (0.031) | (0.039) | (0.047) | (0.043) | |
| Severe social incidents | 0.090** | 0.102*** | −0.014 | 0.137*** | 21,846 |
| | (0.035) | (0.036) | (0.050) | (0.049) | |
| Employment suits | −2.764 | −0.546 | 0.473 | −0.325 | 1,665 |
| | (3.446) | (0.709) | (0.518) | (0.264) | |
| *C. Records of governance work* | | | | | |
| Non-standard audit opinion, $t+1$ (pp)[a] | 0.021 | 0.033 | 0.042 | 0.016 | 21,870 |
| | (0.043) | (0.034) | (0.033) | (0.033) | |
| Material weakness, $t+1$ (pp)[a] | 0.063 | −0.043 | 0.412 | −0.197 | 18,401 |
| | (0.400) | (0.387) | (0.311) | (0.340) | |
| Securities suits | 0.023 | −0.081 | 0.195*** | 0.127** | 16,682 |
| | (0.055) | (0.076) | (0.055) | (0.050) | |
| Class actions | 0.003 | −0.049 | 0.232*** | 0.178*** | 17,226 |
| | (0.056) | (0.057) | (0.054) | (0.049) | |
| Derivative and fraud suits | −0.372 | 0.066 | 0.305*** | 0.157 | 10,678 |
| | (0.227) | (0.122) | (0.095) | (0.128) | |
| Governance incidents | 0.097*** | −0.009 | 0.147** | 0.106** | 22,208 |
| | (0.035) | (0.046) | (0.060) | (0.053) | |
| Severe governance incidents | 0.093*** | 0.006 | 0.184*** | 0.094 | 21,848 |
| | (0.035) | (0.044) | (0.056) | (0.060) | |
| *D. Records of innovation work* | | | | | |
| Patent applications | −0.124 | 0.100 | −0.250*** | 0.378*** | 22,404 |
| | (0.097) | (0.096) | (0.076) | (0.064) | |
| Log R&D, $t+1$[a] | −0.008* | 0.006 | −0.001 | 0.017*** | 14,819 |
| | (0.005) | (0.004) | (0.004) | (0.004) | |
| Value per granted patent (KPSS), $t+1$[a] | 6.563* | −1.167 | −1.866 | 0.755 | 4,125 |
| | (3.863) | (2.459) | (2.238) | (2.188) | |

*Notes.* The specification is that of Table C18. Serious violations are regressed with the logarithm of the number of inspections of the same kind as a control, and log R&D in $t+1$ with log R&D in $t$. The audit outcomes are indicators times 100, so that their coefficients are in percentage points. Columns (2) and (3) repeat column (1) of Tables C8 and C10 for the social and governance records, and column (4) the innovation rows of Table C13, Panel B. Outcomes are defined in Table A1 and in the notes to Tables C8 and C10. Standard errors clustered by firm are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table C20.** Surges in the Shares and the Career Sites behind Them

| | Environmental (1) | Social (2) | Governance (3) | Innovation (4) |
|---|---|---|---|---|
| Firms with a surge | 48 | 108 | 136 | 88 |
| Share in the year before the surge, median (percent) | 3.9 | 5.7 | 15.1 | 11.4 |
| Rise in the share, median (percentage points) | 3.0 | 4.8 | 9.5 | 8.7 |
| Surges with at least half of the rise within continuing sites (percent) | 97.9 | 94.4 | 94.1 | 95.5 |
| Surges at firms with a single career site in both years (percent) | 81.2 | 73.1 | 69.1 | 69.3 |
| *Surges at firms with more than one career site:* | | | | |
| Number | 9 | 29 | 42 | 27 |
| At least half of the rise within continuing sites (percent) | 88.9 | 82.8 | 81.0 | 85.2 |
| Part of the rise within continuing sites, median (percent) | 100 | 100 | 100 | 99 |
| Openings at sites new in the surge year, mean (percent) | 9.5 | 9.1 | 16.1 | 11.6 |

*Notes.* A surge is a rise in a firm's share from $t-1$ to $t$ that lies in the top decile of that year's increases, at a firm with at least 50 openings in both years, after a change from $t-2$ to $t-1$ no larger in absolute value than the year's median absolute change, and with at least half of the rise still in place in $t+1$. The firm must be observed from $t-2$ to $t+1$, and only its first surge is counted. Each surge is decomposed into the change in the share within the career sites that posted openings in both years and the contribution of the sites that entered or left. At a firm whose openings come from a single career site in both years, the whole rise is within that site. Surges occur from 2016 to 2022.

# Appendix D. The Climate-Policy Days

This appendix describes the design of the tests in Sections 6 to 8. Internet Appendix IA.V.1 describes the days of policy news, and Internet Appendix Table IA.38 lists them.

## D.1 Abnormal Returns

For each firm and day, I estimate the loadings of the firm's daily excess return on the five factors of Fama and French (2015) and momentum (Carhart, 1997) over trading days −250 to −30, and I require at least 120 returns in this window. The abnormal return is the excess return minus the return predicted by the loadings, without the intercept. The cumulative abnormal return sums the abnormal returns over trading days −1 to +1, in percentage points. I winsorize it at the 1st and 99th percentiles of each day. The sample of each day contains the firms in Compustat North America with a CRSP security and an environmental share in the year before the event, between 1,264 and 2,717 firms per day.

## D.2 Placebo Tests

Returns on the same day are correlated across firms, and standard errors clustered by firm may overstate the precision of a single day's estimate. The placebo test compares the estimate on each event with estimates of the same regression on days without such news. For each event, the candidate days are all trading days within 24 months of the event that lie more than seven calendar days from any of the climate-policy and regulation days and from eight other days of major news.[29] Daily returns end on 31 December 2024, so that the candidate days of the 2024 election lie mostly before it. The candidate days number 486 for the 2024 election and between 894 and 977 for the other events. On each candidate day, I compute abnormal returns with the loadings estimated for the event itself, for the firms with returns on at least 98 percent of the days of the span. For a single event, the $p$-value is the share of candidate days on which the coefficient is at least as large in absolute value as on the event. For a pooled regression, I draw one candidate day per event, estimate the pooled regression on the drawn days with the signs of the actual events, and repeat this 20,000 times. The $p$-value is the share of draws whose coefficient is at least as large in absolute value as the actual one.

[29]The eight days concern other capabilities measured in the same way: the signing of the CHIPS and Science Act, the release of ChatGPT, Nvidia's results of May 2023, the COVID-19 crash of February 2020, the protests of June 2020, the Russian invasion of Ukraine, the failure of Silicon Valley Bank, and the approval of Nasdaq's board-diversity rule.

## D.3 The 2020 Election

The 2020 election is difficult to isolate. The networks called the race on Saturday, 7 November, and on Monday, 9 November, Pfizer and BioNTech announced the first results of their COVID-19 vaccine trial. The vaccine news moved returns across the cross-section. A firm's abnormal return on 16 November 2020, the day of Moderna's announcement, which carried no election news, is strongly related to its return around 9 November (0.71, $t = 8.7$). I therefore report the 2020 election but do not interpret its coefficient. Alternative windows and controls for the vaccine news are in Appendix Table E5.

## D.4 Analysts

Analysts' revisions compare the last monthly IBES consensus before each event with the first consensus after it, for the same fiscal period. The monthly consensus is computed in the middle of each month, so that the window around each event spans about a month. For the agreement on the IRA, the window runs from 14 July to 18 August 2022. It therefore also contains Senator Manchin's withdrawal and measures the net effect of the episode. Earnings announcements fall in most windows, and the placebo months of Appendix Table E26 share this feature.

# Appendix E. Additional Tables for the Application

This appendix reports the tables behind Sections 6 to 8, in the order in which the text cites them. Unless a table states otherwise, the dependent variable is the cumulative abnormal return over trading days −1 to +1 in percentage points, multiplied by +1 on favorable and −1 on unfavorable days when days are pooled; the shares are standardized among the firms with returns around each event; and every regression includes the event controls of equation (9) and NAICS-3-by-event effects, or NAICS-3 effects for a single day, with standard errors clustered by firm in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

## E.1 Main Results and Timing

Table E1 reports the day-by-day regressions behind the left panel of Figure 2, with the environmental share and with its two parts entered together, and the placebo $p$-value of each day, computed over all the candidate days of Appendix D.2.

**Table E1.** Policy-Day Returns by Day

| | Environmental share | | Clean technology | | Environmental management | |
|---|---|---|---|---|---|---|
| | Coefficient | Placebo $p$ | Coefficient | Placebo $p$ | Coefficient | Placebo $p$ |
| Paris Agreement, 2015 (+) | 0.232 | 0.411 | 0.308* | 0.205 | −0.110 | 0.336 |
| | (0.226) | | (0.181) | | (0.129) | |
| Trump elected, 2016 (−) | −0.820*** | 0.003 | −0.941*** | 0.001 | 0.114 | 0.375 |
| | (0.215) | | (0.162) | | (0.193) | |
| Paris withdrawal, 2017 (−) | −0.053 | 0.962 | 0.017 | 0.901 | −0.124* | 0.634 |
| | (0.120) | | (0.105) | | (0.075) | |
| Biden elected, 2020 (+) | −0.070 | 0.948 | 0.084 | 0.539 | −0.231 | 0.329 |
| | (0.199) | | (0.173) | | (0.199) | |
| Manchin withdraws, 2022 (−) | −0.241* | 0.377 | −0.239** | 0.420 | −0.064 | 0.530 |
| | (0.130) | | (0.119) | | (0.127) | |
| IRA agreement, 2022 (+) | 0.835*** | 0.008 | 0.803*** | 0.013 | 0.259** | 0.036 |
| | (0.139) | | (0.118) | | (0.110) | |
| Trump re-elected, 2024 (−) | −1.146*** | 0.004 | −1.141*** | 0.004 | −0.240 | 0.076 |
| | (0.291) | | (0.302) | | (0.184) | |

*Notes.* One regression per day of the unsigned CAR[−1,+1] on the environmental share (or its two parts), with the event controls and NAICS-3 effects. The sign in parentheses is the direction of the news. Placebo $p$: share of the candidate days of Appendix D.2 with a coefficient at least as large in absolute value.

Table E2 and Figure E1 document the dose-response of Section 6.2. The table reports the quintile coefficients that Figure 3 plots, the same quintiles formed from the share net

of its industry's mean, the slope per percentage point of the share without the firms with the most extreme shares, and the standardized coefficient when the share is winsorized at its 95th percentile or taken in logarithms. The figure shows the relation across twenty bins of the share, net of the controls and the industry-by-event effects.

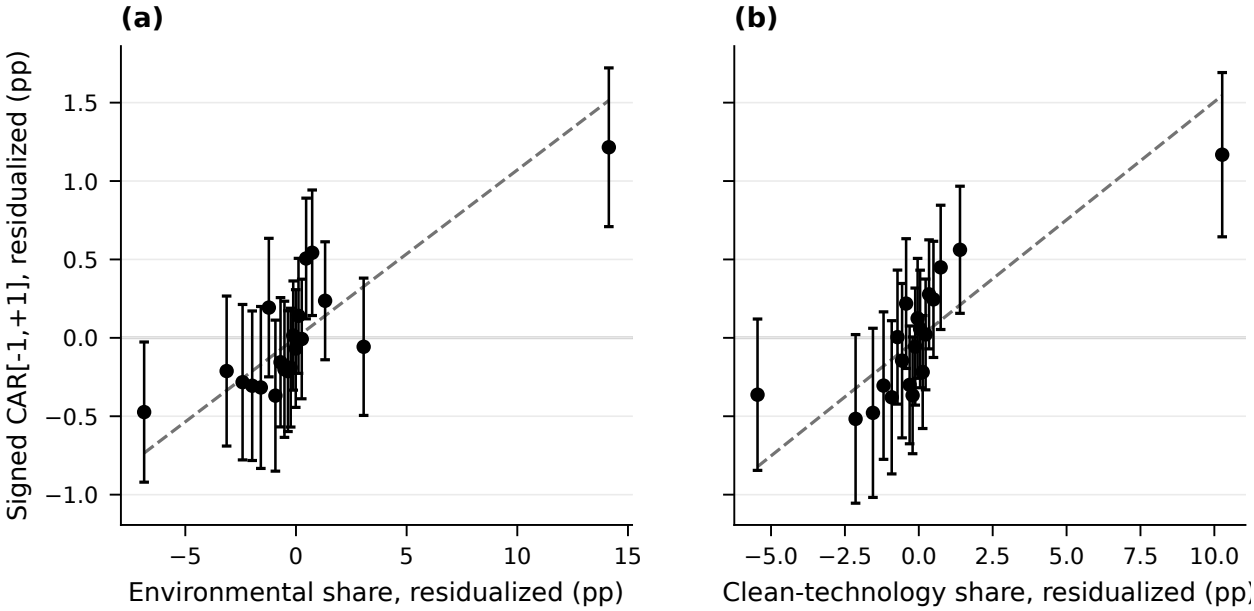



**Figure E1.** Binned Scatter of the Repricing. Twenty equal-count bins of the share net of the controls and industry-by-event effects, against the mean signed CAR[−1,+1] net of the same, on the six days; the line has the slope of the regression.

Table E3 reports the windows of Section 6.3, before and after the three-day window, with placebo $p$-values from the same candidate days.

Table E4 splits the return of the first session after the news on the four days on which the news arrived after the close, the elections of 2016 and 2024, Senator Manchin's withdrawal, and the agreement on the IRA, into the return from the previous close to the open and the return from the open to the close, from the opening and closing prices in CRSP. Because these returns are not measured against a factor model, they enter with the controls and NAICS-3 effects, which absorb the common movement of each industry, as the market-adjusted benchmark of Table E6 does.

Table E5 examines the 2020 election, which cannot be separated from the news about COVID-19 vaccines (Appendix D.3). It reports the two parts of the share over several windows, with and without two controls for the vaccine news: the firm's abnormal return on 16 November 2020, the day of Moderna's announcement, which carried no election news, and its cumulative abnormal return around the COVID-19 crash of 24 February 2020. The Moderna return is strongly related to the return around 9 November in every window that

**Table E2.** The Repricing along the Distribution of the Share

| | Environmental share (1) | Clean technology (2) |
|---|---|---|
| *Quintiles of the rank within the industry (lowest omitted)* | | |
| Quintile 2 | 0.110 | 0.310* |
| | (0.169) | (0.169) |
| Quintile 3 | 0.148 | 0.296* |
| | (0.167) | (0.164) |
| Quintile 4 | 0.259 | 0.531*** |
| | (0.164) | (0.165) |
| Quintile 5 (highest) | 0.670*** | 0.655*** |
| | (0.172) | (0.172) |
| *Quintiles of the industry-demeaned share (lowest omitted)* | | |
| Quintile 2 | 0.215 | 0.616*** |
| | (0.195) | (0.225) |
| Quintile 3 | −0.073 | 0.789*** |
| | (0.222) | (0.255) |
| Quintile 4 | 0.180 | 0.972*** |
| | (0.225) | (0.270) |
| Quintile 5 (highest) | 0.732*** | 1.037*** |
| | (0.191) | (0.211) |
| *Per percentage point of the share* | | |
| All firm-events | 0.107*** | 0.151*** |
| | (0.017) | (0.021) |
| Without the top 1 percent | 0.071*** | 0.128*** |
| | (0.019) | (0.029) |
| Without the top 5 percent | 0.088** | 0.243*** |
| | (0.036) | (0.065) |
| Without the bottom and top 5 percent | 0.071 | 0.346*** |
| | (0.046) | (0.098) |
| *Transformed share* | | |
| Winsorized at the 95th percentile, per pp | 0.178*** | 0.383*** |
| | (0.030) | (0.061) |
| $\log(1 + \text{share in pp})$ | 0.723*** | 0.981*** |
| | (0.121) | (0.137) |

*Notes.* Six days. Quintiles: indicators of quintiles of the share (the lowest omitted), formed from the share's rank within the firm's NAICS-3 industry on each day (industries with at least five firms) or from the share minus its industry mean within each day; column 2 holds the environmental management share fixed. Slopes: the share in percentage points; extremes are defined on the share net of the controls and industry-by-event effects, and the 95th percentile is that of each day.

**Table E3.** Returns before and after the News

| | Environmental share (1) | Placebo $p$ (2) | Clean technology (3) | Placebo $p$ (4) | Firm-events (5) |
|---|---|---|---|---|---|
| Before: days −10 to −2 | 0.168* | 0.641 | 0.087 | 0.831 | 12,407 |
| | (0.098) | | (0.093) | | |
| Before: days −5 to −2 | 0.090 | 0.911 | 0.013 | 0.769 | 12,407 |
| | (0.069) | | (0.062) | | |
| Event: days −1 to +1 | 0.592*** | <0.001 | 0.564*** | <0.001 | 12,407 |
| | (0.090) | | (0.070) | | |
| After: days +2 to +5 | 0.337*** | 0.043 | 0.243*** | 0.138 | 12,395 |
| | (0.066) | | (0.054) | | |
| After: days +2 to +20 | 0.258* | 0.634 | 0.166 | 0.866 | 12,352 |
| | (0.147) | | (0.121) | | |
| After: days +2 to +60 | 0.087 | 0.925 | 0.053 | 0.944 | 12,253 |
| | (0.281) | | (0.258) | | |

*Notes.* Six days; the CAR over the window named in each row, signed, with complete returns in the window. Placebo $p$: the same window on the candidate days of Appendix D.2, with 20,000 draws of one candidate day per event; candidate days end on 31 December 2024, and their windows use returns through June 2025.

**Table E4.** Overnight and Intraday Returns on the First Session after the News

| | Overnight (close to open) (1) | Intraday (open to close) (2) | Full session (3) | Firm-events (4) |
|---|---|---|---|---|
| 2016 election, 9 Nov 2016 (−) | −0.569*** | 0.076 | −0.510*** | 1,639 |
| | (0.116) | (0.103) | (0.168) | |
| Manchin withdraws, 15 Jul 2022 (−) | −0.159*** | −0.144** | −0.275*** | 2,713 |
| | (0.046) | (0.065) | (0.070) | |
| IRA agreement, 28 Jul 2022 (+) | 0.343*** | 0.243*** | 0.635*** | 2,715 |
| | (0.064) | (0.075) | (0.104) | |
| 2024 election, 6 Nov 2024 (−) | −0.800*** | −0.367*** | −1.134*** | 2,610 |
| | (0.144) | (0.122) | (0.211) | |
| The four days, signed | 0.450*** | 0.180*** | 0.629*** | 9,677 |
| | (0.071) | (0.052) | (0.085) | |

*Notes.* The first session after news that arrived after the close: 9 November 2016, 15 July 2022, 28 July 2022, and 6 November 2024. Overnight: from the previous close to the open, with distributions on the day; intraday: from the open to the close; full session: the CRSP daily return; in percent, winsorized at 1/99 within day. One regression per day, as observed, with the event controls and NAICS-3 effects; the last row pools the four days, signed, with NAICS-3-by-event effects.

contains that day, but not to the return of 4 to 6 November, before the networks called the race, and neither hiring share is significant at the 5 percent level in any window or with any control.

**Table E5.** The 2020 Election: Windows and the Vaccine News

| | No control | | Moderna announcement | | | COVID-19 crash | | |
|---|---|---|---|---|---|---|---|---|
| | Clean technology | Environ-mental management | Clean technology | Environ-mental management | AR, 16 Nov | Clean technology | Environ-mental management | CAR, 24 Feb |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| CAR[−1, +1], 6–10 Nov | 0.084 | −0.231 | −0.021 | −0.288 | 0.708*** | 0.081 | −0.238 | −0.015 |
| | (0.173) | (0.199) | (0.167) | (0.219) | (0.082) | (0.183) | (0.200) | (0.042) |
| CAR[0, +1], 9–10 Nov | −0.110 | −0.376* | −0.215 | −0.434* | 0.715*** | −0.087 | −0.387* | −0.039 |
| | (0.177) | (0.201) | (0.169) | (0.226) | (0.076) | (0.184) | (0.203) | (0.038) |
| CAR[0, +4], 9–13 Nov | 0.173 | −0.355* | 0.058 | −0.414* | 0.801*** | 0.179 | −0.366* | −0.073* |
| | (0.261) | (0.214) | (0.240) | (0.244) | (0.081) | (0.276) | (0.216) | (0.040) |
| 4–6 Nov, before the call | 0.237 | 0.059 | 0.244 | 0.062 | −0.057 | 0.163 | 0.070 | 0.036 |
| | (0.180) | (0.155) | (0.182) | (0.157) | (0.061) | (0.171) | (0.155) | (0.030) |

*Notes.* The day of 9 November 2020, the first trading day after the networks called the election on 7 November and the morning of Pfizer and BioNTech's vaccine announcement. Each row is a window of abnormal returns, as observed: trading days −1 to +1 (the window of Table 11), 0 to +1, 0 to +4, and 4 to 6 November, before the call. Each regression includes the clean-technology and environmental management shares, the event controls, and NAICS-3 effects. Columns (3) to (5) add the firm's abnormal return on 16 November 2020, the day of Moderna's vaccine announcement; columns (6) to (8) add its cumulative abnormal return over trading days −1 to +1 around 24 February 2020. Between 2,385 and 2,405 firm-events.

## E.2 Robustness

Table E6 reports the alternative return benchmarks, the price and size screens, the weighted regressions, the samples restricted by the number of openings, the sample reweighted to the population of listed firms, the sample without firm-specific news, and the regression with day-specific controls of Section 6.5. For the reweighting, the population of each day is every CRSP common stock listed on the NYSE, AMEX, or Nasdaq with a Compustat record at the end of the month before the event. A logit of inclusion in the sample on log market equity and its square, log book-to-market, log age, an NYSE indicator, and two-digit industry gives each firm's probability of inclusion, and each firm-event is weighted by the inverse of that probability, trimmed at its 1st percentile. The weights close about three quarters of the difference in log market equity between the sample and the population.

Table E7 drops one main day at a time and adds two further days of news about the policy. Day 0 of these days is 6 January 2021, the first session after the Georgia Senate runoffs of 5 January, which gave Democrats control of the Senate, and 20 December 2021, the first session after Senator Manchin said on 19 December that he could not vote for the Build Back Better Act. Both days also carried other news, since 6 January 2021 is the day of the

**Table E6.** Robustness: Returns, Samples, Weights, and Controls

| | Environmental share (1) | Placebo $p$ (2) | Clean technology (3) | Environmental management (4) | Firm-events (5) |
|---|---|---|---|---|---|
| *Return benchmark* | | | | | |
| Five factors and momentum (baseline) | 0.592*** | <0.001 | 0.564*** | 0.090 | 12,407 |
| | (0.090) | | (0.070) | (0.066) | |
| Market model | 0.508*** | <0.001 | 0.434*** | 0.132* | 12,407 |
| | (0.087) | | (0.072) | (0.069) | |
| Market-adjusted return | 0.520*** | <0.001 | 0.454*** | 0.139* | 12,407 |
| | (0.089) | | (0.074) | (0.072) | |
| *Sample and weights* | | | | | |
| Price at least $5 | 0.592*** | | 0.583*** | 0.130* | 11,254 |
| | (0.099) | | (0.080) | (0.067) | |
| Market equity above the NYSE 20th percentile | 0.551*** | | 0.552*** | 0.110 | 9,355 |
| | (0.112) | | (0.092) | (0.068) | |
| Both screens | 0.516*** | | 0.518*** | 0.103 | 9,197 |
| | (0.111) | | (0.092) | (0.066) | |
| Weighted by market equity | 0.224** | | 0.205** | 0.095 | 12,407 |
| | (0.108) | | (0.088) | (0.060) | |
| Weighted by the square root of market equity | 0.406*** | | 0.385*** | 0.113** | 12,407 |
| | (0.090) | | (0.076) | (0.052) | |
| *Openings in the formation year* | | | | | |
| At least 50 | 0.567*** | | 0.559*** | 0.078 | 10,836 |
| | (0.105) | | (0.090) | (0.069) | |
| At least 100 | 0.730*** | | 0.738*** | 0.128 | 9,293 |
| | (0.126) | | (0.110) | (0.079) | |
| At least 250 | 0.769*** | | 0.861*** | 0.059 | 7,002 |
| | (0.147) | | (0.125) | (0.086) | |
| Weighted by openings | 0.408*** | | 0.675*** | 0.034 | 12,407 |
| | (0.145) | | (0.133) | (0.094) | |
| *Selection* | | | | | |
| Reweighted to the population of listed firms | 0.556*** | | 0.467*** | 0.078 | 12,021 |
| | (0.078) | | (0.060) | (0.073) | |
| *Firm-specific news* | | | | | |
| Without earnings announcements in the window | 0.556*** | | 0.505*** | 0.113* | 10,835 |
| | (0.078) | | (0.062) | (0.060) | |
| *Controls* | | | | | |
| Each control with its own coefficient on each day | 0.573*** | | 0.570*** | 0.074 | 12,407 |
| | (0.095) | | (0.073) | (0.068) | |

*Notes.* Six days. Market model: excess return minus $\alpha + \beta$ times the market excess return, estimated on trading days −250 to −30; market-adjusted: return minus the market return, which with NAICS-3-by-event effects equals using raw returns. Price and market equity at the end of the month before the event. Weights sum to one within each day. Controls: each of the eight event controls interacted with an indicator for each day, so that the controls take 48 coefficients instead of eight.

attack on the Capitol and the week of 20 December 2021 brought news about the Omicron variant.

**Table E7.** Robustness: The Choice of Days

| | Environmental share (1) | Clean technology (2) | Environmental management (3) | Firm-events (4) |
|---|---|---|---|---|
| *Six days without one of them, signed* | | | | |
| Without the Paris Agreement | 0.630*** | 0.598*** | 0.105 | 11,143 |
| | (0.094) | (0.074) | (0.071) | |
| Without the 2016 election | 0.553*** | 0.506*** | 0.101 | 10,766 |
| | (0.096) | (0.079) | (0.069) | |
| Without the Paris withdrawal | 0.677*** | 0.677*** | 0.086 | 10,638 |
| | (0.111) | (0.085) | (0.077) | |
| Without the 2020 election | 0.753*** | 0.702*** | 0.146** | 10,002 |
| | (0.106) | (0.089) | (0.070) | |
| Without the IRA agreement | 0.505*** | 0.488*** | 0.033 | 9,690 |
| | (0.100) | (0.080) | (0.085) | |
| Without the 2024 election | 0.407*** | 0.408*** | 0.060 | 9,796 |
| | (0.082) | (0.062) | (0.069) | |
| *Two further days of policy news* | | | | |
| Georgia runoffs, 6 Jan 2021 (+), as observed | 0.894*** | 0.753*** | 0.398*** | 2,390 |
| | (0.180) | (0.153) | (0.152) | |
| Build Back Better ends, 20 Dec 2021 (−), as observed | −0.534*** | −0.420*** | −0.276** | 2,368 |
| | (0.123) | (0.105) | (0.114) | |
| The two days, signed | 0.771*** | 0.660*** | 0.317*** | 4,758 |
| | (0.136) | (0.116) | (0.107) | |
| The eight days, signed | 0.653*** | | | 17,165 |
| | (0.087) | | | |

*Notes.* Upper rows: the six days without the named day, signed. Lower rows: the two further days of policy news, with the shares of 2020, as observed (one day) or signed (pooled); the eight days add them to the six.

Table E8 asks whether the environmental share simply moves with the market, which rose on each of the six days of climate-policy news. On the days of the largest market moves of each year, listed in Internet Appendix Table IA.40, the coefficient of the share is close to zero, and on the 20 largest moves of the period it is negative.

Table E9 re-estimates the headline coefficients of the paper under two alternative rules for keeping postings. The first keeps every posting removed by 30 September 2023, however long it stayed open, instead of only those open at most 180 days. The second also adds the postings still open on that date, dated 2023. Each estimate uses the specification of the table in which it is reported, with the shares, and the number of openings among the controls, recomputed under the rule. On the policy days, the firm-events are held fixed.

**Table E8.** Repricing on Climate-Policy Days and on Days of Large Market Moves

| | Climate-policy | Days of large market moves | | | |
|---|---|---|---|---|---|
| | days | All | Rises | Falls | Largest |
| | (1) | (2) | (3) | (4) | (5) |
| Environmental share | 0.592*** | 0.018 | 0.034 | 0.002 | -0.117*** |
| | (0.090) | (0.024) | (0.037) | (0.032) | (0.035) |
| Days | 6 | 40 | 20 | 20 | 20 |
| Firm-events | 12,407 | 89,247 | 44,612 | 44,635 | 47,047 |
| Environmental share, (1) minus (2) | | 0.664*** | | | |
| | | (0.092) | | | |

*Notes.* Column (1) is the six-day estimate of Table 11. Columns (2) to (5) apply the same regression to days of large market moves in 2015–2024, each more than seven calendar days from any climate-policy day and from the other days selected: in each year, the two largest rises and the two largest falls of the market excess return (2), the rises (3) and the falls (4) among them, and the 20 largest moves of the period (5). On these days the return is multiplied by +1 when the market rose and by −1 when it fell. The last row is the difference between (1) and (2), from a regression that stacks the two samples and interacts the share with an indicator of climate-policy days. With standard errors clustered by firm and day, the coefficient in (2) is 0.018 (0.049).

The coefficients keep their sign and their size. On the six policy days, the coefficient of the environmental share is 0.592 under the baseline rule, 0.599 without the duration limit, and 0.601 with the open postings added, and most other coefficients move by less than 4 percent. The largest changes are in the regressions for a first green application, which rest on few firms, from 1.64 to 1.30 percentage points for a first Y02 application and from 1.85 to 1.63 for a first IRA-class application. The coefficient for a first Y02 application and the coefficient for other applications, both significant at the 10 percent level under the baseline rule, fall just short of it under the alternative rules.

**Table E9.** The Main Results under Alternative Rules for Keeping Postings

| | Baseline | Any duration | Any duration, open postings added |
|---|---|---|---|
| *A. Policy-day returns: the environmental share (Table 11)* | | | |
| Six days | 0.592*** (0.090) | 0.599*** (0.091) | 0.601*** (0.090) |
| Seven days | 0.527*** (0.081) | 0.531*** (0.081) | 0.532*** (0.080) |
| Favorable days | 0.386*** (0.109) | 0.386*** (0.108) | 0.386*** (0.108) |
| Unfavorable days | 0.762*** (0.136) | 0.777*** (0.137) | 0.789*** (0.136) |
| Elections | 0.665*** (0.145) | 0.671*** (0.144) | 0.678*** (0.144) |
| Other days | 0.498*** (0.092) | 0.499*** (0.094) | 0.499*** (0.094) |
| *B. One outcome of each kind of work (Section 4)* | | | |
| Environmental share: Y02 applications | 0.205*** (0.049) | 0.204*** (0.049) | 0.204*** (0.049) |
| Social share: injury rate in the firm's states | 0.089*** (0.018) | 0.088*** (0.018) | 0.088*** (0.018) |
| Governance share: securities suits | 0.195*** (0.055) | 0.195*** (0.055) | 0.196*** (0.055) |
| Innovation share: R&D over lagged assets (pp) | 3.96*** (0.15) | 4.08*** (0.15) | 4.11*** (0.15) |
| *C. Green technology: the environmental share (Table 7)* | | | |
| Y02 applications | 0.205*** (0.049) | 0.204*** (0.049) | 0.204*** (0.049) |
| IRA-class applications | 0.296*** (0.062) | 0.296*** (0.062) | 0.296*** (0.062) |
| Environmental technologies outside Y02 | 0.214*** (0.082) | 0.220*** (0.081) | 0.220*** (0.081) |
| Green share of applications | 0.076*** (0.007) | 0.077*** (0.007) | 0.077*** (0.007) |
| Other applications | −0.218* (0.127) | −0.201 (0.125) | −0.201 (0.125) |
| First Y02 application (pp) | 1.64* (0.90) | 1.30 (0.82) | 1.30 (0.82) |
| First IRA-class application (pp) | 1.85** (0.80) | 1.63** (0.76) | 1.63** (0.76) |
| *D. Credits and clean facilities: the environmental share (Table 15)* | | | |
| Any clean credit (pp) | 6.81*** (1.17) | 6.87*** (1.15) | 6.87*** (1.15) |
| Manufacturing credit, 45X (pp) | 3.27*** (0.96) | 3.32*** (0.96) | 3.32*** (0.96) |
| Number of credits | 0.157*** (0.035) | 0.156*** (0.034) | 0.156*** (0.034) |
| Any facility (pp) | 1.93** (0.90) | 2.03** (0.90) | 2.03** (0.90) |
| Any manufacturing facility (pp) | 3.13*** (0.90) | 3.12*** (0.90) | 3.12*** (0.90) |
| Announced capex (Poisson) | 0.202** (0.080) | 0.200** (0.081) | 0.200** (0.081) |

*Notes.* Coefficients per standard deviation of the share, with standard errors in parentheses, from the specification of the table cited in each panel; standard errors are clustered by firm, except in Panel D, where each firm is one observation and they are robust to heteroskedasticity. Baseline: postings removed by 30 September 2023 and open at most 180 days, dated by the year of their removal. Any duration: postings removed by 30 September 2023, however many days they stayed open. Any duration, open postings added: in addition, the postings still open on 30 September 2023, dated 2023. Under each rule, the shares and the number of openings are recomputed, the shares are standardized as in the baseline, and the estimation sample keeps the firm-years with at least 20 openings. Panel A holds the baseline firm-events fixed. Panel B, social share: the injury rate at the firm's establishments in a state, across the firm's states in the same year. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

## E.3 Is It the Capability?

Table E10 reports the composition check of Section 6.6, with each of the four shares alone and with the four together.

**Table E10.** The Composition of Hiring: The Four Shares Alone and Together

| | Environmental share (1) | Social share (2) | Governance share (3) | Innovation share (4) |
|---|---|---|---|---|
| Each share alone | 0.592*** | −0.096 | −0.097 | 0.112 |
| | (0.090) | (0.074) | (0.068) | (0.081) |
| The four shares together | 0.612*** | −0.161** | −0.040 | 0.071 |
| | (0.092) | (0.079) | (0.071) | (0.080) |
| Firm-events | 12,407 | 12,407 | 12,407 | 12,407 |

*Notes.* Six days. First row: each share enters alone, in four regressions; the environmental share alone is the estimate of Table 11, column 1. Second row: the four shares enter together.

Tables E11 and E12 report the regressions that Figure 6 summarizes, beside each existing measure and all of them, and in the samples in which some of these measures cannot drive the result, and Table E13 reports the comparison with the clean business lines that firms report. Internet Appendix Table IA.43 describes the segments.

**Table E11.** The Environmental Share beside Existing Measures

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| Environmental share | 0.592*** | 0.588*** | 0.500*** | 0.587*** | 0.566*** | 0.421*** | 0.585*** | 0.351*** | 0.259** |
| | (0.090) | (0.091) | (0.092) | (0.091) | (0.093) | (0.093) | (0.091) | (0.097) | (0.116) |
| Rating and carbon | | Yes | | | | | | Yes | Yes |
| Climate talk | | | Yes | | | | | Yes | Yes |
| Intangible capital | | | | Yes | | | | Yes | Yes |
| Green patents | | | | | Yes | | | Yes | Yes |
| Clean revenue | | | | | | Yes | | Yes | Yes |
| Tax and trade | | | | | | | Yes | Yes | Yes |
| Industry × event | NAICS-3 | NAICS-3 | NAICS-3 | NAICS-3 | NAICS-3 | NAICS-3 | NAICS-3 | NAICS-3 | NAICS-6 |
| Firm-events | 12,407 | 12,407 | 12,407 | 12,407 | 12,407 | 12,407 | 12,407 | 12,407 | 11,121 |

*Notes.* The six days of Table 11 (NAICS-6-by-event effects in column 9). Columns 2 to 7 add, in turn: the LSEG environmental score and carbon intensity (the logarithm of Scope 1 and 2 emissions over market equity); the opportunity, regulatory, and physical climate exposures of Sautner et al. (2023*a*); SG&A and R&D over assets and the knowledge and organization capital of Peters and Taylor (2017) over assets; an indicator of any Y02 or IRA-class application in 2007–2013, the firm's pre-sample mean applications in these classes, and its applications over $t-2$ to $t$; the revenue share of Compustat business segments that sell clean-technology products and an indicator of any such segment; and the cash effective tax rate, the foreign share of sales, and the foreign share of pretax income (Wagner, Zeckhauser and Ziegler, 2018; Ramelli et al., 2021). Columns 8 and 9 include all of them. Controls that are not observed for every firm are standardized within event, set to zero where missing, and accompanied by an indicator of missing values, so that every column keeps the sample of column 1.

**Table E12.** The Environmental Share in Subsamples and by Occupation

| | Complete cases | Complete cases, all controls | No green patent 2007–13 | Strict newcomers | No clean segment | Elections only | Elections, tax and trade | Occup. mix | Within occup. |
|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Environmental share | 0.463*** | 0.334** | 0.626*** | 0.635*** | 0.336*** | 0.665*** | 0.644*** | 0.161** | 0.483*** |
| | (0.162) | (0.148) | (0.139) | (0.162) | (0.097) | (0.145) | (0.146) | (0.080) | (0.082) |
| Firm-events | 5,688 | 5,688 | 9,516 | 9,100 | 12,150 | 6,657 | 6,657 | 12,394 | 12,394 |

*Notes.* The specification of Table E11, column 1. (1) and (2) the firm-events for which every control is observed, without and with the controls; (3) firms with no Y02 or IRA-class application in 2007–2013; (4) the firms of column 3 outside utilities and the tenth of NAICS-6 industries with the highest 2007–2013 Y02 share of applications; (5) firms that report no clean-technology segment; (6) and (7) the three elections, without and with the tax and trade controls; (8) and (9) the two parts of the environmental share in equation (5), the part that the firm's mix of occupations predicts and the part within occupations, each standardized and both in one regression.

**Table E13.** Environmental Hiring beside Reported Clean Business

| | Environmental share | Clean technology | Environmental management | Observations |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Segment sample | 0.653*** | 0.616*** | 0.101 | 12,093 |
| | (0.081) | (0.061) | (0.065) | |
| Beside clean-segment revenue | 0.488*** | 0.471*** | 0.085 | 12,093 |
| | (0.080) | (0.063) | (0.063) | |
| Clean-segment revenue share (per SD) | 0.206** | 0.168** | | 12,093 |
| | (0.082) | (0.083) | | |
| Firms reporting no clean segment | 0.393*** | 0.471*** | 0.065 | 11,836 |
| | (0.084) | (0.072) | (0.063) | |

*Notes.* Six days. Column 1: regressions on the environmental share; columns 2 and 3: regressions on its two parts, each standardized. Firms with Compustat Segments data; a clean segment sells products such as renewable power, solar and wind equipment, electric vehicles, batteries, or biofuels. The second row adds the standardized clean-segment revenue share, whose coefficient is shown below it, and an indicator of any clean segment.

Table E14 compares the environmental share with environmental ratings and carbon intensity on eight days, the six main days and two days of news about the regulation of emissions, as observed rather than signed. Panel B estimates the responses on the firms with both an LSEG score and Trucost emissions, with the share, the score, and carbon intensity in the same regression. Holding the share and emissions fixed, the LSEG score has a negative coefficient on six of the eight days, significant at the 10 percent level or better on all but the 2024 election, and positive and insignificant coefficients on the withdrawal from the Paris Agreement and the 2020 election, so that the score falls on days of both directions. The sample of Panel B, from 660 to 2,309 firms, holds firms larger than the average. Columns (7) and (8) are the regulation days of Section 7.2: on 30 June 2022, the Supreme Court limited the authority of the Environmental Protection Agency to regulate carbon dioxide emissions

from existing power plants (Supreme Court of the United States, 2022), and on 6 March 2024, the Securities and Exchange Commission adopted rules on climate-related disclosure (U.S. Securities and Exchange Commission, 2024). Beside the environmental management share, the clean-technology share moves on neither day. Internet Appendix Table IA.41 reports the estimates of the score and of carbon intensity alone that Figure 2 plots.

**Table E14.** The Environmental Share, Ratings, and Carbon Intensity on Eight Days of Policy News

| | Climate-policy days | | | | | | Regulation days | |
|---|---|---|---|---|---|---|---|---|
| | Paris 2015 (1) | Trump 2016 (2) | Paris exit 2017 (3) | Biden 2020 (4) | IRA 2022 (5) | Trump 2024 (6) | W.Va. v. EPA 2022 (7) | SEC rule 2024 (8) |
| Direction of the news | + | − | − | + | + | − | − | + |
| *A. The environmental share alone* | | | | | | | | |
| Environmental share | 0.232 | −0.820*** | −0.053 | −0.070 | 0.835*** | −1.146*** | −0.188* | −0.035 |
| | (0.226) | (0.215) | (0.120) | (0.199) | (0.139) | (0.291) | (0.110) | (0.153) |
| Firms | 1,264 | 1,641 | 1,769 | 2,405 | 2,717 | 2,611 | 2,726 | 2,697 |
| *B. Beside the LSEG score and carbon intensity* | | | | | | | | |
| Environmental share | −0.008 | −0.140 | −0.103 | 0.081 | 0.844*** | −1.354*** | −0.118 | −0.013 |
| | (0.285) | (0.358) | (0.098) | (0.224) | (0.153) | (0.254) | (0.123) | (0.176) |
| LSEG score | −0.313** | −0.446* | 0.019 | 0.235 | −0.283** | −0.105 | −0.412*** | −0.248* |
| | (0.158) | (0.253) | (0.090) | (0.221) | (0.142) | (0.216) | (0.125) | (0.133) |
| Carbon intensity | 0.048 | −1.090*** | −0.334*** | −1.923*** | 0.285 | 0.654** | 0.075 | −0.075 |
| | (0.228) | (0.327) | (0.120) | (0.367) | (0.180) | (0.283) | (0.164) | (0.188) |
| Firms | 660 | 835 | 1,487 | 2,105 | 2,300 | 1,977 | 2,309 | 2,037 |
| *C. The two parts* | | | | | | | | |
| Clean-technology share | 0.308* | −0.941*** | 0.017 | 0.084 | 0.803*** | −1.141*** | −0.130 | −0.044 |
| | (0.181) | (0.162) | (0.105) | (0.173) | (0.118) | (0.302) | (0.107) | (0.156) |
| Environmental management share | −0.110 | 0.114 | −0.124* | −0.231 | 0.259** | −0.240 | −0.130 | 0.011 |
| | (0.129) | (0.193) | (0.075) | (0.199) | (0.110) | (0.184) | (0.090) | (0.112) |
| Firms | 1,264 | 1,641 | 1,769 | 2,405 | 2,717 | 2,611 | 2,726 | 2,697 |

*Notes.* CAR$[-1,+1]$ in percentage points per standard deviation, as observed rather than signed. Direction: the sign of the news, as in Table IA.38; on the regulation days, it refers to the expected demand for environmental management work. Panel A: all firm-events, as in Table E1. Panel B: firm-events with both an LSEG environmental score and Trucost emissions; carbon intensity is the logarithm of Scope 1 and 2 emissions over market equity (Ramelli et al., 2021); the score and carbon intensity are standardized within event. Panel C: the clean-technology and environmental management shares entered together.

Table E15 enters the emission measures of Bolton and Kacperczyk (2021, 2023) beside the share on the five main days that Trucost covers, and estimates the clean-technology share within terciles of emissions in each industry and day (Section 6.6).

Table E16 restates the event study as portfolios sorted on one characteristic at a time, without controls. On each of the six days, firms are sorted into terciles of each characteristic of the preceding year within NAICS-3 industries, and the return of the portfolio that is long the top tercile and short the bottom tercile is signed by the direction of the news and averaged over the six days. Unlike the hedge portfolios of Engle et al. (2020), which track a monthly series of climate news out of sample, these portfolios are evaluated on the days on

**Table E15.** The Environmental Share beside the Emission Measures of Carbon Risk

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| *A. The share beside the emission measures* | | | | | |
| Environmental share | 0.660*** | 0.678*** | | | |
| | (0.099) | (0.101) | | | |
| Clean technology | | | 0.646*** | 0.661*** | |
| | | | (0.077) | (0.079) | |
| Environmental management | | | 0.099 | 0.106 | |
| | | | (0.072) | (0.071) | |
| Log scope 1 emissions | | 0.432 | | 0.432 | 0.312 |
| | | (0.333) | | (0.333) | (0.338) |
| Log scope 2 emissions | | −0.176 | | −0.128 | −0.115 |
| | | (0.148) | | (0.147) | (0.150) |
| Log scope 1 intensity | | −0.427* | | −0.444* | −0.235 |
| | | (0.254) | | (0.253) | (0.258) |
| Growth of scope 1 emissions | | 0.068 | | 0.067 | 0.053 |
| | | (0.066) | | (0.066) | (0.066) |
| Growth unknown | | −0.080 | | −0.076 | −0.068 |
| | | (0.185) | | (0.185) | (0.185) |
| Emissions disclosed | | 0.315 | | 0.341 | 0.381* |
| | | (0.213) | | (0.212) | (0.215) |
| Firm-events | 9,571 | 9,571 | 9,571 | 9,571 | 9,571 |
| *B. Clean technology within terciles of emissions (low, middle, high)* | | | | | |
| Terciles of scope 1 emissions | 0.580*** | 0.768*** | 0.696*** | | |
| | (0.167) | (0.189) | (0.151) | | |
| Firm-events | 2,985 | 3,111 | 3,258 | | |
| Terciles of scope 1 intensity | 0.713*** | 0.665*** | 0.584*** | | |
| | (0.261) | (0.198) | (0.112) | | |
| Firm-events | 2,935 | 3,184 | 3,239 | | |

*Notes.* Signed CAR$[-1,+1]$ on the five main days other than the Paris Agreement, which Trucost does not cover, for the firm-events with positive Trucost scope 1 emissions, with the controls and effects of Table 11. Panel A: the environmental share without and with the emission measures (columns 1 and 2), its two parts entered together without and with them (3 and 4), and the measures alone (5). The measures are the logarithms of scope 1 and of one plus scope 2 emissions, the logarithm of one plus scope 1 emissions per unit of revenue, and the log change of scope 1 emissions from the previous fiscal year, winsorized at 2.5/97.5 and set to zero with an indicator where unknown, each standardized within event, and an indicator that the firm reports its scope 1 emissions rather than Trucost estimating them. Panel B: the clean-technology share, entered with the environmental management share, within terciles of scope 1 emissions or of their intensity in each NAICS-3 industry and day. Standard errors clustered by firm in parentheses. *, **, ***: 10, 5, 1 percent.

which the news is known.

Table E17 reports the tests of interest-rate, oil, and market exposure. For each firm and day, the rate beta and the oil beta are the coefficients on the daily change in the 10-year Treasury yield, in basis points, and on the daily return of Brent crude oil in a regression of the firm's daily excess return on the factors of Fama and French (2015), momentum (Carhart, 1997), and both series, over trading days −250 to −30, and the market beta is the factor beta of the event controls. Each exposure, and then all three, enters interacted with an indicator for each day, so that its price can differ from day to day. Over the six windows, the 10-year yield changed by +4, +32, −6, +19, −13, and 0 basis points, and the rate beta is priced significantly on the 2020 and 2024 elections, so that the control has power.

**Table E16.** Portfolio Sorts on the Policy Days

| | Equal-weighted | | | Value-weighted | | |
|---|---|---|---|---|---|---|
| Long–short portfolio sorted on | Return | Placebo $p$ | Days | Return | Placebo $p$ | Days |
| Environmental share of hiring | 0.53*** | <0.001 | 4/6 | 0.31 | 0.101 | 4/6 |
| Clean-technology share of hiring | 0.53*** | <0.001 | 5/6 | 0.04 | 0.845 | 2/6 |
| Climate talk on earnings calls | 0.44*** | <0.001 | 4/6 | 0.16 | 0.451 | 3/6 |
| LSEG environmental score | 0.41*** | 0.008 | 5/6 | 0.41* | 0.051 | 5/6 |
| Low carbon intensity | 0.07 | 0.658 | 3/6 | −0.06 | 0.793 | 4/6 |

*Notes.* On each of the six days, firms are sorted into terciles of each characteristic within NAICS-3 industries that have at least three firms with the characteristic; the portfolio is long the side expected to gain from climate policy (higher share, score, or talk; lower carbon intensity) and short the other. Return: the portfolio's cumulative abnormal return over the three days around the event, in percent, signed and averaged over the six days. Placebo $p$: the same portfolios on 20,000 draws of one placebo day per event (Appendix D.2), two-sided. Days: the number of the six days with a positive signed return. Value weights: market equity at the fiscal year-end. Climate talk is the overall exposure of Sautner et al. (2023*a*). *, **, and *** denote placebo $p$-values below 0.10, 0.05, and 0.01.

**Table E17.** Interest Rates, Oil, and the Market on the Policy Days

| | Baseline (1) | + Rate beta × day (2) | + Oil beta × day (3) | + Market beta × day (4) | All three (5) |
|---|---|---|---|---|---|
| Environmental share | 0.592*** | 0.584*** | 0.569*** | 0.574*** | 0.551*** |
| | (0.090) | (0.089) | (0.091) | (0.090) | (0.090) |
| Clean technology | 0.564*** | 0.561*** | 0.554*** | 0.549*** | 0.541*** |
| | (0.070) | (0.069) | (0.070) | (0.069) | (0.068) |
| Environmental management | 0.090 | 0.084 | 0.065 | 0.090 | 0.067 |
| | (0.066) | (0.065) | (0.068) | (0.067) | (0.066) |
| Firm-events | 12,407 | 12,407 | 12,407 | 12,407 | 12,407 |

*Notes.* Six days, signed. Rate and oil betas: coefficients on the daily change in the 10-year Treasury yield (basis points) and on the daily Brent return (percent) in a regression of the firm's daily excess return on the five factors, momentum, and both series over trading days −250 to −30; winsorized at 1/99 and standardized within day. Market beta: the factor beta of the event controls, standardized within day. Columns 2 to 5 interact the standardized betas with an indicator for each day. The rows of the two parts come from regressions on both parts. Regressed on the share with the event controls and NAICS-3-by-event effects, the standardized rate beta has a coefficient of −0.030 (0.012) and the oil beta 0.097 (0.015).

Table E18 reports the two midterm elections of Section 6.6: those of 2018, which changed the majority in the House but carried no identifiable news about climate policy and serve as a placebo, and those of 2022, whose results also bore on the risk to the IRA. Returns are computed as on the policy days, over the three-day window and on each of its days, without and with the firm's rate and oil betas as controls. On both days the polls closed after the markets, so the first session after the results is day 0, the window that identifies the election. On 10 November 2022, the day after, the October consumer price index came in below expectations and long-term yields fell sharply.

**Table E18.** The Midterm Elections of 2018 and 2022

| | Environmental share (1) | With rate and oil betas (2) | Firm-events (3) |
|---|---|---|---|
| *2018 midterm elections (results on the night of 6 Nov)* | | | |
| CAR$[-1,+1]$ | $-0.142$ | $-0.140$ | 2,068 |
| | (0.144) | (0.144) | |
| Election day, before the results | $-0.034$ | $-0.035$ | 2,068 |
| | (0.082) | (0.081) | |
| First session after the results | 0.016 | 0.015 | 2,068 |
| | (0.082) | (0.084) | |
| Following session | $-0.104^{*}$ | $-0.100^{*}$ | 2,068 |
| | (0.058) | (0.058) | |
| *2022 midterm elections (results on the night of 8 Nov)* | | | |
| CAR$[-1,+1]$ | $0.511^{**}$ | $0.467^{**}$ | 2,684 |
| | (0.217) | (0.213) | |
| Election day, before the results | $0.264^{***}$ | $0.253^{***}$ | 2,684 |
| | (0.099) | (0.097) | |
| First session after the results | 0.088 | 0.127 | 2,684 |
| | (0.095) | (0.099) | |
| Following session | 0.178 | 0.114 | 2,684 |
| | (0.128) | (0.127) | |

*Notes.* The 2018 and 2022 midterm elections; day 0 is 7 November 2018 and 9 November 2022. Abnormal returns from the five factors and momentum, as observed, on the environmental share of the year before, with the event controls and NAICS-3 effects. Column 2 adds the firm's rate and oil betas (Table E17).

Table E19 uses the product markets that Hoberg and Phillips (2010, 2016) define from the similarity of firms' product descriptions in their 10-K filings (TNIC-3). A firm's rivals on a given day are its TNIC-3 rivals in the year before the day that have an environmental share, and their exposure is the mean of their shares, weighted by product similarity and standardized within event like the firm's own share. Of the firm-events of Table 11, 10,354 have at least one such rival and every control. Three explanations are consistent with the

positive coefficient of the rivals' share, and the test cannot separate them. Most of the credits are uncapped and paid per unit of clean activity (Bistline, Mehrotra and Wolfram, 2023), so that one firm's claims do not reduce another's. The rivals' share may also measure the exposure of the product market as a whole, which three-digit industries do not capture. Finally, because the firm's own share is measured with error, its rivals' share may partly capture the firm's own exposure, so that the estimates do not rule out a competition effect that the other forces offset.

**Table E19.** Product-Market Rivals on the Climate-Policy Days

| | Own share only (1) | + Rivals' environmental share (2) | Two parts, + rivals' clean (3) |
|---|---|---|---|
| Environmental share | 0.616*** | 0.439*** | |
| | (0.108) | (0.113) | |
| Clean technology | | | 0.370*** |
| | | | (0.101) |
| Environmental management | | | 0.142** |
| | | | (0.070) |
| Rivals' environmental share | | 0.472*** | |
| | | (0.135) | |
| Rivals' clean-technology share | | | 0.466*** |
| | | | (0.141) |
| Firm-events | 10,354 | 10,354 | 10,354 |

*Notes.* Six days, for the firm-events with at least one product-market rival that has an environmental share. Rivals: the firm's TNIC-3 rivals with a positive similarity score in the year before the event. Rivals' share: the similarity-weighted mean of the rivals' environmental share (column 2) or clean-technology share (column 3) of the same year, standardized within event. Column (1) omits the rivals' share on the same sample; column (3) enters the two parts of the firm's environmental share. A competition effect would imply a negative coefficient on the rivals' share.

Table E20 compares, on the 2016 election, the full sample with the 835 firms that have both an LSEG score and Trucost emissions, on which the estimate for the environmental share weakens (Section 6.6). The clean-technology part behaves differently. It is $-0.83$ ($t = -3.5$) on these firms and remains $-0.64$ ($t = -2.5$) beside both measures, whereas the environmental management part is positive on them, so that the weak estimate for the environmental share on the rated firms comes from its environmental management part.

**Table E20.** The 2016 Election: The Full Sample and the Firms with Ratings and Emissions

| | | Firms with an LSEG score and Trucost emissions | | | | |
|---|---|---|---|---|---|---|
| | Full sample | Share alone | + Rating | + Carbon | + Both | Full, + both |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *A. The environmental share* | | | | | | |
| Environmental share | −0.820*** | −0.399 | −0.328 | −0.217 | −0.140 | −0.750*** |
| | (0.215) | (0.351) | (0.348) | (0.361) | (0.358) | (0.216) |
| LSEG environmental score | | | −0.429* | | −0.446* | −0.151 |
| | | | (0.254) | | (0.253) | (0.215) |
| Carbon intensity | | | | −1.076*** | −1.090*** | −0.872*** |
| | | | | (0.327) | (0.327) | (0.251) |
| *B. Its two parts* | | | | | | |
| Clean technology | −0.941*** | −0.830*** | −0.788*** | −0.682*** | −0.637** | −0.882*** |
| | (0.162) | (0.240) | (0.230) | (0.260) | (0.251) | (0.161) |
| Environmental management | 0.114 | 0.325 | 0.388* | 0.398* | 0.465** | 0.143 |
| | (0.193) | (0.227) | (0.228) | (0.228) | (0.229) | (0.193) |
| LSEG environmental score | | | −0.455* | | −0.469* | −0.177 |
| | | | (0.251) | | (0.250) | (0.212) |
| Carbon intensity | | | | −0.963*** | −0.976*** | −0.833*** |
| | | | | (0.332) | (0.332) | (0.250) |
| Firm-events | 1,641 | 835 | 835 | 835 | 835 | 1,641 |

*Notes.* The 2016 election, 9 November 2016; the CAR$[-1,+1]$, as observed, with the event controls and NAICS-3 effects. Panel A enters the environmental share; Panel B its two parts together. Columns (2) to (5): the firms with both an LSEG environmental score and Trucost emissions, with the score, carbon intensity, or both added, each standardized among these firms. Column (6): the full sample, with the score and carbon intensity set to zero where missing and an indicator for each missing value (not shown).

## E.4 What Explains the Repricing

Tables E21 and E22 examine the tax credits and clean facilities of Table 15 (Section 7.2), whose sources Internet Appendix IA.III.7 describes. The association with credits changes little across specifications. For facilities, holding earlier facilities fixed reduces the coefficient on any facility from 4.3 to 1.8 and removes the one on announced capital expenditures, because firms that built before August 2022 continue to build, whereas the coefficient on manufacturing facilities does not depend on this control. Among firms with no earlier credit or facility, the coefficients are smaller and significant at the 10 percent level or better, and among all North American firms all six are significant at the 1 percent level. When the four shares enter together, the coefficients of the environmental share are unchanged, none of the other shares has a significant positive coefficient on any of the six outcomes, and the innovation share is negatively associated with mentions of the manufacturing credit ($-0.59$, $t = -2.3$), so that these outcomes single out environmental work among the four.

Table E23 repeats the identification tests of Tables E11 and E12 with the two parts of environmental work entered together. The clean-technology share carries the result in every column, and the environmental management share is never significant at the 5 percent level.

Table E24 reports the investment of the firms of the estimation sample in the year after

**Table E21.** Credits and Facilities: Robustness

| | Tax credits in 10-K filings, 2023–2026 | | | Clean facilities announced, Aug. 2022–June 2026 | | |
|---|---|---|---|---|---|---|
| | Any clean credit (1) | Manufacturing credit (45X) (2) | Number of credits (3) | Any facility (4) | Any manufacturing facility (5) | Announced capex (6) |
| *A. The environmental share* | | | | | | |
| NAICS-3 effects only | 7.29*** | 3.50*** | 0.165*** | 4.34*** | 3.50*** | 0.314*** |
| | (1.18) | (0.98) | (0.036) | (1.18) | (0.96) | (0.084) |
| + earlier activity | 6.90*** | 3.27*** | 0.157*** | 1.84** | 3.00*** | 0.028 |
| | (1.14) | (0.94) | (0.034) | (0.87) | (0.87) | (0.080) |
| + firm controls (Table 15) | 6.81*** | 3.27*** | 0.157*** | 1.93** | 3.13*** | 0.202** |
| | (1.17) | (0.96) | (0.035) | (0.90) | (0.90) | (0.080) |
| No earlier activity | 4.68*** | 1.79* | 0.075*** | 1.79* | 1.63* | 0.388* |
| | (1.44) | (0.99) | (0.024) | (0.92) | (0.85) | (0.230) |
| All North American firms | 5.36*** | 2.70*** | 0.127*** | 2.24*** | 3.34*** | 0.196*** |
| | (0.94) | (0.77) | (0.029) | (0.84) | (0.81) | (0.075) |
| *B. The clean-technology part, with the environmental management part* | | | | | | |
| NAICS-3 effects only | 7.19*** | 3.87*** | 0.165*** | 5.59*** | 3.50*** | 0.284*** |
| | (1.05) | (0.92) | (0.034) | (1.03) | (0.92) | (0.065) |
| + earlier activity | 6.87*** | 3.69*** | 0.158*** | 2.58*** | 3.02*** | 0.027 |
| | (1.04) | (0.90) | (0.033) | (0.92) | (0.90) | (0.060) |
| + firm controls (Table 15) | 6.92*** | 3.73*** | 0.161*** | 2.84*** | 3.22*** | 0.192*** |
| | (1.05) | (0.91) | (0.033) | (0.94) | (0.93) | (0.072) |
| No earlier activity | 5.55*** | 3.12** | 0.092*** | 2.80** | 1.98* | 0.673** |
| | (1.75) | (1.26) | (0.031) | (1.19) | (1.04) | (0.341) |
| All North American firms | 5.46*** | 3.15*** | 0.133*** | 2.91*** | 3.44*** | 0.188*** |
| | (0.90) | (0.77) | (0.029) | (0.87) | (0.83) | (0.068) |
| Firms, row 4 | 2,158 | 2,158 | 2,158 | 2,283 | 2,283 | 774 |
| Firms, row 5 | 3,006 | 3,006 | 3,006 | 3,006 | 3,006 | 1,259 |

*Notes.* Each cell reports the coefficient of the share in one regression, with the dependent variables and estimation of Table 15. Panel A reports the environmental share, and Panel B the clean-technology part, from regressions that also include the environmental management part. Rows 1 to 3 successively add NAICS-3 effects, activity before August 2022, and the firm controls, so that row 3 reproduces Table 15. Row 4 keeps, with the firm controls, the firms with no 10-K mention of production or investment tax credits or of section 45Q before August 2022 (columns 1 to 3) or with no facility announced before August 2022 (columns 4 to 6). Row 5 adds the firms of Compustat North America located outside the United States, with all controls. Rows 1 and 2 cover 2,462 firms (978 in column 6), and the last rows of the table give the firms of rows 4 and 5. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

each year of news (Section 7.4). I code the news of each year from 2015 to 2024 as +1 in the years of the Paris Agreement, the 2020 election, and the agreement on the IRA, as −1 in the years of the 2016 election, the withdrawal from the Paris Agreement, and the 2024 election, and as zero in the other years, and I regress each outcome in year $t+1$ on the firm's share in year $t-1$ and its interaction with the news of year $t$. Because the industry-by-year effects absorb the news itself, the interaction compares, within an industry and year, firms whose hiring was oriented toward clean work to a greater or lesser degree before the news arrived. For capital expenditure, fixed assets, and employment, the interaction with an annual version of the index of growth measures of Table E25 is insignificant.

Table E25 asks whether the repricing depends on how large a part of firm value growth

**Table E22.** Credits and Facilities: The Four Shares

| | Tax credits in 10-K filings, 2023–2026 | | | Clean facilities announced, Aug. 2022–June 2026 | | |
|---|---|---|---|---|---|---|
| | Any clean credit (1) | Manufacturing credit (45X) (2) | Number of credits (3) | Any facility (4) | Any manufacturing facility (5) | Announced capex (6) |
| Environmental share | 6.89*** | 3.37*** | 0.159*** | 2.00** | 3.19*** | 0.197** |
| | (1.17) | (0.97) | (0.035) | (0.91) | (0.92) | (0.081) |
| Social share | −0.67 | −0.85* | −0.019* | −0.61 | −0.53* | −0.029 |
| | (0.76) | (0.51) | (0.011) | (0.60) | (0.28) | (0.171) |
| Governance share | −0.45 | 0.24 | −0.010 | −0.14 | 0.19 | −0.319 |
| | (0.50) | (0.37) | (0.007) | (0.33) | (0.23) | (0.408) |
| Innovation share | −0.48 | −0.59** | −0.014* | −0.24 | −0.30 | −0.156 |
| | (0.37) | (0.26) | (0.007) | (0.25) | (0.24) | (0.209) |
| Activity before August 2022 | Yes | Yes | Yes | Yes | Yes | Yes |
| Firm controls | Yes | Yes | Yes | Yes | Yes | Yes |
| NAICS-3 effects | Yes | Yes | Yes | Yes | Yes | Yes |
| Mean of dependent variable | 2.95 | 1.09 | 0.044 | 3.28 | 1.05 | 33.7 |
| Firms | 2,370 | 2,370 | 2,370 | 2,370 | 2,370 | 974 |

*Notes.* The four shares of 2019–2021 enter together, each standardized across North American firms, and the innovation share therefore leaves the firm controls. The specification is otherwise that of Table 15. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

**Table E23.** Identification with the Two Parts of Environmental Work

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| *A. Adding each existing measure, then all of them (columns as in Table E11, Panel A)* | | | | | | | | | |
| Clean technology | 0.564*** | 0.559*** | 0.495*** | 0.565*** | 0.554*** | 0.408*** | 0.562*** | 0.367*** | 0.316*** |
| | (0.070) | (0.070) | (0.078) | (0.069) | (0.075) | (0.079) | (0.070) | (0.089) | (0.098) |
| Environmental management | 0.090 | 0.088 | 0.079 | 0.076 | 0.088 | 0.071 | 0.083 | 0.039 | −0.120 |
| | (0.066) | (0.066) | (0.068) | (0.066) | (0.067) | (0.066) | (0.067) | (0.068) | (0.088) |
| Firm-events | 12,407 | 12,407 | 12,407 | 12,407 | 12,407 | 12,407 | 12,407 | 12,407 | 11,121 |
| *B. Samples* | | | | | | | | | |
| | Complete cases | Complete cases, all controls | No green patent 2007–13 | Strict newcomers | No clean segment | Elections only | Elections, tax and trade | Trump 2016, tax and trade | |
| Clean technology | 0.488*** | 0.359** | 0.636*** | 0.682*** | 0.398*** | 0.688*** | 0.673*** | 0.945*** | |
| | (0.149) | (0.145) | (0.131) | (0.168) | (0.090) | (0.119) | (0.119) | (0.160) | |
| Environmental management | 0.090 | 0.071 | 0.140 | 0.147 | 0.059 | 0.028 | 0.017 | −0.146 | |
| | (0.087) | (0.092) | (0.101) | (0.121) | (0.065) | (0.115) | (0.116) | (0.193) | |
| Firm-events | 5,688 | 5,688 | 9,516 | 9,100 | 12,150 | 6,657 | 6,657 | 1,641 | |

*Notes.* The regressions of Tables E11 and E12, with the clean-technology and environmental management shares entered together in place of the environmental share. Panel A: columns as in Table E11. Panel B: complete cases are firm-events with every existing measure; strict newcomers exclude utilities and the tenth of industries whose patenting was most green before 2014; column (8) is the 2016 election alone, with the tax and trade controls of Wagner, Zeckhauser and Ziegler (2018).

**Table E24.** Investment after the News

| | Capital expenditure (1) | Growth of PP&E (2) | Employment growth (3) | R&D (4) | Equity issuance (5) | Clean facility (6) |
|---|---|---|---|---|---|---|
| *A. News of the six years* | | | | | | |
| Clean technology × news | 0.094*** | 0.916*** | 0.558** | 0.019 | 0.217 | 1.022** |
| | (0.031) | (0.287) | (0.256) | (0.023) | (0.204) | (0.405) |
| Environmental share × news | 0.084*** | 0.662** | 0.315 | 0.042* | 0.204 | 0.928** |
| | (0.032) | (0.310) | (0.275) | (0.023) | (0.193) | (0.380) |
| Environmental management × news | −0.001 | −0.248 | −0.289* | 0.041** | 0.029 | 0.007 |
| | (0.026) | (0.176) | (0.173) | (0.017) | (0.086) | (0.134) |
| *B. Clean technology × news, one year at a time* | | | | | | |
| IRA agreement, 2022 | 0.240*** | 1.980*** | 0.405 | 0.043 | −0.354 | 1.765* |
| | (0.079) | (0.438) | (0.453) | (0.049) | (0.332) | (0.933) |
| Trump re-elected, 2024 | 0.052 | 0.874 | 0.351 | 0.039 | −0.018 | 1.102** |
| | (0.052) | (0.638) | (0.476) | (0.049) | (0.302) | (0.554) |
| Firm-years | 22,888 | 22,760 | 22,574 | 22,978 | 22,127 | 19,970 |
| Mean of the outcome | 3.32 | 6.71 | 1.49 | 4.27 | 3.21 | 1.64 |

*Notes.* Annual panel of the North American firms of the estimation sample, with outcomes in fiscal years 2016–2025. Each cell is the coefficient of the interaction between the firm's share in year $t-1$, standardized over the panel, and the news of year $t$ in a regression of the outcome in year $t+1$ on the share, the interaction, and Tobin's $q$, the logarithm of assets, return on assets, and leverage in year $t-1$, with firm and NAICS-3-by-year effects. Each share enters a separate regression. In Panel A the news is $+1$ in 2015, 2020, and 2022, $-1$ in 2016, 2017, and 2024, and zero otherwise. In Panel B it is 1 in 2022 only, or $-1$ in 2024 only, so that a positive coefficient is a response in the direction of the news. Outcomes, in percent: capital expenditure and R&D (zero where not reported) over lagged assets, the log growth of net property, plant, and equipment and of employees, the sale of stock over lagged assets, each winsorized at 1/99 within year, and 100 if the firm announces a facility in the Clean Investment Monitor in year $t+1$, which is observed from 2018, so that column (6) uses the news of 2017, 2020, 2022, and 2024. Standard errors clustered by firm in parentheses. *, **, ***: 10, 5, 1 percent.

opportunities represent in general or on organization capital (Section 7.1), and whether it is confined to suppliers of capital goods or absorbed by the firm's stocks of patent applications (Section 6.6).

Table E26 reports the revisions of analysts' forecasts day by day, as observed rather than signed. Forecasts of long-term growth move with the clean-technology share on the agreement on the IRA and with the environmental share on the 2024 election, in the direction of the news, and the revisions on the other days are small and imprecise. The last row of each panel places the revisions around each day among those between all 115 pairs of consecutive monthly IBES consensus forecasts of 2015–2024. For each pair, I regress the revision on the environmental share in a simpler specification and report the share of pairs whose coefficient, signed by the direction of the day, is smaller than that of the day. On the four days other than the IRA and the 2024 election, the revisions of long-term growth lie between the 24th and the 78th percentiles.

Table E27 reports the test of tax capacity of Section 7.4, Table E28 the option-implied expected returns of Section 7.1, and Table E29 the ownership by funds that invest by

**Table E25.** Growth in General, Organization Capital, Suppliers of Capital Goods, and Patent Stocks

| | Environmental share | | Clean technology | | Firm- |
|---|---|---|---|---|---|
| | Share (1) | × rank (2) | Share (3) | × rank (4) | events (5) |
| *A. Interaction with the rank of a measure of growth* | | | | | |
| Tobin's $q$ | 0.597*** | 0.030 | 0.581*** | 0.210 | 12,394 |
| | (0.089) | (0.241) | (0.066) | (0.247) | |
| Total $q$ | 0.590*** | 0.267 | 0.581*** | 0.433* | 10,843 |
| | (0.091) | (0.250) | (0.068) | (0.236) | |
| Market-to-book | 0.584*** | −0.379* | 0.537*** | −0.382* | 12,407 |
| | (0.088) | (0.216) | (0.065) | (0.225) | |
| Minus the forward earnings yield | 0.600*** | 0.136 | 0.595*** | 0.045 | 11,786 |
| | (0.093) | (0.211) | (0.073) | (0.198) | |
| Minus PP&E over assets | 0.587*** | 0.213 | 0.559*** | 0.118 | 12,133 |
| | (0.090) | (0.246) | (0.070) | (0.272) | |
| Index of the four | 0.595*** | −0.035 | 0.569*** | 0.007 | 12,407 |
| | (0.090) | (0.234) | (0.066) | (0.253) | |
| *B. Terciles of the growth index* | | | | | |
| Low | 0.678*** | | 0.626*** | | 3,992 |
| | (0.190) | | (0.143) | | |
| Middle | 0.639*** | | 0.529*** | | 4,059 |
| | (0.159) | | (0.143) | | |
| High | 0.472*** | | 0.511*** | | 4,134 |
| | (0.142) | | (0.132) | | |
| *C. Interaction with the rank of organization capital* | | | | | |
| Capitalized SG&A over assets | 0.652*** | −0.269 | 0.649*** | −0.374 | 10,932 |
| | (0.097) | (0.259) | (0.078) | (0.281) | |
| Peters–Taylor, over assets | 0.583*** | −0.277 | 0.550*** | −0.253 | 12,132 |
| | (0.089) | (0.240) | (0.067) | (0.245) | |
| *D. Suppliers of capital goods and other firms* | | | | | |
| Suppliers of capital goods | 0.796*** | | 0.552*** | | 2,064 |
| | (0.145) | | (0.094) | | |
| All other firms | 0.384*** | | 0.509*** | | 10,343 |
| | (0.099) | | (0.111) | | |
| *E. Stocks of green and other patent applications* | | | | | |
| With both stocks | 0.575*** | | 0.554*** | | 12,407 |
| | (0.091) | | (0.071) | | |

*Notes.* Signed CAR$[-1,+1]$ on the six days, with the controls and effects of Table 11. Columns 1 and 2: the environmental share; columns 3 and 4: the clean-technology share, entered with the environmental management share. Panels A and C add the firm's rank on the measure within its NAICS-3 industry and day, centred to run from −0.5 to 0.5, and its interaction with the share. Tobin's $q$ as in Table A1; total $q$ of Peters and Taylor (2017); market-to-book, minus the logarithm of book-to-market; forward earnings yield, the IBES consensus for the first fiscal year at the last monthly consensus before the day over the price at the end of the previous month, on the same share basis; the index, the mean of the available ranks on $q$, market-to-book, the forward earnings yield, and minus PP&E over assets (at least two), ranked again. Panel B: the share within terciles of the index. Panel C: organization capital over assets, from SG&A capitalized by perpetual inventory since 1975 with a depreciation rate of 15 percent, following Eisfeldt and Papanikolaou (2013) but in nominal terms, or the organization capital of Peters and Taylor (2017). Panel D: suppliers of capital goods are NAICS-3 industries 333, 334, 335, and 336 except motor vehicles (3361), and construction (23). Panel E: the logarithms of one plus the firm's Y02 and other patent applications from 2007 to the year before the share, standardized within event; without them, the coefficients are 0.59 and 0.56. Standard errors clustered by firm in parentheses. *, **, ***: 10, 5, 1 percent.

**Table E26.** Revisions of Analysts' Forecasts by Day

| | Paris 2015 (1) | Trump 2016 (2) | Paris exit 2017 (3) | Biden 2020 (4) | IRA 2022 (5) | Trump 2024 (6) |
|---|---|---|---|---|---|---|
| *A. Long-term growth (pp)* | | | | | | |
| Environmental share | −0.073 | −0.052 | −0.061 | 0.028 | 0.428 | −0.529** |
| | (0.094) | (0.238) | (0.122) | (0.100) | (0.267) | (0.230) |
| Clean technology | −0.077 | −0.099 | −0.063 | −0.083 | 0.539** | −0.362* |
| | (0.106) | (0.221) | (0.121) | (0.088) | (0.230) | (0.199) |
| Environmental management | −0.025 | 0.052 | −0.010 | 0.135 | −0.098 | −0.465 |
| | (0.063) | (0.195) | (0.060) | (0.090) | (0.165) | (0.287) |
| Firms | 956 | 1,187 | 1,218 | 930 | 795 | 857 |
| Percentile among all months | 23.5 | 78.3 | 67.0 | 67.0 | 96.5 | 98.3 |
| *B. EPS, fiscal year 2 (percent of price)* | | | | | | |
| Environmental share | −0.018 | 0.073 | 0.007 | −0.089** | −0.053 | −0.098 |
| | (0.014) | (0.112) | (0.019) | (0.035) | (0.047) | (0.065) |
| Clean technology | −0.021** | 0.057 | 0.009 | −0.045* | −0.003 | −0.028 |
| | (0.010) | (0.103) | (0.017) | (0.025) | (0.040) | (0.072) |
| Environmental management | 0.004 | 0.032 | −0.003 | −0.085* | −0.086** | −0.136*** |
| | (0.014) | (0.037) | (0.008) | (0.046) | (0.043) | (0.046) |
| Firms | 1,200 | 1,524 | 1,662 | 2,158 | 2,450 | 2,393 |
| Percentile among all months | 16.5 | 2.6 | 32.2 | 4.3 | 11.3 | 97.4 |

*Notes.* The revision of the IBES consensus between the last monthly consensus before the event and the first after it (Appendix D.4), regressed on the environmental share or on its two parts entered together, with the controls of Table 11 except market beta and idiosyncratic volatility, and NAICS-3 effects; as observed, not signed. The window of the IRA, 14 July to 18 August 2022, also contains Senator Manchin's withdrawal. Panel A: long-term growth, in percentage points. Panel B: earnings per share for the second fiscal year, in percent of the share price. Percentile: the percentage of the 115 pairs of consecutive months of 2015–2024 whose coefficient on the environmental share, with the logarithms of market equity, openings, employees, and assets and NAICS-3 effects and multiplied by the direction of the day, is smaller than that of the day.

environmental, social, and governance criteria (Section 7.3). Internet Appendix IA.V.9, IA.V.8, and IA.V.7 document the three measures.

Table E30 replaces ownership with the funds' overweight of the stock relative to the market (Section 7.3).

**Table E27.** Tax Capacity and the Response to Policy News

| | IRA days (1) | 2024 election (2) | Pre-IRA days (3) | IRA minus pre-IRA (4) |
|---|---|---|---|---|
| Clean-technology share × low tax capacity | 0.458*** | 0.761** | 0.431*** | 0.026 |
| | (0.127) | (0.384) | (0.163) | (0.215) |
| with size and age interactions | 0.608*** | 1.199*** | 0.293 | 0.315 |
| | (0.161) | (0.411) | (0.225) | (0.282) |
| Loss firms only: IRA minus pre-IRA | | | | −0.047 |
| | | | | (0.265) |
| Firm-events | 5,432 | 2,611 | 4,673 | 10,105 |

*Notes.* Low tax capacity: a tax loss carryforward or negative pretax income in the fiscal year before the event. IRA days: 14 and 27 July 2022; pre-IRA days: the Paris Agreement, the 2016 election, and the Paris withdrawal. Column 4 interacts every term with an indicator of the IRA days. Size and age interactions as in Section 8.

**Table E28.** Option-Implied Expected Returns

| | 30-day options (1) | 91-day options (2) |
|---|---|---|
| Environmental share | −0.210 | −0.036 |
| | (0.209) | (0.181) |
| Clean-technology share | −0.185 | −0.069 |
| | (0.174) | (0.134) |
| Environmental management share | −0.062 | 0.118 |
| | (0.163) | (0.206) |
| Firm-events | 11,701 | 11,701 |

*Notes.* Six days, signed. Dependent variable: change in one half of the stock's SVIX index, the cross-sectional component of the expected excess return of Martin and Wagner (2019), in percentage points a year, from trading days −5 to −2 to days +2 to +5, from OptionMetrics surfaces.

**Table E29.** ESG-Fund Ownership and the Response to Policy News

| | Six days (1) | From 2020 (2) | Before 2020 (3) | IRA agreement (4) | 2024 election (5) |
|---|---|---|---|---|---|
| Environmental share × ESG-fund ownership | 0.038* | 0.093*** | 0.017 | 0.120*** | 0.138** |
| | (0.022) | (0.032) | (0.020) | (0.038) | (0.055) |
| with curvature and size | 0.052** | 0.100*** | 0.032 | 0.129*** | 0.153** |
| | (0.025) | (0.033) | (0.023) | (0.039) | (0.060) |
| Environmental share | 0.364*** | 0.408*** | 0.188 | 0.427*** | 0.715** |
| | (0.097) | (0.136) | (0.115) | (0.133) | (0.296) |
| Firm-events | 12,407 | 7,733 | 4,674 | 2,717 | 2,611 |

*Notes.* ESG-fund ownership: shares held by mutual funds whose names mark them as environmental, social, and governance funds at their last report in the six months before each day, over shares outstanding, standardized within day. Second row adds the squared share and its interaction with log market equity.

**Table E30.** ESG Funds' Overweights and the Response to Policy News

| | Six days | | 2015–2017 | | 2020–2024 | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Environmental share | 0.605*** | 0.530*** | 0.402*** | 0.399*** | 0.715*** | 0.587*** |
| | (0.091) | (0.094) | (0.111) | (0.117) | (0.133) | (0.137) |
| Rank of the ESG-fund overweight | −0.287 | −0.333 | −0.372 | −0.374 | −0.215 | −0.295 |
| | (0.204) | (0.203) | (0.235) | (0.234) | (0.309) | (0.307) |
| Environmental share × rank | | 0.389* | | 0.013 | | 0.667** |
| | | (0.216) | | (0.241) | | (0.319) |
| Firm-events | 12,407 | 12,407 | 4,674 | 4,674 | 7,733 | 7,733 |

*Notes.* Signed CAR$[-1,+1]$, with the controls and effects of Table 11. Overweight: the stock's weight in the combined holdings of the ESG funds of Table E29 at their last report before the day, valued at the end of the previous month, minus its weight in the market value of CRSP stocks in that month, over the latter; winsorized at 1/99 within day and ranked within NAICS-3 industry and day, centred to run from −0.5 to 0.5. Standard errors clustered by firm in parentheses. *, **, ***: 10, 5, 1 percent.

## E.5 Where the Repricing Is Largest

Table E31 reports the subgroup estimates that Figure 7 plots and the regression with all interactions. Table E32 compares tangible intensity in the year of the event with its mean over fiscal 2011–2013, before the sample period. With the current ratio, the coefficient of the clean-technology share falls with tangible intensity and is largest in the lowest tercile, whereas with the pre-sample ratio the interaction is smaller and insignificant. Part of the difference comes from the sample, since the 1,735 firm-events without Compustat data in

2011–2013 drop out of the pre-sample version and their clean-technology coefficient is large (last block of the table). They include the firms that entered Compustat after 2013, which are young by the paper's measure of age. The table also splits the sample by age; Table E31 splits it by size.

**Table E31.** Where the Repricing Is Largest

| | Environmental share (1) | Clean technology (2) | Firm-events (3) |
|---|---|---|---|
| *A. Weight of the capability in the firm* | | | |
| Low PP&E/assets, young firms | 1.507*** | 1.543*** | 2,236 |
| | (0.408) | (0.389) | |
| Low PP&E/assets, old firms | 0.610** | 0.922** | 1,764 |
| | (0.266) | (0.454) | |
| Middle PP&E/assets, young firms | 0.612*** | 0.554*** | 1,873 |
| | (0.202) | (0.135) | |
| Middle PP&E/assets, old firms | 0.481* | 0.382 | 2,031 |
| | (0.251) | (0.247) | |
| High PP&E/assets, young firms | 0.421* | 0.410** | 1,460 |
| | (0.217) | (0.188) | |
| High PP&E/assets, old firms | 0.377** | 0.423** | 2,327 |
| | (0.166) | (0.164) | |
| Small firms | 0.703*** | 0.663*** | 6,066 |
| | (0.135) | (0.119) | |
| Large firms | 0.413*** | 0.391*** | 6,253 |
| | (0.113) | (0.103) | |
| *B. Visibility* | | | |
| High analyst coverage | 0.857*** | 0.868*** | 6,713 |
| | (0.139) | (0.099) | |
| Low analyst coverage | 0.334*** | 0.259** | 5,621 |
| | (0.121) | (0.111) | |
| *C. One regression with all interactions* | | | |
| Environmental share | 0.756*** | | 12,133 |
| | (0.143) | | |
| × PP&E/assets (per SD) | −0.199*** | | |
| | (0.076) | | |
| × small | 0.523*** | | |
| | (0.157) | | |
| × low analyst coverage | −0.800*** | | |
| | (0.187) | | |
| × young | 0.014 | | |
| | (0.118) | | |

*Notes.* The six days, within groups of firms. Column 1 reports regressions on the environmental share and column 2 the clean-technology share from regressions on both of its parts. Panel A: terciles of property, plant, and equipment over assets within event; young firms are younger than the event's median firm, counting from the first Compustat record; small firms have market equity below the event's median. Panel B: the number of analysts with an IBES forecast, zero when none, above or below the event's median. Panel C: all firm-events with property, plant, and equipment and age observed; the environmental share is interacted with property, plant, and equipment over assets, standardized within event, and with the indicators of small, low-coverage, and young firms, which also enter alone (not shown).

**Table E32.** Tangible Intensity before the Sample, Pre-Sample Data, and Age

| | Clean technology (1) | Environmental management (2) | Firm-events (3) |
|---|---|---|---|
| *Interaction with tangible intensity* | | | |
| Tangible intensity in the year | −0.150** | −0.008 | 12,133 |
| | (0.070) | (0.046) | |
| Tangible intensity in 2011–2013 | −0.055 | 0.029 | 10,363 |
| | (0.067) | (0.045) | |
| *By tercile of tangible intensity* | | | |
| Lowest tercile, in the year | 1.405*** | 0.118 | 4,042 |
| | (0.269) | (0.158) | |
| Middle tercile, in the year | 0.494*** | −0.197 | 3,990 |
| | (0.112) | (0.140) | |
| Highest tercile, in the year | 0.432*** | 0.084 | 3,920 |
| | (0.111) | (0.081) | |
| Lowest tercile, in 2011–2013 | 0.473 | −0.000 | 3,437 |
| | (0.343) | (0.112) | |
| Middle tercile, in 2011–2013 | 0.620*** | −0.026 | 3,398 |
| | (0.082) | (0.211) | |
| Highest tercile, in 2011–2013 | 0.353*** | 0.095 | 3,350 |
| | (0.082) | (0.071) | |
| *Pre-sample data and firm age* | | | |
| With Compustat data in 2011–2013 | 0.549*** | 0.095 | 10,330 |
| | (0.062) | (0.062) | |
| Without Compustat data in 2011–2013 | 0.773** | −0.131 | 1,735 |
| | (0.358) | (0.485) | |
| Young (below the event's median age) | 0.629*** | −0.037 | 5,729 |
| | (0.095) | (0.142) | |
| Old | 0.472*** | 0.142** | 6,328 |
| | (0.127) | (0.064) | |

*Notes.* Six days, on the clean-technology and environmental management shares entered together. Tangible intensity: property, plant, and equipment over assets in the year of the shares or its mean over fiscal 2011–2013 in Compustat North America, standardized within event. First block: the interactions of each part with tangible intensity; second block: the coefficients within terciles of tangible intensity, formed within event; third block: groups of firm-events with tangible intensity in the year, by whether the firm has Compustat data in 2011–2013 and by age relative to the event's median. Standard errors clustered by firm in parentheses. *, **, ***: 10, 5, 1 percent.

# Appendix F. Market Evidence beyond Event Returns

Section 6 shows that firm value moves with the orientation of hiring when news about the policy arrives. This appendix reports further market evidence. Sections F.1 and F.2, which Section 7.1 summarizes, ask whether prices fully reflect the information in the measure, that is, whether the market is predictably surprised when earnings reveal it and whether firms with higher shares earn different average returns.

## F.1 Earnings Announcements

If prices do not fully reflect the information in a measure of work that no account records, the market may be surprised when earnings reveal that work. Edmans (2011) shows this pattern for the "100 Best Companies to Work For in America", which exhibit more positive earnings surprises and announcement returns. I apply the same test. For every quarterly earnings announcement in the two years after the share is measured, I compute the market-adjusted return over the three days around the announcement and relate it to the share. Table F1, Panel A, shows no such pattern. Announcement returns are unrelated to the environmental share in the first year (0.00, standard error 0.05) and in the second ($-0.01$), and so is the sum of a year's announcements. The clean-technology share is unrelated to them as well ($-0.03$), and the environmental management share only marginally (0.07, $t = 1.6$).

The absence of announcement returns is not due to a lack of news in earnings. Firms with a higher environmental share report earnings below the analysts' consensus in the following two years, by 0.10 and 0.16 standard deviations of the forecasts per standard deviation of the share ($t$-statistics of $-3.0$ and $-3.7$; Table F2). Analysts thus overestimate the near-term earnings of these firms, but prices do not respond when the shortfall is announced. I find no evidence of the predictable underreaction that Edmans (2011) documents.

**Table F1.** Earnings Announcements and Average Returns

| | Environmental share (1) | Clean technology (2) | Environmental management (3) | Observations (4) |
|---|---|---|---|---|
| *A. Earnings-announcement returns, CAR[−1, +1] in percent per SD* | | | | |
| Year $t+1$, each announcement | 0.002 | −0.028 | 0.069 | 93,681 |
| | (0.051) | (0.050) | (0.044) | |
| Year $t+2$, each announcement | −0.006 | 0.004 | −0.004 | 77,986 |
| | (0.062) | (0.060) | (0.042) | |
| Year $t+1$, annual sum | −0.049 | −0.165 | 0.266 | 23,203 |
| | (0.210) | (0.208) | (0.178) | |
| Year $t+2$, annual sum | −0.047 | −0.014 | −0.013 | 19,312 |
| | (0.251) | (0.245) | (0.167) | |
| *B. Monthly excess returns, Fama–MacBeth (percent per SD)* | | | | |
| Environmental share | −0.022 | | | 114 months |
| | (0.083) | | | |

*Notes.* Panel A (Edmans, 2011): market-adjusted return over the three days around each quarterly earnings announcement in year $t+1$ or $t+2$, in percent, winsorized at the 1st and 99th percentiles within year, on the standardized environmental share of year $t$ (column 1) or its two parts (columns 2 and 3), the logarithms of market equity, assets, employees, and openings, and Tobin's $q$, with NAICS-3-by-quarter effects (by year for the annual sum, scaled to four announcements); standard errors clustered by firm. Panel B: monthly excess returns from July of $Y+1$ to June of $Y+2$ on the environmental share of year $Y$, log market equity, log book-to-market, and momentum; mean of 114 monthly cross-sections, with Newey–West standard errors (six lags). $^{*}$, $^{**}$, and $^{***}$ denote significance at the 10, 5, and 1 percent levels.

**Table F2.** Is the Information in Prices?

*A. Earnings surprises and forecast revisions*

| | Scale controls | | | Full controls | | |
|---|---|---|---|---|---|---|
| | SUE, $t+1$ | SUE, $t+2$ | Revision, $t+1$ | SUE, $t+1$ | SUE, $t+2$ | Revision, $t+1$ |
| Environmental share | -0.102*** | -0.157*** | -0.001 | -0.064* | -0.116*** | -0.000 |
| | (0.034) | (0.042) | (0.001) | (0.034) | (0.042) | (0.001) |
| Firm-years | 18,346 | 14,645 | 19,702 | 17,530 | 14,045 | 18,840 |

*B. Interaction with low analyst coverage (scale controls)*

| | SUE, $t+1$ | SUE, $t+2$ | Revision, $t+1$ |
|---|---|---|---|
| Environmental share | -0.122*** | -0.173*** | -0.001 |
| | (0.037) | (0.045) | (0.001) |
| × low coverage | 0.046 | 0.039 | 0.001 |
| | (0.050) | (0.057) | (0.001) |

*C. Monthly returns*

| | Quintiles, all firms | | Quintiles within industry-year | | Fama–MacBeth | |
|---|---|---|---|---|---|---|
| | High minus low | FF5 + MOM alpha | High minus low | FF5 + MOM alpha | Share | × low coverage |
| Environmental | 0.11 | 0.09 | -0.06 | 0.02 | -0.022 | -0.069 |
| | [0.45] | [0.39] | [-0.35] | [0.12] | (0.083) | (0.070) |
| Clean technology | 0.58** | 0.44* | -0.02 | 0.10 | | |
| | [2.39] | [1.70] | [-0.11] | [0.75] | | |
| Environmental management | -0.74** | -0.65** | 0.09 | 0.02 | | |
| | [-2.28] | [-2.32] | [0.47] | [0.08] | | |
| LSEG environmental score | 0.17 | -0.06 | 0.15 | -0.04 | | |
| | [0.41] | [-0.23] | [0.58] | [-0.26] | | |
| Months | | | 114 | | 114 | |

*Notes.* Market sample; the environmental share and its parts as in Section 3. Panel A: standardized unexpected earnings (SUE), the mean over the quarters of year $t+1$ or $t+2$ of actual minus consensus earnings in standard deviations of forecasts, controlling for the SUE of year $t$, and the revision of the consensus for fiscal year $t+1$, scaled by price. Scale controls: log assets, employees, and openings, with NAICS-3-by-year effects; full controls add leverage, PP&E, capital expenditures, R&D (with a missing indicator), and SG&A over assets, return on assets, sales growth, log age, and a dividend indicator. Panel B: interactions with below-median analyst coverage. Panel C: value-weighted top-minus-bottom quintile portfolios, formed each June and held a year, monthly percent, alphas against the five factors of Fama and French (2015) and momentum (Carhart, 1997), Newey–West $t$-statistics in brackets (Newey and West, 1987), ranked across all firms or within NAICS-3-by-year; and monthly Fama–MacBeth regressions with log market equity, book-to-market, and momentum. Standard errors clustered by firm in parentheses unless noted. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

## F.2 Average Returns

If prices did not fully reflect the information in the share, or if the orientation of hiring carried a risk premium, firms with a higher share would earn different average returns. In monthly Fama–MacBeth regressions over 2015–2024, the coefficient of the environmental share is −0.02 (standard error 0.08; Table F1, Panel B). A portfolio long the top and short the bottom quintile of the share earns 0.11 percent a month, with a five-factor-and-momentum alpha of 0.09 percent, and neither is significant (Table F2). Sorted on the clean-technology share across all firms, the portfolio earns 0.58 percent a month ($t = 2.4$), but this spread reflects the industries toward which the share tilts, since it disappears when firms are ranked within industry-years (−0.02). Nor does the share predict returns differently among firms with below-median analyst coverage (interaction of −0.07, $t = -1.0$).

Ben Dor et al. (2022) find that firms with high abnormal ESG hiring in 2015 outperformed, with a value-weighted five-factor alpha of 4.18 percent a year over 2016–2020 ($t = 1.93$) among firms sensitive to environmental issues. Table F3 applies their design to the environmental share. I rank the 900 largest firms by the share in 2015 minus its average in earlier years, within GICS sectors, and hold the high-minus-low portfolio over 2016–2020. Since the environmental pillar weights they use are not available to me, I approximate their environment-sensitive sectors by energy, materials, industrials, and utilities. The 2015 cohort also earns a positive alpha, of 7.7 percent a year, and it does not come from the climate-policy days, since it is unchanged when their months are dropped. The alpha is fragile, however. ExxonMobil and General Electric alone account for 39 percent of the bottom tercile's value weight, and their excess returns over 2016–2020 were −36 and −62 percent. With equal weights, the alpha is −1.0 percent, and when the portfolio is formed anew each December from 2015 to 2023 and held for the following year, the value-weighted alpha is −4.1 percent a year ($t = -2.0$). Across all sectors, the 2015 cohort earns 3.1 percent ($t = 1.9$) and the yearly cohorts −1.7 percent.

I find no robust evidence that average returns differ with the share in this period, and hence no sign that prices fail to reflect the information in the measure or that it carries a premium. A decade of returns is short for estimating a premium, however, and realized returns can reflect unexpected shifts rather than expected returns, as Pástor, Stambaugh and Taylor (2022) show for green stocks.

**Table F3.** Abnormal Environmental Hiring and Returns

| | Environmental share (1) | Clean technology (2) | Environmental management (3) | Observations (4) |
|---|---|---|---|---|
| *High-minus-low tercile, five-factor alpha (percent a year)* | | | | |
| *Environment-sensitive sectors* | | | | |
| 2015 cohort, value-weighted | 7.72*** | 6.27*** | 5.29*** | 60 months |
| | (1.69) | (1.86) | (1.91) | |
| without the policy months | 7.57*** | 5.56*** | 5.88*** | 57 months |
| | (1.79) | (2.05) | (1.97) | |
| 2015 cohort, equal-weighted | −1.01 | 1.04 | −2.29 | 60 months |
| | (1.56) | (1.40) | (1.66) | |
| Yearly cohorts, value-weighted | −4.11** | −3.16* | 0.32 | 108 months |
| | (2.09) | (1.82) | (2.23) | |
| Yearly cohorts, equal-weighted | −1.41 | 0.04 | −0.78 | 108 months |
| | (1.05) | (1.05) | (0.99) | |
| *All sectors* | | | | |
| 2015 cohort, value-weighted | 3.07* | 3.18** | 0.95 | 60 months |
| | (1.62) | (1.49) | (1.76) | |
| Yearly cohorts, value-weighted | −1.70 | 0.17 | −0.66 | 108 months |
| | (1.76) | (1.48) | (1.29) | |
| Yearly cohorts, equal-weighted | −1.14 | 1.40* | −0.75 | 108 months |
| | (0.85) | (0.79) | (0.58) | |

*Notes.* The design of Ben Dor et al. (2022, Exhibit 11). Among the 900 largest firms, the abnormal share is the year's share minus the firm's mean over previous years; the portfolio is long the top and short the bottom tercile within GICS sectors. Environment-sensitive sectors: energy, materials, industrials, and utilities. The 2015 cohort is held from 2016 to 2020, and its second row drops the months of the climate-policy days; yearly cohorts are formed each December from 2015 to 2023 and held for the next year. Alphas against the five factors of Fama and French (2015), in percent a year, with Newey–West standard errors (three lags) in parentheses; ExxonMobil and General Electric hold 39 percent of the 2015 bottom tercile's value weight. *, **, and *** denote significance at the 10, 5, and 1 percent levels.